Social Security Design and Its Political Support*

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Abstract

The design of social security systems shapes the political support for intergenerational transfers. In an overlapping generations model with heterogeneous households and probabilistic voting, we study how contributive vs. universal pensions and the link between benefits and earnings affect the social security tax rate, labor supply and capital accumulation. Taxes are lower when benefits are earnings-related, and a universal pension raises them when informal retirees are poorer than formal ones. Calibrated to Argentina, the model accounts for about three fifths of the rise in pension spending over the 2005–2010 expansion of coverage. Calibrated to the U.S., we show that pension spending is primarily determined by ageing and the earnings link. Testing it against 21 OECD countries, we show that with a given pension design the model correctly predicts that pension spending increases with ageing and not with income inequality. The model with a given pension design is seemingly at odds with the data in predicting that earnings-related benefits are associated with lower taxes. We reconcile this by letting the electorate also choose the design: we show that an interior choice requires redistribution through pensions to be costly unless the rich vote much more often than the poor. Testing it on the 21 OECD countries, we show that spending and the earnings link in the model rise together as population growth falls and are still unrelated to inequality, as in the data.

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1 Introduction

Pensions are the largest transfer program of most governments. In 2020 public pension spending averaged about 8% of GDP across the OECD and close to 4% in emerging and middle-income economies.[1] Systems differ along two key dimensions. First, some systems condition benefits on a contribution record while others offer universal entitlements. Second, Bismarckian systems tie benefits and past earnings closely together, whereas Beveridgean systems pay flat benefits. Both dimensions shape the labor supply of workers, the redistribution embedded in the system, and the coalition of voters that supports it.

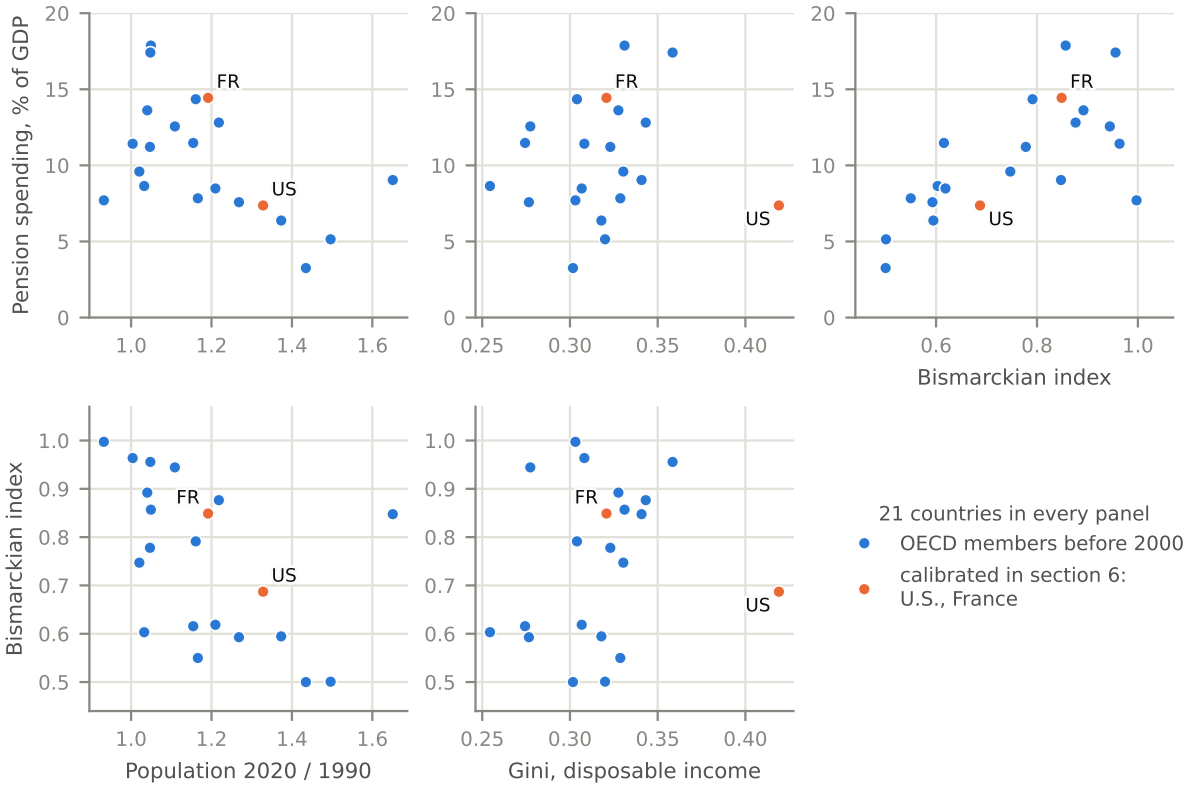
Given their size, it is no surprise that the design of social security systems remains a central concern in policy debates. As populations age, policymakers must navigate difficult trade-offs among intergenerational equity, fiscal sustainability, and labor market efficiency. Moreover, pension arrangements are not determined by actuarial considerations alone: they are the outcome of political processes in which heterogeneous voters and organized interests shape the level and composition of benefits and contributions. A political-economy perspective is therefore critical for understanding why countries differ in the size of their systems and in their design. Figure 1.1 shows how pension spending, population ageing, income inequality, and a Bismarckian index correlate for 21 OECD countries. Both pension spending and the index rise as population growth falls, and they rise together. Neither shows a strong correlation with income inequality. We build a dynamic politico-economic model to explain the size of the system and its design, and their relation to ageing and inequality.

We study the question in an overlapping generations model with capital accumulation and heterogeneous households. Households consume, save, and supply labor, and as voters, they choose among policies on labor income taxes and retirement benefits. Policy preferences vary by income and age; we model the resolution of these conflicts through probabilistic voting with elections taking place every period. Policy decisions not only impact economic outcomes, but also affect future policy decisions. Voters take these effects into account. We study Markov equilibria in which only fundamental state variables enter. Households differ in their productivity and propensity to vote. In addition, a share of households work in an informal sector or get income from home production, where labor is not taxed and savings do not enter the capital stock. Preferences are those of Greenwood et al. (1988), so that labor supply depends on the after-tax wage and how current effort raises future benefits. Benefits combine a flat component with an earnings-related one and informal households receive a universal pension.

We show that when utility is logarithmic, politico-economic equilibria retain analytical tractability despite the presence of heterogeneity and institutional richness. With

[1] The dispersion is wide even among rich countries: a bit more than 3% of GDP in Ireland and Chile and more than 17% in Italy and Greece (OECD 2021).

Figure 1.1: Population growth, income inequality, pension spending and design



Note: 21 OECD countries, selected as in appendix G.1. Pension spending is public cash spending on old age and survivors in 2020, in % of GDP; population growth is the population of 2020 relative to 1990; the Gini is that of disposable income. The Bismarckian index is $I = 1/(2-\theta)$, with θ the earnings-linked share of the public pension net of minimum top-ups and tested benefits. Sources: OECD (SOCX, *Pensions at a Glance 2021* and population statistics), World Bank WDI and UNU-WIDER WIID.

hand-to-mouth informal households, the equilibrium tax is a function of parameters and demographics alone. The analytical frame clarifies how population ageing, retirees' political influence, pension design, and income inequality shape the equilibrium tax rate. A more Bismarckian benefit formula lowers the equilibrium tax, since it shifts benefits from poor retirees, who value them more at the margin, to rich ones, so that retirees as a group support a smaller system. The effect of a more universal pension is ambiguous: it raises the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. When utility is constant relative risk aversion (CRRA), we show that the state space for equilibrium policies can be reduced to a single aggregate state and provide numerical solutions.[2]

We present two empirical applications of the model. First, we study Argentina as the case in which the design changed: from 2005 a series of tax amnesties raised the coverage of the elderly from 68% to over 90%, and pension spending rose by 1.9% of GDP by 2010. In the calibrated model the reform accounts for about three fifths of that rise. The model also predicts modest declines in labor supply and savings, and a fall in

[2]When informal households can save, the state space includes two states.

the savings of the newly covered households relative to other households, consistent with what Gonzalez-Rozada and Ruffo (2016) estimate for the amnesties. Second, initially taking the design as given, we calibrate the model to the U.S. and explore how income distribution, voting profile, pension design, ageing, and leisure preferences shape the politico-economic equilibrium. We find that, at intertemporal elasticities of substitution up to one, the size of the pension system increases strongly with ageing and with flat benefits, the latter measured from one corner of the design to the other, and responds much less to other characteristics. We then construct a set of 21 synthetic countries based on our calibrated U.S. model, each with the observed design, demography, income distribution and propensities to vote of one of the countries in figure 1.1. The model predicts pension spending that falls with population growth about as steeply as in the data and, like spending in the data, is unrelated to income inequality. The model with a given design is at odds with the data in one important respect: it predicts that a more earnings-related pension design lowers the pension spending, the opposite of what figure 1.1 shows.

We then let the electorate choose the earnings link as well as the tax. Without a cost of redistribution the choice is a corner. The flat component of benefits is a pure transfer within the retired cohort, from retirees above mean income to those below, and an electorate that weighs retirees by their marginal utility wants all of it: we prove that unless propensities to vote rise sharply with income, the design chosen with the tax is fully Beveridgean, whatever the tax rate, the demography and the income distribution.[3] An interior design requires flat benefits to be more expensive than earnings-related ones. We introduce a deadweight cost of the redistribution the flat component performs, read as the loss from the implicit tax it levies on earnings (Summers 1989) and rising with the square of that tax as in Harberger (1964). A single cost parameter is calibrated so that the U.S. electorate re-elects its observed design. Taken to the same 21 countries at the U.S. cost of redistribution, this resolves the wrong correlation between pension design and spending from the model with a given design: an ageing population chooses both a more Bismarckian design and higher pension spending, which is sufficient to create the positive correlation we see in figure 1.1. Similarly in line with data, neither the design nor spending is related to inequality.

The paper belongs to the literature on politico-economic equilibria of social security (e.g. Cooley and Soares 1999; Galasso 1999; Forni 2005; Gonzalez-Eiras and Niepelt 2008; Song 2011), surveyed by Galasso and Profeta (2002). As in Gonzalez-Eiras and Niepelt (2008) and Song (2011), in our model under log-GHH preferences and with hand-to-mouth informal households the equilibrium has no strategic effects and taxes increase with the

[3]The corner survives the timing of section 7: at the U.S. calibration under log preferences the electorate also picks the Beveridgean corner when it chooses the design one period ahead of the benefits it governs (appendix C.1).

political power of retirees and with ageing, and as in Song (2011) they increase with income inequality when propensities to vote are equal; as in Uchida and Ono (2024), the equilibrium tax is a function of parameters and demographics alone; and as in Gonzalez-Eiras and Niepelt (2005), the effect of ageing under CRRA preferences is close to the logarithmic one after recalibration. What we add is a pension system that differs in two dimensions, in coverage and in the earnings link, and workers who differ in productivity and in their propensity to vote. Neither adds a state when informal households are hand-to-mouth, and the calibrated model quantifies what each feature of the design does to the tax rate, labor supply and savings.

On the design of the system, Galasso and Profeta (2002) leave open why some systems are Beveridgean and others Bismarckian, and why the Bismarckian ones are larger, two questions that a vote on the size of the system alone cannot answer. Casamatta et al. (2000) study the political support of Bismarckian against Beveridgean benefits, Conde-Ruiz and Profeta (2007) the joint choice of size and design, Koethenbuerger et al. (2008) the trade-off by which more redistributive systems are smaller, and Cremer et al. (2007) the choice of type and generosity. In Conde-Ruiz and Profeta (2007) majority voting ties the design to inequality: high inequality leads to a small Beveridgean system and low inequality to a larger Bismarckian one. We embed the choice in a dynamic equilibrium with capital accumulation and labor supply, in which the design is chosen every period, one period ahead of the benefits it governs, and take it to the data. Across the countries of figure 1.1 the design is unrelated to inequality, and so is the design that our model predicts under probabilistic voting.

Section 2 presents the model and the equilibrium concepts. Section 3 derives the equilibrium under log-GHH preferences, in which the tax is a function of parameters and demographics alone, and its comparative statics. Section 4 describes how the equilibria are computed outside that case. Section 5 calibrates the model to Argentina and compares its prediction with the observed reform. Section 6 calibrates it to the U.S., ranks the determinants of pension spending and tests the spending it predicts for 21 synthetic OECD countries at their observed designs. Section 7 lets the electorate choose the design: it introduces the cost of redistribution, calibrates it on the U.S., ranks the determinants of the design there and tests the designs that the 21 synthetic countries choose (section 7.3). Section 8 concludes. The appendix collects proofs, derivations, calibration details, and selected robustness checks. The online appendix[4] collects auxiliary tables and figures and presents additional robustness checks; a technical documentation accompanies the paper (Berg and Gonzalez-Eiras 2026).

[4] championape.github.io/SSDclaude

2 The Model

2.1 Demographics, Inequality, and Institutions

We consider an economy populated by overlapping generations of workers and retirees. Within each generational cohort, we consider $J+1$ household types indexed $j = 0, \ldots, J$. The $j > 0$ household types allocate time between labor and leisure, pay taxes on labor income, consume, and save. When they reach retirement, they spend the return on their savings and any social security benefits they may receive, and die. Type $j = 0$ households similarly allocate time between labor and leisure, consume, and save for retirement. However, they supply labor to an informal labor market and are thus not subject to labor income taxes, and they save for retirement using an informal savings technology. The distribution of the formal household types $j > 0$ is given by γ_j with $\sum_{j>0} \gamma_j = 1$, and each cohort has γ_0 informal households, of type $j = 0$, per formal household, a share $\gamma_0/(1 + \gamma_0)$ of the cohort. The ratio of formal workers to formal retirees in period t follows a deterministic process, and is given by ν_t; since γ_0 is constant, it is also the ratio of all workers to all retirees.

The government runs a pay-as-you-go (PAYG) pension system with a balanced budget. Each period, the government imposes a tax on labor supplied in the formal sector and transfers the collected amount directly to retirees. We consider social security systems that differ along two dimensions: whether workers need to qualify to receive full benefits or not and the type of benefits provided, either tied to earnings or flat-rate. We describe the different systems in section 2.4.

Policy decisions are taken by a government that acts in the interest of voters, but lacks commitment.

2.2 Technology

In the formal sector, a continuum of competitive firms transform aggregate capital and labor into output. We define the aggregate formal labor supply as

$$H_t = N_t h_t, \qquad h_t \equiv \sum_{i>0} \gamma_i \eta_i h_{t,i}, \tag{1a}$$

where N_t is the number of formal workers at time t, $h_{t,i}$ is time allocated to labor by household type i, and η_i is their labor productivity. Capital is owned by formal retirees and fully depreciates after a period. The economy-wide capital stock per formal worker, k_t, therefore corresponds to the economy-wide per capita savings of formal workers in the

previous period, s_{t-1}, normalized by ν_t. We define the aggregate capital stock from

$$K_t = N_{t-1} s_{t-1}, \qquad s_{t-1} \equiv \sum_{i>0} \gamma_i s_{t-1,i}. \tag{1b}$$

The representative firm produces output using labor and capital and a conventional Cobb-Douglas technology

$$Y_t = K_t^{\alpha} H_t^{1-\alpha} \tag{2}$$

In equilibrium this implies the following factor prices:

$$R_t = \alpha \left(\frac{s_{t-1}}{\nu_t h_t} \right)^{\alpha-1}, \qquad w_t = (1-\alpha) \left(\frac{s_{t-1}}{\nu_t h_t} \right)^{\alpha}. \tag{3}$$

Informal households produce with a linear technology, and we index their factor prices to the formal sector's: the informal wage w_t^0 is the formal wage that would prevail at the same capital per formal worker, s_{t-1}/ν_t, without the pension system, and the return on informal savings is proportional to the formal return, $R_t^0 = r_0 R_t$ (appendix A). At a given capital stock the informal wage therefore does not respond to taxes: a payroll tax that lowers formal hours raises the formal wage, but not informal earnings or the old-age endowment of the hand-to-mouth informal households of section 3.

2.3 Preferences and Household Choices

Households value consumption and leisure, and discount the future at factor $\beta \in (0,1)$. Representing utility by a function $u(c,h)$ with $u'_c \geq 0, u'_h \leq 0, u''_{cc} \leq 0, u''_{hh} \leq 0$, the welfare of a young household of type i born in period t is defined by

$$u\left(c_{1,t}^i, h_{t,i}\right) + \beta u\left(c_{2,t+1}^i, 0\right). \tag{4}$$

The budget constraints for formal and informal workers are given by:

$$c_{1,t}^i = w_t \eta_i h_{t,i}(1-\tau_t) - s_{t,i}, \qquad i > 0, \tag{5a}$$
$$c_{1,t}^0 = w_t^0 \eta_0 h_{t,0} - s_{t,0}, \tag{5b}$$

where τ_t is the social security tax rate. As retirees, households receive a pension b_{t+1}^i that may depend on the individual household's contribution, and thus on their labor supply, $h_{t,i}$:

$$c_{2,t+1}^i = s_{t,i} R_{t+1} + b_{t+1}^i(h_{t,i}), \qquad i > 0, \tag{5c}$$
$$c_{2,t+1}^0 = s_{t,0} R_{t+1}^0 + b_{t+1}^0(h_{t,0}). \tag{5d}$$

Optimization yields the first order conditions that represent a leisure/consumption tradeoff and a conventional Euler condition for intertemporal consumption smoothing:[5]

$$-\frac{\partial u_i}{\partial h_{t,i}} = \frac{\partial u_i}{\partial c^i_{1,t}}\left(w_t(1-\tau_t)\eta_i + \frac{\partial b^i_{t+1}/\partial h_{t,i}}{R_{t+1}}\right), \qquad i>0 \tag{6a}$$

$$-\frac{\partial u_0}{\partial h_{t,0}} = \frac{\partial u_0}{\partial c^0_{1,t}}\left(w^0_t\eta_0 + \frac{\partial b^0_{t+1}/\partial h_{t,0}}{R^0_{t+1}}\right), \tag{6b}$$

$$\frac{\partial u_i}{\partial c^i_{1,t}} = \beta R_{t+1}\frac{\partial u_i}{\partial c^i_{2,t+1}}, \qquad i>0 \tag{6c}$$

$$\frac{\partial u_0}{\partial c^0_{1,t}} = \beta R^0_{t+1}\frac{\partial u_0}{\partial c^0_{2,t+1}}. \tag{6d}$$

Optimal savings and labor supply decisions can, in general, be written as

$$h_{t,i} = h_{t,i}\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right) \tag{7a}$$

$$s_{t,i} = s_{t,i}\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right), \tag{7b}$$

where the dependence on h_{t+1}, τ_{t+1} comes from the pension system.

2.4 Social Security System

The government runs a pay-as-you-go pension system with a balanced budget. Every period, the government taxes labor income in the formal sector and pays pensions to current retirees.

We consider systems that differ along two dimensions. First, in *contributive* systems workers have to meet a contribution requirement to receive full pension benefits. In contrast, in a *universal* system all workers are eligible to receive pensions upon retirement, regardless of whether they contributed to it or not.

Second, pension schemes also differ according to the relationship between the level of contributions and benefits. *Bismarckian* systems provide earnings-related benefits; this gives workers an additional incentive to work as they perceive this increases their future pensions. In contrast, *Beveridgean* systems offer flat payments and lack this incentive. We model this by making benefits consist of a common amount for all formal workers and a second component that is proportional to past labor income.[6]

Benefits are thus given by:

$$b^i_t = \left[\theta h_{t-1,i}\eta_i + (1-\theta)h_{t-1}\right]\bar{b}_t \tag{8a}$$

$$b^0_t = \epsilon h_{t-1}\bar{b}_t \tag{8b}$$

[5] We are using shorthand notation $u_i \equiv u(c_i, h_i)$ with timing implicit by the variable we take derivatives with respect to.

[6] This formulation has been frequently used in the study of redistributive social security, see e.g. Casamatta et al. (2000).

where ϵ determines the level of universal pensions received by informal workers, θ measures how Bismarckian the benefits of formal workers are, and $\bar{b}_t$ is the pension rate per unit of past effective labor, so that the average formal pension is $h_{t-1}\bar{b}_t$. With $\epsilon = 0$ the system is purely contributive and there are no pensions for informal workers. With $\epsilon = 1 - \theta$ they receive the basic common pension of formal workers, and with $\epsilon = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ the system is purely universal in the sense that informal workers receive the same pensions as the type of formal working household with the lowest productivity.

The balanced budget requirement implies that

$$\nu_t w_t \tau_t h_t = \sum_{j \geq 0} \gamma_j b_t^j$$

This implies that the pension rate $\bar{b}_t$ is:

$$\bar{b}_t = \frac{\nu_t w_t h_t \tau_t}{h_{t-1}(1 + \gamma_0 \epsilon)}. \tag{8c}$$

2.5 Elections

Elections take place at the beginning of each period. We assume that preferences are aggregated through probabilistic voting.[7] Thus, policy maximizes a convex combination of the objective functions of all groups of voters, where the weights reflect the groups' sizes and their responsiveness to policy changes. We allow for age-related variation in responsiveness, reflected in a per capita political influence weight of unity for young voters and a per capita weight of $\omega \geq 0$ for retired voters.

2.6 Equilibrium

Economic equilibrium. In economic equilibrium, firms and households optimize given factor prices, policies, and a vector of aggregate states z_t, which includes exogenous demographics as well as the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}_j$. In a rational expectations equilibrium, the path of policies $\{\tau_s\}_{t \leq s \leq T}$ fully determines an allocation and price system.

Conditional on some initial states z_0 and a policy sequence $\{\tau_t\}_{0 \leq t \leq T}$, a competitive equilibrium is given by an allocation and price system such that

i. households optimize;

ii. aggregate states are defined according to (1), markets clear, and factor prices are determined according to (3) for all t; and

[7] See Lindbeck and Weibull (1987).

iii. the government budget constraints in (8) are satisfied for all t.

Politico-economic equilibrium. In politico-economic equilibrium political decision makers optimally choose tax rates, taking all implications of their actions into account and forming rational expectations about future policy choices. We assume that these choices are Markov, i.e. they are functions of the fundamental state variables. The decision maker at date t takes the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}$ as well as the continuation policy $\tau_t^{+1}(z_{t+1})$ as given. Furthermore, given the continuation tax function, the policymaker takes as given the following law of motion for the state variables as a function of current policy choices:

$$z_{t+1} = \zeta_t \left(z_t, \tau_t, \tau_t^{+1} \right). \tag{9}$$

Subject to (3), (8), (9), the policymaker chooses τ_t to maximize

$$\mathcal{W}_t \left(z_t, \tau_t; \tau_t^{+1}(z_{t+1}) \right) \equiv \sum_j \gamma_j \mu_j \left[\omega \mathcal{O}_j(z_t, \tau_t) + \nu_t \mathcal{Y}_j \left(z_t, \tau_t; \tau_t^{+1}(z_{t+1}) \right) \right],$$

where the objective function is the weighted sum of indirect utilities from retirees $\mathcal{O}_j$ and workers $\mathcal{Y}_j$, with weights reflecting relative population sizes through ν_t and γ_j, relative political weight of retirees ω, and propensity to vote μ_j.

Definition 1. *A* politico-economic equilibrium *as of period t conditional on z_t consists of a sequence of tax functions, $\{\tau_\iota(\cdot)\}_{\iota \geq t}$; a sequence of continuation tax functions, $\{\tau_\iota^{+1}(\cdot)\}_{\iota \geq t}$; a sequence of laws of motion for the state variables, $\{\zeta_\iota(\cdot)\}_{\iota \geq t}$; policy choices, $\{\tau_\iota^\star\}_{\iota \geq t}$; and a competitive equilibrium allocation such that*

i. tax functions are optimal subject to continuation tax functions:

$$\tau_\iota(z_\iota) \in \arg\max_{\tau_\iota} \mathcal{W}_\iota(z_\iota, \tau_\iota; \tau_\iota^{+1}(\cdot)) \text{ for all } z_\iota, \text{ s.t. } (3), (8), (9), \ \iota \geq t;$$

ii. continuation tax functions are consistent with tax functions:[8]

$$\tau_{\iota-1}^{+1}(z_\iota) = \tau_\iota \left(z_\iota, \tau_\iota^{+1}(z_{\iota+1}(\cdot)) \right) \text{ for all } z_\iota, \ \iota \geq t;$$

iii. laws of motion are consistent with the policy and continuation policy functions according to (9);

iv. equilibrium tax choices are generated by the continuation tax function,

$$\tau_\iota^\star = \tau_{\iota-1}^{+1}(z_\iota) \text{ for all } \iota \geq t,$$

[8] With a finite horizon, $\tau_T^{+1}(\cdot) = \emptyset$.

and $\{\tau_\iota^\star\}_{\iota \geq t}$ *implements a competitive equilibrium allocation.*

For infinite horizon and a recursive time-autonomous structure, the policy and continuation policy functions are time-autonomous functions of the state as well, and conditions i. and ii. above are combined in a fixed point requirement.

3 Logarithmic-GHH Preferences

This section derives the model's analytical results. We follow Greenwood et al. (1988) and assume

$$u(c_{1,t}^i, h_{t,i}) = \ln\left(\tilde{c}_{1,t}^i\right), \qquad \tilde{c}_{1,t}^i \equiv c_{1,t}^i - \frac{X_i \xi}{1+\xi}\left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \tag{10}$$

where ξ is the Frisch elasticity of labor supply and X_i is a type-specific taste for leisure. Proposition 1 characterizes the economic equilibrium under these log-GHH preferences. If informal households are in addition hand-to-mouth, the politico-economic equilibrium carries no state (proposition 2) and the tax rate is a function of parameters and demographics alone. The quantitative sections keep log-GHH preferences as the benchmark and let informal households save for retirement. They also report results under CRRA-GHH preferences, $u(c_{1,t}^i, h_{t,i}) = \left((\tilde{c}_{1,t}^i)^{1-1/\rho} - 1\right)/(1-1/\rho)$, in which ρ is the intertemporal elasticity of substitution and $\rho = 1$ is the logarithmic case.

3.1 Economic Equilibrium

With log-GHH preferences, formal households i supply labor and save according to

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i}\right)^{\xi} \tag{11a}$$

$$s_{t,i} = \frac{\beta}{1+\beta}\left[w_t\left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi}\left(h_{t,i}\right)^{\frac{1+\xi}{\xi}}\right] - \frac{b_{t+1}^i / R_{t+1}}{1+\beta}. \tag{11b}$$

Appendix A shows that in economic equilibrium every relevant variable x (savings, labor supply and consumption) is a power of capital per worker times a coefficient,

$$x_t = \Theta_{x,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_x},$$

where the coefficient $\Theta_{x,t}$ depends on parameters and taxes and the exponent σ_x on parameters only.

Proposition 1. *In economic equilibrium we have the following:*

i. Aggregate labor is decreasing in τ_t, ϵ and γ_0 and increasing in τ_{t+1} and θ. Aggregate savings are decreasing in τ_t, τ_{t+1}, and θ and are increasing in ϵ and γ_0.

ii. Labor income inequality and relative labor supply are independent of pension policy and given by

$$\frac{\eta_i h_{t,i}}{h_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h}, \qquad \Gamma_h \equiv \sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{12}$$

iii. Savings of formal households relative to average savings are independent of τ_t and linear in τ_{t+1}:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}\frac{1-\theta}{1+\beta}\left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} - 1\right). \tag{13}$$

iv. When $\tau_{t+1} = 0$ or with fully Bismarckian pensions ($\theta = 1$), the heterogeneity in savings exactly matches labor income inequality. An increase in τ_{t+1} or a decrease in θ amplifies the spread in private savings.

v. Informal savings increase in τ_t and θ and decrease in τ_{t+1} and ϵ; current taxes also raise their ratio to formal savings.

Part i of proposition 1 states that current taxes reduce labor supply and savings, while future taxes increase labor supply, since higher expected benefits make work more attractive, and reduce the need to save. A higher θ increases labor supply for the same reason; with GHH preferences this raises the marginal utility of $\tilde{c}_{1,t}$ and so, through the Euler equation, lowers aggregate savings. A higher ϵ shifts future benefits toward informal households and lowers the benefit formal households expect, so that they save more and work less, and a larger share of informal households has the same effects.

Parts ii and iii reduce the state space of the politico-economic equilibrium, as the next subsection shows. Part iv implies that higher future taxes or a lower θ, when pensions are redistributive, widen the dispersion of savings and thus of private wealth. Part v works through the universal pension, since informal earnings do not respond to pension policy. The pension's present value to an informal household is proportional to formal savings, with a factor that rises with future taxes and ϵ: current taxes and a higher θ, which lower formal savings, therefore raise informal savings, while future taxes and a higher ϵ raise the present value of the pension and lower them.

3.2 Politico-Economic Equilibrium

In a Markov equilibrium the distributions of past savings and past labor supply are in principle state variables of the political choice of the tax rate. Part ii of proposition 1

removes the distribution of labor supply, since taxes do not move relative labor supply.[9] If informal households save, their past savings are a further state, and one that both current and future taxes move. For the analytical results we therefore take informal households to be hand-to-mouth: they do not save, so that their Euler condition (6d) does not apply, and when old they receive an endowment tied to the informal wage, which can be read as work in old age at a lower productivity.[10]

With log-GHH preferences and hand-to-mouth informal households, the policymaker maximizes

$$\begin{aligned}\mathcal{W}_t(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) = \sum_i \gamma_i \mu_i \left[\omega \ln\left(c_{2,t}^i\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^i\right) + \beta \ln\left(c_{2,t+1}^i\right)\right)\right] \\ + \gamma_0 \mu_0 \left[\omega \ln\left(\tilde{c}_{2,t}^0\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^0\right) + \beta \ln\left(\tilde{c}_{2,t+1}^0\right)\right)\right] \qquad (14)\end{aligned}$$

Proposition 2. *In politico-economic equilibrium with hand-to-mouth informal households, all consumption functions are log-separable in the endogenous state variables, making policy only a function of parameters, including system characteristics and demographics. Politico-economic equilibrium taxes are increasing in ω and decreasing in ν_t, the latter whenever the aggregate marginal support of formal retirees for the tax is positive at the equilibrium, as it is without a universal pension. With equal propensities to vote, and with retirees' savings consistent with the tax and the design they anticipated, taxes are increasing in income inequality and, if no formal type borrows against its pension, decreasing in θ, given the continuation policy when there is a universal pension. The effect of an unanticipated change in ϵ is ambiguous and depends on the pension's share of consumption of informal and formal retirees.*

The economic equilibrium is log-separable in the aggregate states (appendix A.2), and the one remaining vector of states, the distribution of formal households' savings $s_{t-1,i}/s_{t-1}$, is a function of parameters and the current tax alone; by the same token, the distribution carried into the next period does not depend on the current choice (appendix B.1). Despite two dimensions of pension design and heterogeneous households, the politico-economic equilibrium therefore shares the property of Gonzalez-Eiras and Niepelt (2008), Gonzalez-Eiras and Niepelt (2012), Song (2011), and Uchida and Ono (2024), among others, that policy is a function of parameters and demographics only, although the equilibrium conditions are more complex and in most cases can be solved only numerically. Appendix B.1 shows that the equilibrium tax is unique for any Frisch elasticity up to $1/(1-2\alpha)$ (proposition 3), and that with a single type of formal household and no informal households it has a closed form, which with inelastic labor is the tax of Gonzalez-Eiras and Niepelt (2008).

[9] This property of GHH preferences also holds under CRRA preferences (appendix A.1).

[10] We model old-age income as an endowment of goods to keep the model as symmetric with the general case as possible.

Notwithstanding the orthogonality of current policy with state variables, the trade-offs underlying the equilibrium tax rates are dynamic in nature as they relate contemporaneous policy decisions to capital accumulation and thus future wages and tax revenue. The tractability of the model comes from specifying functional forms that render the factor price elasticities and the derivatives of the indirect utility functions orthogonal to the capital stock, and relative labor supply and relative savings decisions independent of current taxes.

The comparative statics in ω and ν_t mirror those in Gonzalez-Eiras and Niepelt (2008) and Song (2011), and the latter also shows that the equilibrium tax rate increases with income inequality. A distinct contribution to the politico-economic literature on social security is the prediction that a more Bismarckian benefit formula weakens political support for taxation. The reason for this is that a higher θ shifts benefits from low-income to high-income retirees, and the former, with their higher marginal utility, weigh more in the political objective, so that retirees' support for taxes declines in the aggregate.[11]

A higher ϵ shifts benefits from formal to informal retirees. With log utility a retiree's support for the tax is the pension's share of their consumption, so the shift raises the support of informal retirees and lowers that of formal ones (appendix B.1 gives the exact condition for the sign). Taxes rise when informal retirees are poorer than formal ones, by more than the small elasticity of the tax base with respect to the tax rate, and the pension is a comparable share of their consumption, as in Argentina (section 5), because their smaller pension then rises in a larger proportion than formal pensions fall.

When informal households save or preferences are CRRA, the tax rate depends on the state and the equilibrium has to be solved numerically; section 4 describes how.

4 Numerical Methods

This section describes how the politico-economic equilibria of sections 5–7 are computed, with the algorithms and the code in the technical documentation (Berg and Gonzalez-Eiras 2026). The quantitative politico-economic literature computes a time-invariant policy function (Krusell and Ríos-Rull 1999; Forni 2005; Song 2011) or compares steady states across demographic scenarios (Galasso 1999). Along a demographic transition, however, the Markov policy function is date-specific, since the policymaker at t faces its own mix of cohorts, ν_t, and a continuation policy set further along the transition. The distinction is immaterial under log-GHH preferences without informal households (proposition 3), but not with a universal pension, under CRRA preferences or with informal savers, where the continuation policy enters the tax condition, or in the design choice,

[11] This holds along a politico-economic equilibrium path where retirees correctly anticipate the change in θ (appendix B)

where a design chosen at t is weighed against the electorate of $t+1$.[12] We therefore compute date-specific policy functions by backward induction from a date T, the end of a finite horizon or the start of a steady state in an infinite one. For a large T the two agree away from T. Appendix D confirms that our results are robust to our choice of T, 2200 for the rich economies and 2190 for Argentina.

In the general case with CRRA preferences, the distribution of labor supply remains independent of policy (part ii of proposition 1) and the distribution of past savings is pinned down by the current tax together with past aggregate savings (appendix A.1). The distribution of past savings matters for the political problem, but since it depends only on the current tax and past aggregate savings we can eliminate it from the state space, keeping only the level of past savings s_{t-1}: we work with the first order condition in τ_t derived with the distribution of past savings held fixed, and then evaluate it at the one consistent distribution for each candidate tax and for each state s_{t-1}. Informal savers add to the state space the ratio $s_{t-1,0}/s_{t-1}$ of informal to formal savings, which also depends on τ_{t-1} (appendix B.2). When the policy function features one or more states, the current tax moves the states and affects future policy choices. We measure the marginal effect by numerical differentiation of the $t+1$ solution grid, which the backward induction has already computed.

Gonzalez-Eiras and Niepelt (2008) establish uniqueness in the limit of finite-horizon economies, but their argument does not carry over: their model has a representative household in each cohort, and so no distribution of savings to make consistent with the tax, and neither informal savers nor a choice of design. Under log-GHH preferences, with no informal households, or with hand-to-mouth ones and $\xi(1-2\alpha) \leq 1$, which our calibration satisfies by far, exactly one candidate passes at every date (proposition 3 in appendix B). At the distribution of savings consistent with a higher tax, retirees below mean income have saved less, which strengthens their support for the tax, but the income from their savings falls by less than their pension rises, so that the first order condition evaluated at consistent distributions is strictly decreasing in the tax. Under CRRA preferences or with informal savers we check uniqueness numerically.[13]

Each reported path starts from the steady state under the first period's policy. Parameters whose targets are equilibrium outcomes, such as the discount factor and the political weight of the old, are calibrated by re-solving the equilibrium path at each trial

[12] In our applications, the distinction has a modest effect on results: a time-invariant policy function, solved at each date's own ν_t and applied along the projected path, misses the tax rate by up to about 0.2 p.p. before the period that precedes T, and by up to 0.3 p.p. in the Argentine calibration at the lowest intertemporal elasticity, and misplaces the design by up to 0.074 (online appendix).

[13] We record at every state and date the number of solution candidates. The candidates are the corners and every root of the first order condition on a grid of taxes, and a candidate passes if it maximizes the political objective with its own consistent distribution of savings held fixed. Several passing candidates would be ranked by payoff, an assumption about equilibrium selection that none of our reported equilibria requires: exactly one passes at every state and date of every solve behind the paper's results (appendix D).

value until the targets hold in the calibration year.

Under log preferences the choice of the design in section 7 adds no state either. The design chosen at t applies from $t+1$ and depends neither on the design chosen in the previous period, which sets current benefits, nor on aggregate savings (proposition 5 in appendix C.4), and the tax at each candidate design is found as above, without a state. Each date's choice is therefore a search over a grid of designs. Under CRRA preferences neither property survives: the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t).

5 Pension System Reform in Argentina

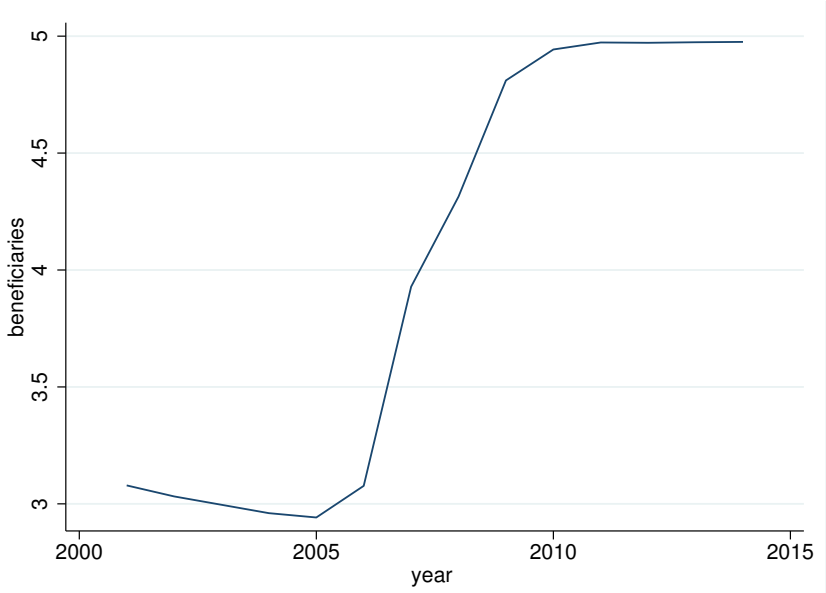
Argentina is a middle-income country with high labor informality. In 1994 the social security system was reformed, tying full benefits to 30 years of contributions, which left about a third of the elderly without a pension by 2004.[14] From 2005 a series of tax amnesties let workers with shorter contribution records draw a pension. They added 2.7 million beneficiaries by 2011 (Rofman and Apella 2015), raising the number of pensioners from about 3 to 5 million (figure 5.1) and the coverage of the elderly from 68% to over 90% (Rofman and Oliveri 2012). Thus, in a few years, Argentina experienced a significant increase in non-contributive pensions: between 2005 and 2010 social security spending increased by 1.9% of GDP with most of this increase, 1.8% of GDP, due to the tax amnesties (Cetrángolo and Grushka 2020).

Appendix E holds the data, the identifying formulas and the sources of the calibration, and the online appendix shows how the design (ϵ, θ) shapes equilibrium outcomes and how other calibrations affect the results.

Calibration. We calibrate the model with informal savers to Argentina before the reforms and read the tax amnesties as an increase in ϵ. Informal households work when young and save for retirement, facing the same rate of return on savings as formal households. A model period is thirty years, the calibration year is 2010, and every household has the same propensity to vote. Table 1 reports the parameters. The pre-reform $\epsilon = 0.21$ is the basic pension of retirees without a full contribution record, 70% of the benefit of the least productive formal workers, in units of the average formal benefit and discounted because it starts five to ten years after the formal retirement age, and $\gamma_0 = 0.47$, informal households per formal household, puts informal households at 32% of all, the share of the population over 65 without a pension before the reform (Rofman and Oliveri 2012). Since some of these households would never have drawn the basic pension, which required ten years of contributions, we also report the reform from a pre-reform $\epsilon = 0$ (appendix

[14] The reform also let workers opt for a fully funded system, whose funds were nationalized in December 2008.

Figure 5.1: Pension coverage in Argentina



Note: Beneficiaries of the national pension system (SIPA) and of non-contributive old-age pensions, annual averages in millions. Source: Cetrángolo and Grushka (2020, table 6), from the Boletín Estadístico de la Seguridad Social.

E). The design $\theta = 0.839$ matches the observed ratio of replacement rates at mean and half-mean income. The reform raises ϵ to the universal level $\epsilon^U = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ at which informal retirees receive the benefit of the least productive formal ones. We impose a formal-sector capital share of $\alpha = 0.35$, a Frisch elasticity of $\xi = 0.30$, and a ratio of workers to retirees ν_t from thirty-year gross growth of the total population as in Gonzalez-Eiras and Niepelt (2008).

The remaining parameters are identified jointly with the equilibrium path. The formal productivities η_i match the income distribution across the four formal quartiles of the household survey (EPH, appendix E), with a taste for leisure X common to them whose level is pinned by the observed formal workweek of 42.5 hours.[15] Four parameters are solved against four targets in 2010: β against a capital-output ratio of 3.23, ω against an equivalent social security tax rate of 11.8%, which is pension spending of 7.1% of GDP before the 2001 crisis divided by formal labor income, 60% of GDP once GDP includes the output of informal households, and η_0 and X_0 against the informal households' income and hours relative to the formal average.

The reform. Table 2 reports the effect in 2010 of an unanticipated and permanent rise of ϵ to ϵ^U under log-GHH preferences, first with the tax rate held at its pre-reform path and then chosen politically. Since the rise is unanticipated, the retirees of 2010 enter it

[15]Identifying one X_i per quartile instead takes the survey's relative hours as data and leaves the level of hours unidentified (appendix E.5). Since the productivity distribution reaches the aggregates only through a sum that both calibrations normalize to one, only η_i and X_i differ, and every result of this section is the same under both.

Table 1: Calibration, Argentina

Parameter	Value	Target
ϵ (*pre*-reform)	0.21	Basic pension, 70% of the minimum, discounted
θ	0.839	Replacement rate dispersion
α	0.35	Factor income shares
ν_t	$[1.65, 0.96]$	30-year gross population growth rates
ξ	0.30	Elasticity of labor supply
β	0.67	Capital–output ratio of 3.23
X	1.92	Average formal workweek of 42.5 hours
η_i	$[0.64, 0.90, 1.14, 1.87]$	Income distribution
γ_0	0.47	Elderly without a pension, 32% of households
ω	1.50	Social security tax of 11.8%
η_0	0.340	Informal relative income, eq. (E.1)
X_0	0.795	Informal relative working hours

Note: The baseline is $\rho = 1$.

with the savings they made under the pre-reform system. We find that taxes increase, which drives down savings and labor supply. With taxes held fixed, the new pensions are paid for by cutting formal benefits by 12.7%, and formal households save more and work marginally less, since the reform lowers the benefit they expect (proposition 1); informal households, who now hold a claim on the pension system, save less.

Table 2: Pension system reform, year 2010

Scenario	Tax rate	Savings rate	Avg. workweek (hours)	Formal benefits
Baseline	11.81%	13.33%	42.54	–
Economic Equilibrium	11.81%	13.64%	42.46	−12.70%
Full effect	13.76%	12.96%	42.30	+1.46%

Note: The reform of section 5, $\rho = 1$. Economic equilibrium: the new ϵ with taxes as in the baseline; full effect: taxes set in politico-economic equilibrium. The savings rate is savings relative to GDP, formal plus informal output; formal benefits are the change in the benefit of every formal retiree against the baseline.

In 2010 the model predicts that the social security tax rate rises by 2.0 p.p., enough to pay for the new pensions and to raise formal benefits by 1.5%, so that none of the expansion is financed by cuts. The reason for this is that the reform moves benefits toward informal retirees, who are poorer than formal ones and value benefits more at the

margin.[16] The tax response builds over time, to 3.5 p.p. above the pre-reform path by 2040, because informal households, now covered, save less for retirement and as retirees demand more taxation. In 2010 the savings rate falls by 0.4 p.p. of GDP and the average workweek by a quarter of an hour.

Two moments of the episode can be set against the model, the rise in pension spending and the response of household savings. Pension spending rises by 1.2% of GDP, and spending on informal retirees, which corresponds to the cost of the amnesties, by 1.1% of GDP, in each case about three fifths of its observed counterpart: the 1.9% of GDP by which spending rose between 2005 and 2010 and the 1.8% of GDP that the amnesties added.[17] The comparison depends on how the pre-reform system is read. If no informal retiree drew a pension before the reform, so that the model is recalibrated at $\epsilon = 0$ and the reform raises ϵ from zero, the tax rises by 4.2 p.p., pension spending by 2.5% of GDP and spending on informal retirees by 2.0% of GDP, more than observed: the two readings bracket the data. Gonzalez-Rozada and Ruffo (2016) find that the amnesties lowered the savings of the households likely to draw the new pensions. This is consistent with the model, in which the reform lowers an informal household's savings from 33% to 27% of the average formal household's in 2010.

The effect of the intertemporal elasticity of substitution. Since the environment has no uncertainty, we read the parameter ρ of CRRA preferences as the intertemporal elasticity of substitution (IES). The evidence on its size is inconclusive,[18] and we keep logarithmic utility as the benchmark. Figure 5.2 repeats the reform with the pre-reform economy calibrated at each ρ and traces the effects in 2010 and 2040 for taxes, savings rates, average workweek, and informal savings.

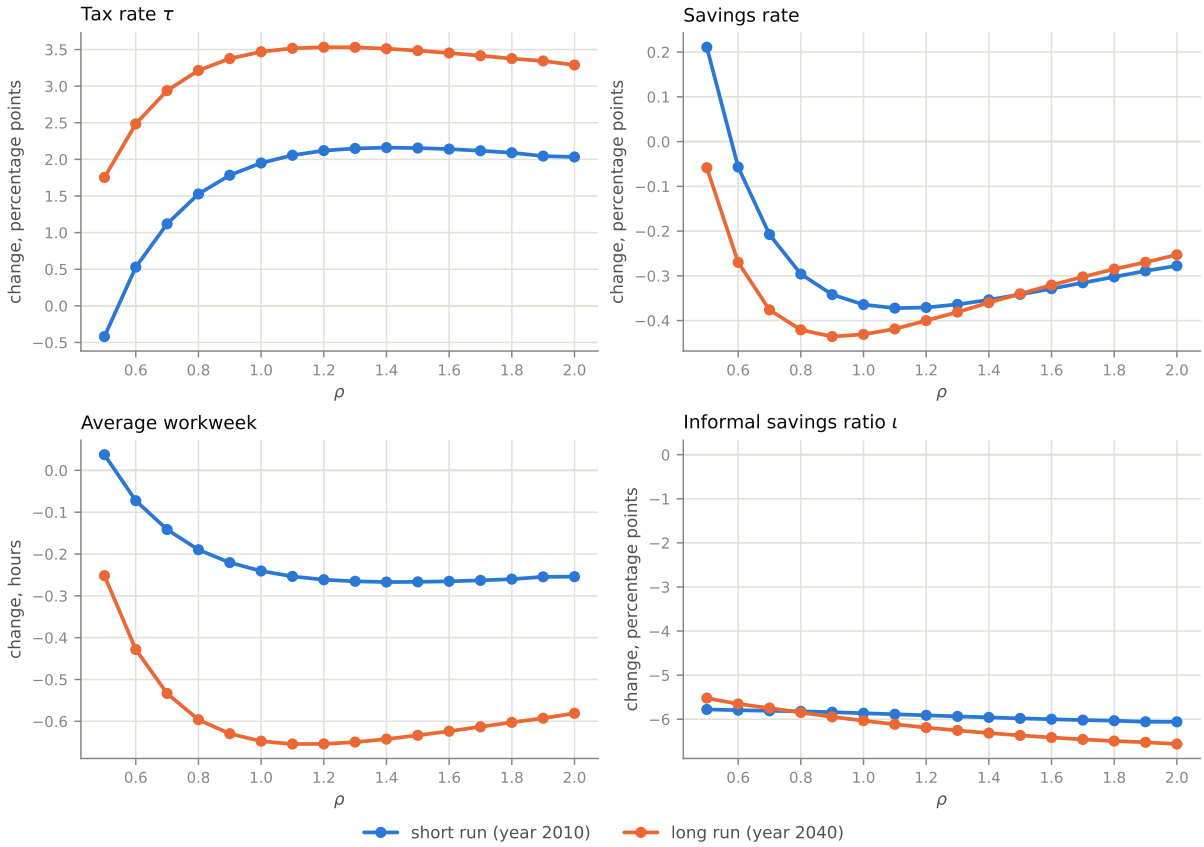
The effects of the pension reform are nonlinear in ρ. For an IES between 1 and 2 they stay close to the log benchmark, in 2010 as in 2040, when the effects on the tax rate and the workweek are larger. Below log utility the responses of the tax rate, savings and hours shrink, and the tax response reverses just above $\rho = 0.5$: at $\rho = 0.5$ the tax rate falls by 0.4 p.p. on impact, although by 2040 it is 1.8 p.p. above its pre-reform path. Only the informal savings ratio falls by about the same, 6 p.p., at every ρ.

[16] The tax also rises when informal households are hand-to-mouth, consistent with condition (B.9) in appendix B.1.

[17] The observed rise starts from 2005, when spending had fallen to 5.2% of GDP because only minimum pensions were adjusted for inflation after the 2001 crisis, while the model's pre-reform economy is calibrated to the pre-crisis 7.1% (appendix E). The nationalization of the funded system in December 2008 moved its members and their contributions into the public system in the same window; it bears on the comparison only through spending, and the 1.8% of GDP counts the beneficiaries of the amnesties alone. Finally, a model period is thirty years, so the model's response in 2010 is a shift in spending that lasts a generation, which we compare with the shift reached within the five years of the amnesties.

[18] Estimates from aggregate consumption data are low, which Guvenen (2006) reconciles with higher values from output and wealth data through limited asset market participation, and asset-pricing models with Epstein–Zin preferences and long-run or disaster risk require an IES above one (Bansal and Yaron 2004; Barro 2009).

Figure 5.2: Short- and long-run effects of the pension reform as a function of ρ



Note: Each panel is the change against the pre-reform path of the same ρ. The savings rate is savings relative to GDP, formal plus informal output, and the informal savings ratio ι is the savings of an informal household relative to the average savings of formal households.

A decomposition of the 2010 tax condition by group shows that the reversal does not come from the formal young, whose resistance to a higher tax is about the same at $\rho = 0.5$ as at log, but from informal households.[19] With a low IES the informal retirees' marginal utility falls faster as their pension rises, so they ask for a smaller rise in the tax. And the informal young, who will now draw a pension, withdraw more of their support for a higher tax than at log, since a higher tax lowers savings and with them the value of that pension. At $\rho = 0.5$ the second force offsets nearly all of the first, and the formal retirees, whose benefits fall at a given tax, tip the balance.

6 Social Security in Rich OECD Countries

We now calibrate the model to the U.S. and use it as a laboratory, in which counterfactual economies change one characteristic of the U.S. calibration at a time. We take the characteristics from France, an older and more equal economy than the U.S., in which

[19] Each group's contribution is the change that the reform makes to its term of the first order condition at the pre-reform tax, divided by the slope of the post-reform condition between the two taxes, so that the contributions sum to the tax response.

the poor vote about as often as the rich and households work shorter hours. In the U.S., as in France, nearly every worker qualifies for a contributive pension, so we set informality aside and the design reduces to one dimension, the extent to which benefits are earnings-related.[20] The design is given throughout the section, which closes by testing the model's predictions for pension spending against the OECD data of figure 1.1. Appendix F and the online appendix give the details of the calibration and further results.[21]

Calibration. We calibrate the U.S. to 2020 in the same way as Argentina in section 5, with a few differences. The discount factor β matches the U.S. thirty-year interest rate rather than a capital–output ratio, and where the Argentine calibration gives every household the same propensity to vote, the political objective here weights each income group by its observed propensity. The tax rate is calibrated to public cash spending on pensions and the design to the earnings-linked share of the public pension, the measures of figure 1.1. France is calibrated in the same way, with the U.S. β and with its households grouped at the U.S. income percentiles, which puts the two countries' household types in one-to-one correspondence for the counterfactuals.[22]

Table 3 reports the parameters. France has fewer workers per retiree than the U.S., a flatter productivity profile and a taste for leisure 1.6 times the U.S. level, and its propensity to vote is nearly flat across income groups, where in the U.S. it rises from 0.47 to 0.77 (tables 8 and 9). The political weight of the old is calibrated to each country's pension spending, and the counterfactuals hold it at its U.S. value.[23]

Determinants of pensions, savings, and labor supply. Figure 6.1 summarizes the counterfactuals. Each row changes one characteristic of the U.S. calibration: its pension design, set at either corner, its demography, in two ageing scenarios, or France's income distribution, voting profile or taste for leisure. The three panels report the effects on the equilibrium tax rate, the savings rate and the workweek in 2020, at three values of the IES. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout.

The tax panel identifies the pension design and ageing as the main determinants of pension spending, with the effect of the design measured from corner to corner. Going

[20] Whitman et al. (2011) estimate that about 4% of individuals aged 62–84 in the U.S. in 2010 would never collect Social Security, most of them for lack of the 40 quarters of covered earnings needed to qualify.

[21] The online appendix holds the calibrations across the intertemporal elasticity of substitution and the counterfactuals of this section across the Frisch elasticity, each figure with the data behind it.

[22] France's design, $\theta = 0.822$, is measured in the same way as the U.S. one and does not depend on where the groups are cut.

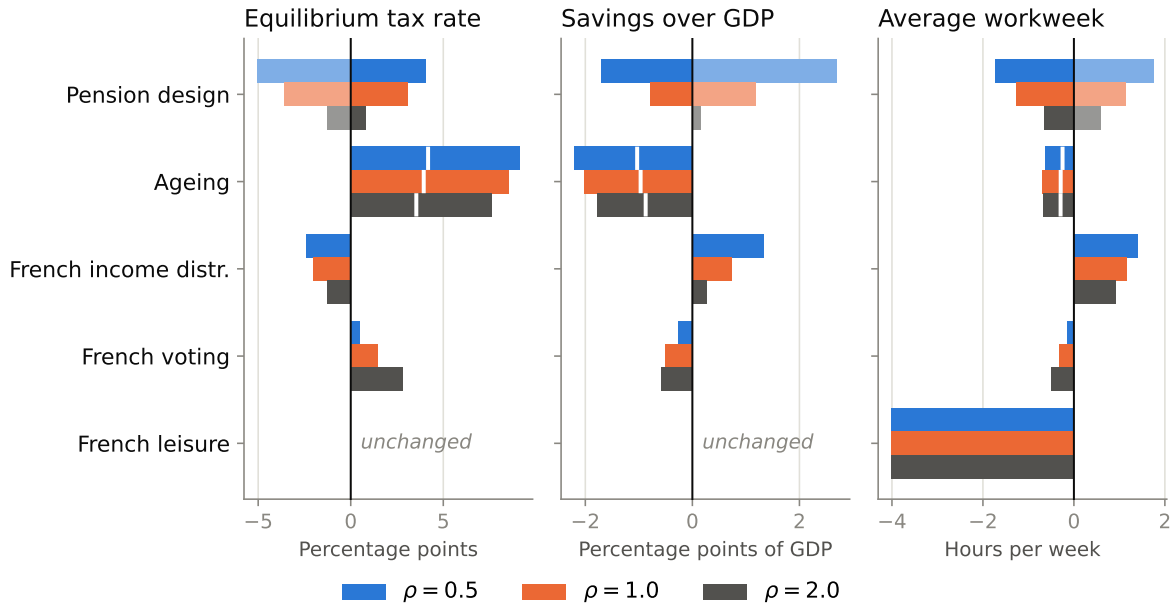
[23] The calibrated parameters absorb institutions that the model does not contain. The political weight of the old, ω, stands for the way legislation is introduced, debated and passed, and for constitutional arrangements such as whether the political system is presidential or parliamentary. The taste for leisure, X, stands for preferences and also for institutions that shape labor supply, such as unionization, collective bargaining, labor-market regulation and statutory holidays, so the large difference in X between France and the U.S. includes these institutional differences.

Table 3: Calibration, the U.S. and France

Parameter	U.S.	France	Identified by
θ	0.544	0.822	Earnings-linked share of the public pension
ω	1.17	1.39	$\tau^{US} = 10.5\%$, $\tau^{FR} = 20.6\%$
X	4.07	6.48	Average workweek
ν_{2020}	1.34	1.04	30-year gross population growth
η_H/η_L	3.08	2.29	Relative productivity, high to low income
β	0.65		30-year interest rate (U.S.)
ρ	1		Log-GHH preferences
ξ	0.30		Frisch elasticity of labor supply
α	0.30		Capital income share

Note: X is the population-weighted mean of X_i; its level is the hours unit, pinned for France by targeting average hours relative to the U.S. rather than in levels. The lower panel is common to the two countries; β is calibrated for the U.S. and imposed on France.

Figure 6.1: The effect of various counterfactuals in the U.S.



Note: The online appendix gives the data behind the figure. In the pension-design row each bar shows both corners, measured from the baseline: $\theta = 0$ in full color and $\theta = 1$ in the light tone. In the ageing row the bar is acute ageing and the white line across it mild ageing.

from a fully Bismarckian system to a fully Beveridgean one raises the equilibrium tax rate by 6.7 p.p. at $\rho = 1$, lowers the savings rate by 2.0 p.p. of GDP and shortens the workweek by 2.4 hours, 1.3 of them between the observed design and the Beveridgean corner. The reason for this is that of proposition 2: flatter benefits direct more of the revenue to poorer retirees, whose higher marginal utility gives them more weight in the

political objective, so retirees support a larger system.

Ageing is the determinant that previous studies of the politico-economic equilibrium of pension spending have highlighted. We consider two scenarios. In *mild ageing* we halve current and projected U.S. population growth, and in *acute ageing* the population is constant, $\nu_t = 1$, close to France's demography. Acute ageing raises the tax rate by 8.6 p.p. and mild ageing by 3.9 p.p., the latter close to the rise of about 3.6 p.p. in U.S. spending that Gonzalez-Eiras and Niepelt (2008) find for a similar fall in population growth.[24] The higher contributions crowd out private savings, by 2.0 p.p. of GDP under acute ageing at $\rho = 1$, while the workweek falls by less than an hour, since capital deepening raises the wage and partly offsets the higher tax.

The other characteristics that set France apart move the tax rate much less. At $\rho = 1$, France's more equal income distribution, imposed with θ held at its U.S. value, lowers it by 2.0 p.p., and its flatter voting profile raises it by 1.5 p.p. France's taste for leisure shortens the workweek by nearly the whole gap to France's and leaves every rate at the baseline, since a proportional change in X_i scales hours, output and savings together. The determinants of pension spending are thus not the determinants of hours worked: leisure preferences move hours most, and ageing, the largest effect on taxes, moves them less than the income distribution does.

The effect of the intertemporal elasticity of substitution. These conclusions hold at every IES we consider, apart from the design's rank at $\rho = 2$. The effects on the tax rate and the savings rate are larger at lower values of ρ, modestly for ageing and much more for the income distribution, with the voting profile the one exception. The IES matters most for the pension design: going from $\theta = 1$ to $\theta = 0$ raises the tax rate by 9.1 p.p. at $\rho = 0.5$, as much as acute ageing, and by only 2.0 p.p. at $\rho = 2$, less than the 2.8 p.p. by which France's voting profile raises it there. The reason for this is that θ moves the tax rate through the difference in marginal utility between poor and rich retirees (proposition 2), and a higher IES makes marginal utility fall less steeply with consumption, so that this difference counts for less.

The OECD cross-section. To test these predictions against the data of figure 1.1, we construct 21 synthetic countries, one for each country of the figure. Each is the U.S. model under log preferences, with the demography, income distribution, pension design and propensities to vote of its country.[25] The synthetic countries thus differ only in demography, the income distribution and the design, the three dimensions of figure 1.1,

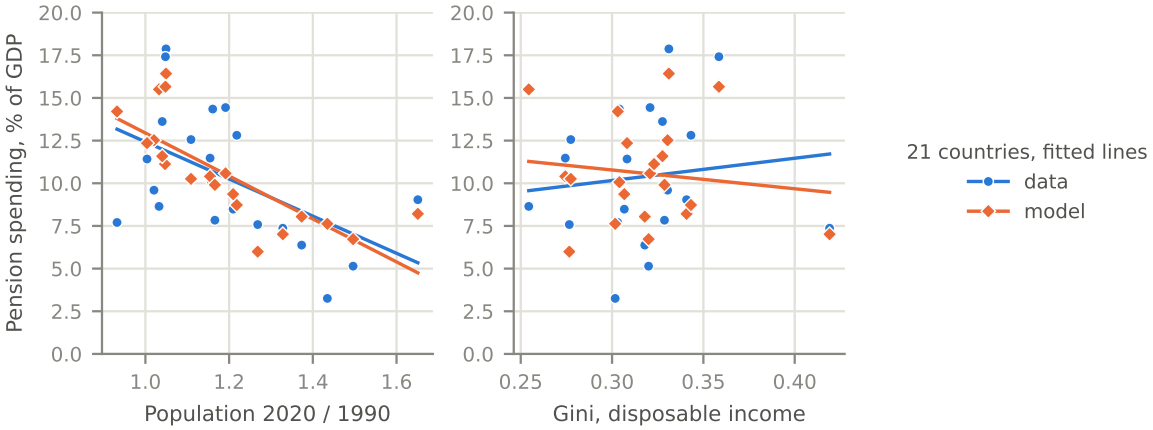
[24] Gonzalez-Eiras and Niepelt (2008) look at the U.S. economy with population growth falling from $\nu_{2000} = 1.38$ to the projected $\nu_{2080} = 1.23$, a fall of about 0.15, similar to the fall of 0.17 in 2020 in the mild ageing scenario.

[25] The propensities to vote by income group are from the Comparative Study of Electoral Systems (CSES). Luxembourg, which has no CSES survey and enforces compulsory voting, votes at the same rate in every income group (appendix G).

and in their propensities to vote. Demography is the number of workers per retiree, corrected for each country's gap to the U.S. in the normal retirement age, as France's is in appendix F. Every synthetic country takes the same political weight of the old, the mean of the weights that reproduce each country's pension spending at its observed design, 1.17.[26]

In figure 6.2, spending falls with population growth in the model as in the data, at a slope within one standard error of the data's, −12.59 against −10.89. The model reproduces the slope but not the dispersion around it, since only demography, the income distribution, the design and turnout differ across its countries. Neither the model nor the data relate spending to the Gini. With the design given, the model thus reproduces the two relations, as the counterfactuals above suggest: demography is the main determinant of the tax rate, and the income distribution moves it little.

Figure 6.2: Pension spending in the data and in the model at the observed designs



Note: 21 countries. The model's spending is $\tau(1-\alpha)$ in 2020 at the observed design, every country at the U.S. $\beta = 0.65$ and at $\omega = 1.17$, the mean of the political weights that reproduce each country's spending without the cost. Turnout: each country's own propensities to vote by income group, from the CSES; Luxembourg, which has no CSES survey and enforces compulsory voting, at a flat profile. The lines are least-squares fits.

The design, however, enters this test as data, and its relation to spending poses a puzzle. At a given demography a more Bismarckian design lowers the tax rate (proposition 2), by 6.7 p.p. from corner to corner at $\rho = 1$ (figure 6.1). In figure 1.1, by contrast, spending rises with the Bismarckian index, and the index rises as population growth falls, which suggests that countries respond to ageing by tying benefits more closely to earnings. A model that holds θ fixed cannot address this response, and section 7 takes it up by letting the electorate choose θ as well as τ.

[26]The weight that reproduces U.S. spending with the U.S. income distribution and propensities to vote measured as for every other country is 1.19, against the 1.17 of table 3 (appendix G.2).

7 Endogenous Pension Design

Up to this point the electorate chooses the size of the pension system while its design is calibrated. In this section it chooses the design θ as well as the tax rate, with the same political objective $\mathcal{W}_t$, in the economy of section 6 without informal households. The choice resolves the puzzle with which section 6 closes. With the deadweight cost of redistribution that we introduce, the U.S. tax rate under log preferences is hump-shaped in θ at a given demography, with its peak at $\theta = 0.58$, close to the observed design of 0.544 that the electorate chooses: at flatter designs a more Bismarckian design raises the tax, since less of the revenue is lost, and at more Bismarckian ones it lowers the tax, as without the cost. Ageing raises both the tax rate and the design the electorate chooses, so that spending can rise with the index across countries that differ in demography. Appendix C holds the derivations, the online appendix the UK and France calibrated in full, appendix G the data and the robustness of the cross-country test that closes the section, and the technical documentation the algorithms (Berg and Gonzalez-Eiras 2026).

7.1 The deadweight cost of redistribution

Left to choose the design in the model as it stands, the electorate chooses flat benefits. For a given revenue, flattening benefits shifts income from retirees who earned more than the average when young to those who earned less, and the political objective, which weighs retirees by their marginal utility, counts the gains of the poorer for more than the losses of the richer. Proposition 4 in appendix C.1 makes the argument exact: if the rich are no more likely to vote than the poor, an electorate that sets the design and the tax at the same time picks the Beveridgean corner, $\theta_t = 0$, whatever the tax rate, the demography and the income distribution. The same appendix extends the result to propensities to vote that rise with income by proportionally less than consumption in retirement. In the U.S. the rich do vote more often, which could in principle reverse the result, but at the U.S. calibration the electorate still picks the Beveridgean corner, at every IES we consider when it sets the design together with the tax, and under log preferences when it sets the design one period ahead of the benefits it governs (appendix C.1).

The model does contain a cost of flat benefits. A flat benefit does not reward an extra hour of work, and at the Beveridgean corner the workweek is about 1.3 hours shorter than at the observed U.S. design (section 6). When the design is chosen ahead of the benefits it governs, the electorate weighs this cost, and at a given anticipated tax retirees, whose benefits are levied on current labor, favor an earnings-related design because of it. Under log preferences and at U.S. inequality the cost is too weak to hold the electorate back. What the model misses is a mechanism through which flat benefits cost more than earnings-related ones on margins other than hours.

The empirical literature points to such margins. Disney (2004) splits public pension contributions in OECD countries into a part matched by benefit entitlements and a part that is not, and finds that the second lowers labor force participation while the first raises it. Kumler et al. (2020) find that after Mexico's 1997 pension reform tied benefits more closely to the wages reported to the social security agency, the under-reporting of wages fell among the younger workers the reform affected. Benefits that are loosely tied to contributions thus erode the base on which contributions are levied, through participation in the formal labor market and through the reporting of earnings.

Summers (1989) explains why the link matters. The part of a payroll contribution that buys the contributor a benefit is closer to deferred pay than to a tax, while the part that buys nothing distorts like a tax. Specifically, with $y_i \equiv \eta_i h_{t-1,i}/h_{t-1}$ the earnings of type i relative to the mean, constant over time by part ii of proposition 1, a retiree of type i receives $\nu_t w_t h_t \tau_t \left[\theta_t y_i + 1 - \theta_t\right]$ on past earnings $w_{t-1} y_i h_{t-1}$, that is

$$\Pi_t \, \tau_t \left[\theta_t + \frac{1-\theta_t}{y_i}\right] \text{ per unit of past earnings,} \qquad \Pi_t \equiv \frac{\nu_t w_t h_t}{w_{t-1} h_{t-1}},$$

where Π_t is the pay-as-you-go return factor: a purely earnings-related system paying the same revenue returns $\Pi_t \tau_t$ per unit of earnings to every type. The flat component moves type i away from that benchmark by

$$t_i \equiv \Pi_t \, \tau_t \, (1-\theta_t) \left(1 - \frac{1}{y_i}\right) \tag{15}$$

per unit of earnings.

We do not model participation and formality explicitly. Instead, we borrow a principle from the literature on the deadweight cost of taxation, that the excess burden of a tax rises with the square of its rate (Harberger 1964). Applied to the implicit tax of each type, it gives a loss that we take out of the revenue that reaches retirees, as a revenue-equivalent excess burden,[27] so that the budget becomes (see appendix C.2 for the derivation)

$$\nu_t w_t \tau_t h_t \, f(\theta_t, \tau_t) = \sum_i \gamma_i b_t^i, \tag{16}$$

$$f(\theta_t, \tau_t) = \exp\!\left(-\tfrac{1}{2} \, \lambda \, \tau_t \, \tilde{V} \, (1-\theta_t)^2\right), \qquad \tilde{V} \equiv \sum_i \gamma_i \frac{(y_i - 1)^2}{y_i},$$

where $\lambda \geq 0$ is a cost parameter to be calibrated, $1 - f$ the share of revenue lost and

[27] The implicit taxes are a pure transfer in the aggregate, $\sum_i \gamma_i y_i t_i = 0$, so with a common elasticity the first-order responses of the contribution bases of taxed and subsidized types cancel in revenue. What remains is the excess burden, a welfare cost that subsidized types bear as well as taxed ones; charging it against the revenue that reaches retirees is a choice of reduced form rather than a consequence of the argument.

$\tilde{V}$ the dispersion of relative incomes.[28] We see this form as a reasonable place to start rather than as an exact account of participation and formality: it has the properties that the principles above suggest, and it leaves the size of the loss to the calibration. The loss vanishes when the system redistributes nothing: at $\theta_t = 1$, and when all incomes are equal, since flat and earnings-related benefits then pay everyone the same. Its share of revenue grows with the tax rate, so each unit of redistribution costs more in a larger system. And since the loss is quadratic in the flat share, the first unit of redistribution costs nothing at the margin: under log preferences the slope of the political objective at the Bismarckian corner, and the tax chosen there, are the same at every value of λ, so an electorate whose objective falls towards that corner without the cost does not choose it at any value of λ (appendix C.2).

The electorate chooses the design one period ahead: at t it sets θ_{t+1}, which governs the benefits of the current young, and the loss that the design causes is taken out of those benefits.[29] A design chosen together with the tax would only re-split benefits that current retirees have already earned, and so would move no decision. Reforms of benefit formulas, moreover, usually apply to benefits not yet in payment. Formally, the cost only replaces θ_t and $1 - \theta_t$ by $f\theta_t$ and $f(1 - \theta_t)$ in the equilibrium of the previous sections (appendix C.3). Since it scales both components of the benefit alike, it cancels from the earnings-linked share of the public pension that identifies the design in section 6, and the observed U.S. design is still $\theta^* = 0.544$. In the tax condition it adds one factor: a larger system loses a larger share of its revenue, so a retiree gains less from a higher tax than the revenue it raises. And at a given anticipated tax, a more earnings-related design now also enlarges the benefits the young will receive, because less of the revenue that pays them is lost; this gain is what can stop the electorate short of the Beveridgean corner.[30] Under log preferences, the design chosen at t depends neither on the design inherited nor on savings (proposition 5 in appendix C.4).

[28] The exponential keeps f between zero and one for any size of the system and gives back the model without the cost at $\lambda = 0$; to first order, $1 - f$ is the quadratic loss itself.

[29] The participation and formality responses that the loss stands for occur earlier, when the young work, and erode the base that pays the retirees of that period. Booked there, the loss of a design would fall on current retirees rather than on the cohort whose benefits it governs, and each period's budget would depend on the design and the tax of the next period. We book it in the period of the benefits, where the implicit tax (15) falls, so that each budget depends on the policy of its own period. The choice bears on the weight the political objective gives the loss, since current retirees and the young enter it with different weights.

[30] The anticipated tax responds to the design as well, and the equilibrium accounts for the response (appendix C.3). At the U.S. choice under log preferences it adds to the gain: of what offsets the pull of the design towards flat benefits at a given loss share and tax, the saving on the loss supplies about 70% and the response of the anticipated tax about 30%.

7.2 Calibration and the determinants of the design

Calibration. We calibrate the cost parameter so that the electorate re-elects the observed design. Concretely, the target is θ_{2020}, set by the electorate of 1990 on an equilibrium path along which every design from the first period on is chosen, and β and ω are recalibrated at every trial value of λ, so that the targets of section 6 still hold with the cost in place. Only the design of 2020 is targeted; every other design along the path is an outcome of the model (online appendix).

Table 4 reports the calibration at each IES. At the calibrated design the share of revenue lost is 6.2% at $\rho = 0.5$ and less than 3% at higher values of the IES, and a flat benefit would lose four to five times as much. Most observed designs lie where the loss is small to modest: the observed design exceeds 0.5 in 13 of the 21 countries of figure 1.1. Across the three values of the IES, the calibrated cost parameter and the share of revenue that a flat benefit would lose both vary by close to an order of magnitude. The cost parameter falls as the IES rises, because a higher IES strengthens the forward-looking stakes in an earnings-related design, and less cost is needed to keep the electorate at the observed level. At $\rho = 0.5$ and 1, λ lies more than an order of magnitude above what participation and taxable-income elasticities alone would imply, and at $\rho = 2$ still 6 to 25 times above it, so the cost also stands in for administrative, evasion and political costs of redistribution that the model does not contain (appendix C.2).

Table 4: The calibrated cost of redistribution for the U.S.

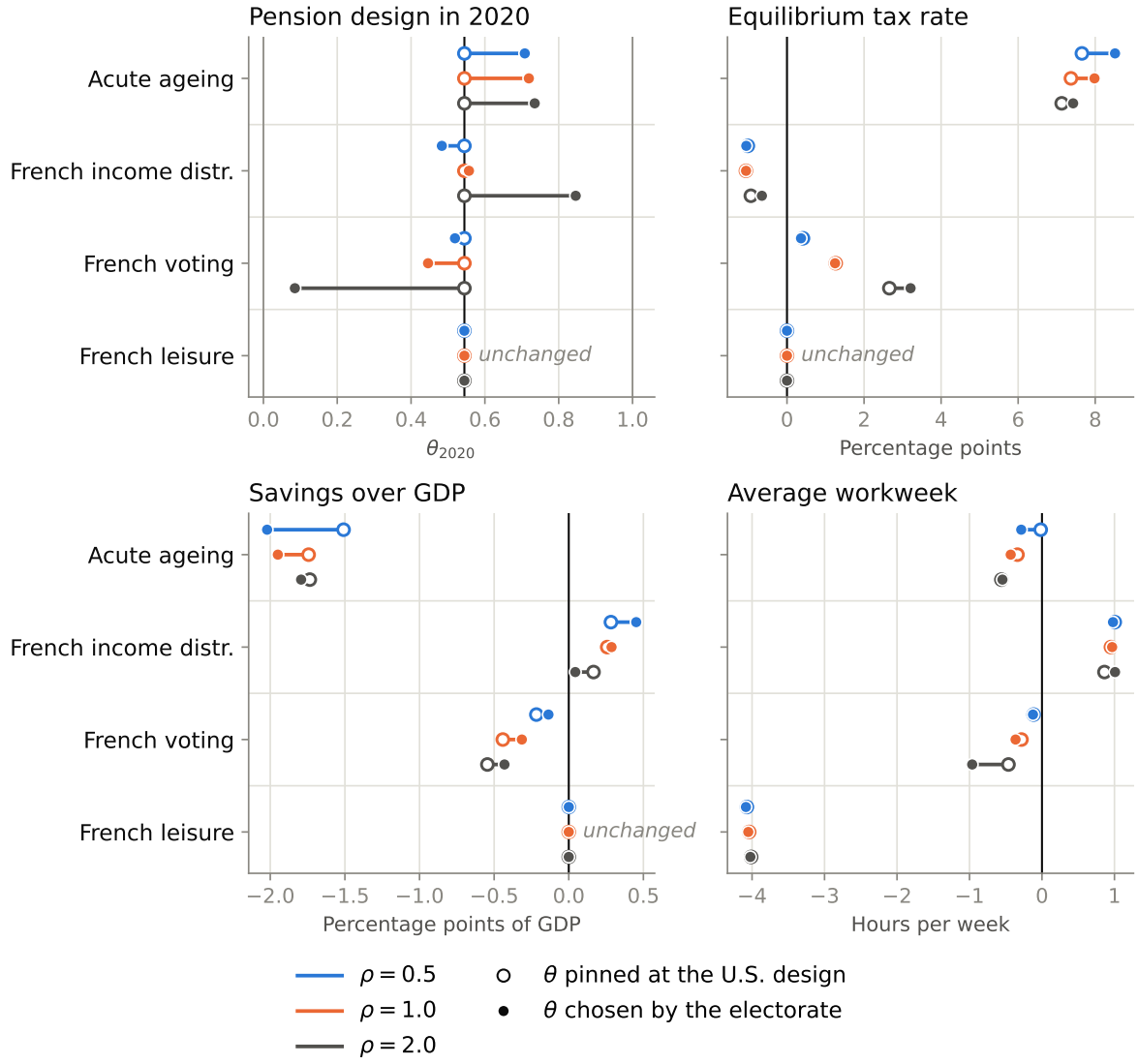
ρ	λ	ω	$1 - f(\theta^*)$	$1 - f(0)$
0.5	13.47	1.73	6.2%	26.6%
1.0	6.25	1.25	2.9%	13.3%
2.0	1.41	1.11	0.7%	3.2%

Note: The cost is that of equation (16), with $\theta^* = 0.544$ and $\tilde{V} = 0.436$ at every ρ. $1 - f(\theta^*)$ and $1 - f(0)$ are the shares of revenue lost in 2020 at the observed design and at a flat benefit, both at the 2020 tax rate.

Determinants of the pension design. Figure 7.1 takes the counterfactuals of section 6 to the economy with the endogenous pension design: acute ageing and France's income distribution, voting patterns and taste for leisure, each imposed on the U.S. calibration over the whole horizon and read in 2020. Every counterfactual is read twice, with the design pinned at θ^* and with the design chosen. Because the cost is present in both, the pinned reading departs from figure 6.1. Ageing still moves the tax rate most, more than five times as much as any other characteristic at $\rho \leq 1$ and more than twice as much

at $\rho = 2$.[31] France's income distribution, however, lowers it by only 1.0 to 1.1 p.p. at $\rho = 0.5$ and 1, about half as much as in figure 6.1, and at $\rho = 1$ by less than France's voting profile raises it.

Figure 7.1: The effect of various counterfactuals with the design chosen



Note: The model is recalibrated for each ρ under the cost equation (16).

First, the results from section 6 about the determinants of the size of the pension system, savings and labor supply remain mostly unchanged. Ageing, however, crowds out private savings more with the design chosen than with it pinned, in particular for low values of ρ. Second, the panel on the pension design shows that ageing consistently ties benefits more closely to earnings, that the French voting profile moves the design towards flatter benefits, and that the direction in which a more equal income distribution moves the design depends on the IES.

[31] With the design pinned at $\rho = 1$, acute ageing raises the tax rate by 7.4 p.p. rather than 8.6 p.p., since the cost makes a larger system worth less to retirees at the margin.

Ageing consistently ties benefits more closely to earnings through two forces. First, ageing gives retirees more weight in the political objective, and at a given anticipated tax retirees prefer a design that raises current labor supply, on which their benefits are levied, which is the more earnings-related one. Second, the higher tax rate associated with ageing raises the loss on every unit of redistribution, since the share of revenue lost grows with the size of the system. Thus, the higher tax increases the incentive to make pension benefits more earnings-related, consistent with the evidence from figure 1.1.

A more equal income distribution moves the design through three forces, whose balance depends on the IES. First, compressing incomes shrinks the gain that poorer young voters expect from flat benefits in their own retirement, while the stake in an earnings-related design that works through labor supply does not depend on the dispersion of incomes, which pushes the design towards more Bismarckian benefits. Second, compressing incomes makes flat benefits cheaper, since the share of revenue lost scales with the dispersion of relative incomes, which France's distribution halves, from 0.436 to 0.218. Third, it lowers the tax rate, and with it the share of revenue that every unit of redistribution loses. The second and third forces push the design towards more Beveridgean benefits and are strongest at low values of the IES, where the calibrated cost parameter is largest (table 4), and the first is strongest at high values, where the stake through labor supply is strongest. A more equal distribution therefore lowers the design by 0.061 at $\rho = 0.5$, where the first two forces nearly offset each other and the third supplies most of the fall, raises it slightly at $\rho = 1$, and raises it by much more at $\rho = 2$.[32]

7.3 The OECD cross-section: the design

We return to the 21 synthetic countries of section 6 to test whether the political process of section 7.2 reproduces the relations of figure 1.1: pension spending and the Bismarckian index both rise as population growth falls, and so rise together, while neither is related to income inequality. Each synthetic country now also chooses its design, at the U.S. cost of redistribution.[33]

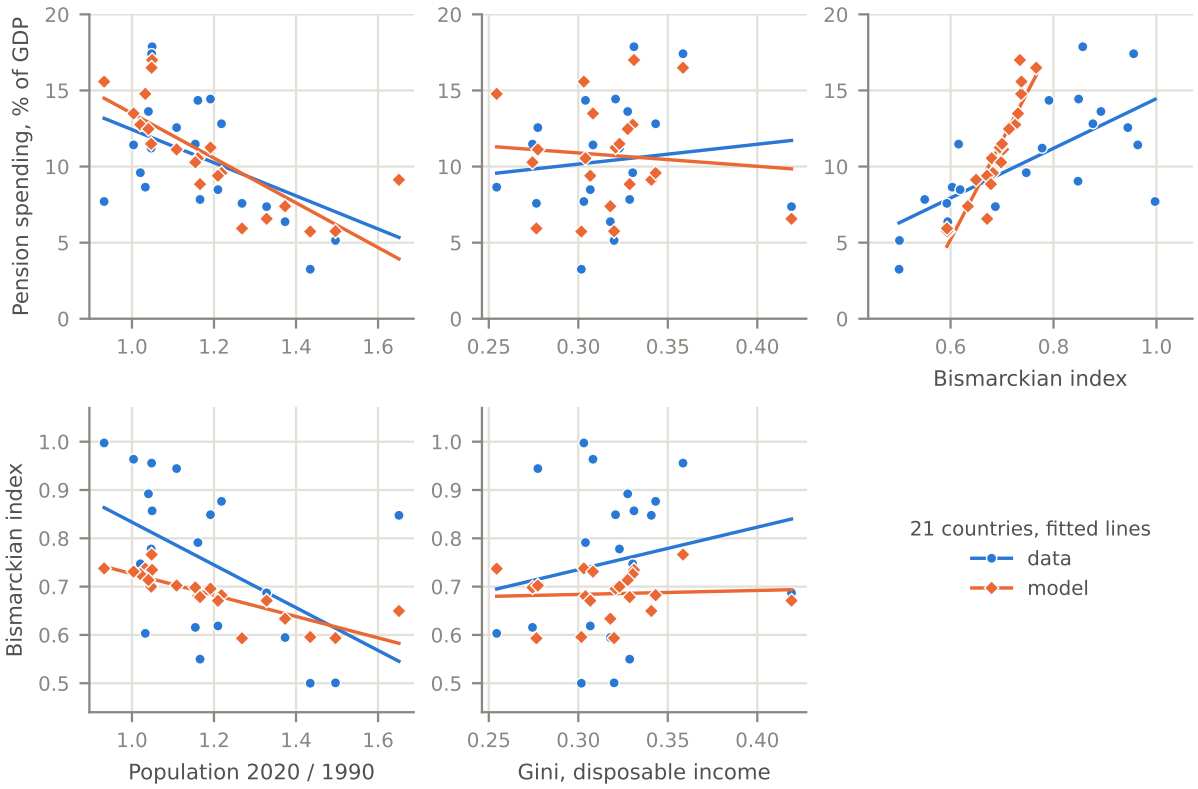
We solve for the politico-economic equilibrium paths of the 21 synthetic countries, the calibrated U.S. economy with each country's demography, income distribution and propensities to vote, at the U.S. discount factor, $\beta = 0.65$, and at the U.S. cost measured from the same data as every other country, OECD earnings data and CSES turnout,

[32]Imposing France's distribution in three steps separates the forces: first everywhere but in the cost, whose share of revenue lost at each design is held at its U.S. value, then on the dispersion of relative incomes in the cost, and last on the tax rate in the cost. The first step raises the design by 0.200, 0.239 and 0.371 at $\rho = 0.5$, 1 and 2. The second lowers it by 0.216, 0.188 and 0.062, and the third by 0.045, 0.039 and 0.007. The split depends on the order of the steps: imposed alone, France's dispersion in the cost lowers the U.S. design by 0.278 at $\rho = 1$.

[33]Specifically, with ε the elasticity of the Harberger loss in appendix C.2, we write the U.S. cost parameter as $\lambda_{US,t} = \varepsilon \Pi_{US,t}/R_{US,t}$, which falls as a population ages, and let country n face $\lambda_{n,t} = \varepsilon \Pi_{n,t}/R_{n,t}$ at its own ratio.

$\varepsilon = 13.09$ (appendix G.2).[34] Every country takes the same political weight of the old, $\omega = 1.20$, the mean of the weights that make the model match each country's pension spending at its observed design (appendix G.3), so that the countries differ only in the three dimensions of figure 1.1 and in their propensities to vote.

Figure 7.2: The relations of figure 1.1 in the data and in the model



Note: 21 countries, as in figure 6.2, in the data and in the model (section 7.3): every country at the U.S. β and ε and at $\omega = 1.20$, the mean of the political weights that match each country's spending at its observed design (appendix G.3), at its cost $\lambda_{n,t}$ and at its own CSES propensities to vote, Luxembourg, which has none, at a flat profile. The model's index is $1/(2-\theta)$ of the design chosen in 1990 for 2020, its spending $\tau(1-\alpha)$ in 2020 on the path of that design. The lines are least-squares fits; table 5 reports their slopes.

Figure 7.2 sets the model beside the data, and table 5 reports the slopes. The model reproduces the signs of the three relations that are significant in the data. Spending, now on the path of the design each country chooses, falls with population growth within one standard error of the data's slope, -14.70 against -10.89. The index falls with population growth too, but about half as steeply, -0.22 against -0.44. Spending therefore rises with the index, as in the data, but about four times as steeply, 64.85 against 16.33.[35]

In the model the joint rise runs through demography: an older country chooses both a larger system and a more Bismarckian design, as acute ageing does for the U.S. in

[34] At the U.S. ratio of 2020 this ε gives $\lambda = 7.27$, where the calibration of table 4, on the households and turnout of the CPS, has $\lambda = 6.25$. The U.S. political weight at these data is 1.26, against 1.25 there.

[35] Luxembourg, whose population grew fastest, flattens the data's slope of the index on population growth. On the 20 countries without it the slope is -0.77 in the data and -0.29 in the model (online appendix).

Table 5: Figure 1.1's relations in the data and in the model

Relation	**Data**	**Model**
Pension spending on population growth	−10.89	−14.70
	(4.11)	(2.71)
Pension spending on the Gini	13.04	−8.77
	(25.40)	(22.90)
Pension spending on the Bismarckian index	16.33	64.85
	(3.91)	(5.86)
Bismarckian index on population growth	−0.44	−0.22
	(0.18)	(0.04)
Bismarckian index on the Gini	0.88	0.08
	(1.06)	(0.33)

Note: Least-squares slopes, standard errors in parentheses, 21 countries. The model as in figure 7.2.

figure 7.1. At a given demography the cost also reverses the relation between the design and the tax at flat designs. In the U.S. calibration with the design pinned, the tax rate of 2020 falls without the cost from 13.6% at $\theta = 0$ to 6.9% at $\theta = 1$, the 6.7 p.p. of section 6. With the cost it rises from 8.6% at $\theta = 0$ to a peak of 10.5% at $\theta = 0.58$, close to the design the electorate chooses, and falls only above it, to 8.7% at $\theta = 1$ (online appendix).[36] Near the chosen design the tax is thus nearly flat in θ, and across countries spending and the index move together with demography.

Figure 7.3 sets each country's predicted design against its observed one. Beside the model it shows a variant in which each country keeps its own political weight of the old, *own* ω for short, calibrated so that the model matches the country's pension spending at its observed design (appendix G.3). The model orders the countries much as the data do: without the U.S., where the cost is calibrated, the correlation between predicted and observed designs is 0.72, and 0.77 with own ω. The root mean squared errors, 0.26 and 0.24, are below the 0.32 of a prediction at the sample mean. The predicted designs, however, spread only a third to a half as much as the observed ones, with standard deviations of 0.11 and 0.16 against 0.32. The compression is also why spending rises more steeply with the index than in the data: in the model spending responds to population growth at least as much as in the data, and the index about half as much. With own ω the slopes keep these signs and magnitudes (the starred rows of table 11).

Neither spending nor the index is related to the Gini in the model, as in the data (table 5). Appendix G repeats the test with workers per retiree uncorrected for the retirement age or corrected at the effective age of labor-market exit, and under two other

[36] The two curves belong to separate calibrations, each through the observed tax rate of 10.5% at θ^*. With the cost β and ω are recalibrated, ω to 1.25 against 1.17 without it (tables 4 and 3), so the curves differ at $\theta = 1$ although the cost vanishes there and, at given β and ω, leaves the tax chosen at that corner unchanged (section 7.1).

Figure 7.3: The predicted and the observed design



Note: The design θ of each country's model for 2020, chosen in 1990, against the observed one. Left, the model of figure 7.2; right, the model with each country's own political weight (own ω), calibrated to its spending at its observed design (appendix G.3). The U.S. is the anchor; in the right panel it lies on the 45° line by construction. The solid line is the least-squares fit over the other 20 countries, and each panel reports, without the U.S., the correlation r and the RMSE of the prediction, beside the RMSE of a prediction at the sample mean.

measures of the design (table 11). The relations that are significant in the data keep their signs throughout, and the slopes on the Gini stay insignificant. The correlation between predicted and observed designs falls markedly only where one small country enters with a predicted design close to the Beveridgean corner: Luxembourg with workers per retiree uncorrected, and Iceland under the measure over all mandatory tiers (figures G.1 and G.2).

8 Conclusions

This paper studies how two features of a pension system, whether benefits require a contribution record and how closely they follow past earnings, shape its political support and through it taxes, savings and labor supply. The setting is an overlapping generations model with capital, heterogeneous households, an informal sector and probabilistic voting, first over the size of the system and then over its design as well.

With log-GHH preferences and hand-to-mouth informal households the equilibrium tax is a function of parameters and demographics alone. It rises with the political weight of the old, with ageing and, when propensities to vote are equal, with income inequality, and it falls when benefits are more closely tied to earnings, since earnings-related benefits move revenue from poor retirees to rich ones. Calibrated to Argentina, the model

accounts for about three fifths of the rise in pension spending that accompanied the 2005–2010 expansion of coverage. In the model calibrated to the U.S., ageing to a constant population and the earnings link are the main determinants of the tax rate, except at an intertemporal elasticity of substitution of two, at which France's flatter voting profile moves the tax more than the earnings link does. In 21 synthetic OECD countries, each at its own propensities to vote, pension spending falls with population growth about as steeply as in the data and, as in the data, is unrelated to income inequality.

When the electorate chooses the earnings link as well, costless redistribution puts the choice at the Beveridgean corner unless the rich vote much more often than the poor, and at the U.S. calibration under log preferences it still does. An interior design needs flat benefits to be costlier than earnings-related ones, and we read the flat component as an implicit tax whose deadweight loss rises with the square of its rate. With this cost calibrated on the U.S. design, the tax rate under log preferences and at a given demography peaks near the design the electorate chooses rather than at flat benefits. Ageing raises both the tax rate and the chosen design at every intertemporal elasticity of substitution, while the effect of a more equal income distribution on the design depends on that elasticity. The 21 synthetic countries reproduce the joint rise of spending and the earnings link as population growth falls, though the earnings link responds about half as much as in the data, and the designs they choose, correlated with the observed ones, spread only a third to a half as much. Neither their designs nor their spending follows inequality, as in the data.

Three limitations point to future work. The model has no risk; in Berg and Gonzalez-Eiras (2023) we study a related environment in which health risk may prevent households from qualifying for a contributive pension. Every quantitative result rests on GHH preferences, which rule out income effects on labor supply, while both the hours responses and the retirees' stake in an earnings-related design work through labor supply. And a cost quadratic in the redistribution performed cannot deliver a fully Bismarckian system; a linear term in the cost, or a formality margin such as the informal sector of our Argentine model, would allow it.

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Appendices

A Economic equilibrium

The appendix derives results for the economic equilibrium with CRRA-GHH and log-GHH preferences and where informal workers' endowment is tied to wages in the formal sector absent taxes.

A.1 With CRRA-GHH preferences

Household solutions. With CRRA-GHH preferences, the first order conditions in (6) can be written as

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^{\xi}$$
$$c^i_{2,t+1} = \tilde{c}^i_{1,t} \left(\beta R_{t+1}\right)^{\rho}.$$

Using the budget constraints and rearranging slightly, we can write individual savings as

$$s_{t,i} = \frac{B_{t+1}}{1+B_{t+1}} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^i_{t+1}/R_{t+1}}{1+B_{t+1}},$$

where we have used the shorthand notation $B_{t+1} \equiv (\beta)^{\rho} \left(R_{t+1}\right)^{\rho-1}$.

When informal households save using informal technology, similar conditions apply:[37]

$$h_{t,0} = \left(\frac{\eta_0 w^0_t}{X_0} \right)^{\xi} \tag{A.1}$$
$$c^0_{2,t+1} = \tilde{c}^0_{1,t} \left(\beta R^0_{t+1}\right)^{\rho}. \tag{A.2}$$

As for formal households, we can derive the following for informal savings:

$$s_{t,0} = \frac{B^0_{t+1}}{1+B^0_{t+1}} \left[w^0_t h_{t,0} \eta_0 - \frac{X_0 \xi}{1+\xi} \left(h_{t,0}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^0_{t+1}/R^0_{t+1}}{1+B^0_{t+1}},$$

where we have used $B^0_{t+1} \equiv (\beta)^{\rho} \left(R^0_{t+1}\right)^{\rho-1}$. When informal households are hand-to-mouth and receive an endowment when old, the solution for labor supply is given by (A.1).

Savings and labor supply in equilibrium. Next, in economic equilibrium, we impose conditions for factor prices in equations (3) and the specification of pension benefits

[37] Note that given our assumption on universal pensions, informal workers do not perceive a link between their work effort and the benefits they might receive.

including the balanced budget condition in equations (8). We can then rewrite type i's labor supply decision as

$$h_{t,i} = \left(\frac{\eta_i}{X_i}\right)^{\xi} \left(\frac{1}{h_t}\right)^{\xi} \left((1-\alpha)(1-\tau_t)\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha} + \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1}\right)^{\xi}.$$

The only type-specific term is $(\eta_i/X_i)^{\xi}$; this shows that the ratio $h_{t,i}/h_t$ is only a function of parameters, namely that

$$\frac{h_{t,i}}{h_t} = \frac{(\eta_i/X_i)^{\xi}}{\Gamma_h}, \qquad \Gamma_h \equiv \sum_{i>0} \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{A.3}$$

Next, we can use the individual labor decision and wage rates to write

$$(1-\alpha)(1-\tau_t) h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} = \frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma_h^{1/\xi}} - \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1}.$$

Using this and the discounted pension benefits in equilibrium, we have that

$$\begin{aligned} s_{t,i} =& \frac{1}{1+B_{t+1}} \left[B_{t+1} \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \frac{1}{1+\xi} - \frac{1-\alpha}{\alpha}\frac{1-\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \right] \\ & - \frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \end{aligned} \tag{A.4}$$

Using the definition of the aggregate state (1) and rearranging, we then have:

$$s_t = \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}, \qquad \Gamma_{s,t} \equiv \frac{1}{1+\xi} \frac{\Gamma_h B_{t+1}}{1 + B_{t+1} + \frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta B_{t+1})} \tag{A.5}$$

Furthermore, note that we can combine equations for s_t, h_t to yield closed-form expressions *given* values of B_{t+1}. From the definition of h_t we use that s_t is proportional to $h_t^{(1+\xi)/\xi}$ to derive:

$$h_t = \Theta_{h,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}}, \qquad \Theta_{h,t} \equiv (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}} \tag{A.6}$$

Using this, an alternative expression for s_t is:

$$s_t = \Theta_{s,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}, \qquad \Theta_{s,t} \equiv \left(\frac{\Theta_{h,t}}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}. \tag{A.7}$$

Using the relation between s_t and $h_t^{\frac{1+\xi}{\xi}}$ in (A.5), the ratio of individual (A.4) to aggregate

savings satisfies:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} \frac{1-\theta}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} - 1 \right). \tag{A.8}$$

For informal households, we assume that informal wages are proportional to formal wages absent taxes and scaled appropriately by informal productivity η_0. Furthermore, we assume that the return to informal savings is proportional to the formal return on savings $R_t^0 = r_0 R_t$.

Specifically, we assume that informal wages are given by:

$$w_t^0 = \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha}{1+\alpha\xi}}$$

Informal labor supply then follows directly from the derivations above for formal workers; in the case with informal savings, the ratio of informal to aggregate savings is similarly given by

$$\frac{s_{t,0}}{s_t} = \frac{B_{t+1}^0}{1+B_{t+1}^0} \frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \frac{1}{\Theta_{s,t}} - \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} \tag{A.9}$$

Consumption functions. It is straightforward to derive relevant equilibrium consumption functions $(c_{1,t}^j, \tilde{c}_{1,t}^j, c_{2,t}^j)$ given the results above. Imposing household budget constraints (5), factor prices (3), pension benefits and balanced budget condition (8), and labor and savings decisions from above, we have for formal households i:

$$c_{1,t}^i = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{s_t}{\Gamma_{s,t}} \left(1 - \frac{B_{t+1}}{(1+B_{t+1})(1+\xi)} \right) + \frac{s_t}{1+B_{t+1}} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \tag{A.10}$$

$$c_{2,t}^i = \alpha\nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} \left(\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta \left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} - 1 \right] \right) \right) \tag{A.11}$$

$$\tilde{c}_{1,t}^i = \frac{s_t}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{(1+\xi)\Gamma_{s,t}} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \right) \tag{A.12}$$

When informal households save using informal technology, equilibrium consumption functions are given by

$$\tilde{c}_{1,t}^0 = \frac{1}{1+B_{t+1}^0} \left[\frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^{\xi}} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} s_t \right] \tag{A.13}$$

$$c_{2,t}^0 = r_0\alpha\nu_t \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} h_t^{1-\alpha} \left(\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_t}{1+\gamma_0\epsilon} \right). \tag{A.14}$$

When informal households are hand-to-mouth, relevant equilibrium consumption func-

tions are given by[38]

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi}\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{A.15}$$

$$\tilde{c}^0_{2,t} = \frac{1}{1+\xi}\frac{(\chi\eta_0)^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha}. \tag{A.16}$$

A.2 With log-GHH preferences

With log-preferences, the economic equilibrium is identical to the CRRA equations outlined in appendix A.1, except for the discount function $B_t = B_t^0 = \beta$. All relevant variables $x_t \in \{s_t, h_t, s_{t,i}, h_{t,j}, \tilde{c}^j_{1,t}, c^i_{2,t}, \tilde{c}^0_{2,t}, c^i_{2,t+1}, \tilde{c}^0_{2,t+1}\}$ can be written in the form $x_t = \Theta_{x,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_x}$ where σ_x are parameter functions

$$\begin{aligned}
\sigma_h = \sigma_h^j &= \frac{\alpha\xi}{1+\alpha\xi} \\
\sigma^j_{\tilde{c}_1,t} = \sigma^i_{c_2,t} = \sigma^0_{\tilde{c}_2,t} = \sigma_{s,t} &= \sigma^i_{s,t} = \frac{\alpha(1+\xi)}{1+\alpha\xi} \\
\sigma^i_{c_2,t+1} = \sigma^0_{\tilde{c}_2,t+1} &= \left(\frac{\alpha(1+\xi)}{1+\alpha\xi}\right)^2,
\end{aligned}$$

and where the functions $\Theta_{x,t}$ depend only on parameters and policy.

Proof of proposition 1. Part *i.* outlines the marginal effect on s_t, h_t from policy and pension characteristics. Given the solution structure, the effect on s_t, h_t can be found by differentiating equations (A.5) and (A.6) with respect to the relevant variable. For τ_t the effect is straightforward:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \tau_t} &= -\frac{\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0 \\
\frac{\partial \ln(s_t)}{\partial \tau_t} &= -\frac{1+\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0.
\end{aligned}$$

We can also show that h_t is unambiguously decreasing in ϵ:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \epsilon} &= \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\frac{1}{1+\gamma_0\epsilon}\left((1+\gamma_0\epsilon)\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} - \gamma_0\right) \leq 0 \\
\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} &= \frac{\gamma_0}{1+\gamma_0\epsilon}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}.
\end{aligned}$$

[38] Recall informal workers are assumed to receive an endowment proportional to wages of formal workers when they retire. For tractability we define this as if they were making a labor supply decision. The relevant parameter determining the size of the endowment is χ.

However, the effect on s_t is harder to sign.

$$\frac{\partial \ln(s_t)}{\partial \epsilon} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \epsilon} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon}.$$

Replacing from the expressions above, and neglecting common factors, the sign of the derivative is given by the sign of

$$-\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} + 1 + \theta\beta.$$

This can be seen to be positive for all $\theta \in [0,1]$. Note that the effect from γ_0 is equivalent to that of ϵ. We now consider the effect of future taxes on labor supply and savings.

$$\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}\right) \geq 0$$

$$\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in τ_{t+1}. For savings the effect is:

$$\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}.$$

As before the sign of this derivative can be shown to be the same as the sign of the following term:

$$\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - (1+\theta\beta).$$

This is the negative of the term above. Thus, for all values of θ and τ_{t+1} we have that an increase in τ_{t+1} reduces savings.

Taking the derivative with respect to θ, we see a similar condition as with the effect of τ_{t+1}, namely:

$$\frac{\partial \ln(h_t)}{\partial \theta} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}\right) \geq 0,$$

$$\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\theta\beta}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in θ. For savings, we have:

$$\frac{\partial \ln(s_t)}{\partial \theta} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \theta} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}.$$

Proceeding as before we find that this is unambiguously negative.[39]

Part *ii.* follows from earlier derivations, specifically equation (A.3). Part *iii.* similarly follows from equation (A.8) with $B_{t+1} = \beta$ in the log case. Finally, part *iv.* follows straightforwardly, comparing (A.8) and (A.3): an average type i household has $\eta_i^{1+\xi}/X_i^{\xi} = \Gamma_h$. Thus, for $\tau_{t+1} = 0$ or $\theta = 1$, we simply have $s_{t,i}/s_t = \eta_i h_{t,i}/h_t$. The spread in savings increases in τ_{t+1} and decreases in θ in the sense that these parameter changes increase $s_{t,i}/s_t$ for $s_{t,i}/s_t > 1$ and decrease for $s_{t,i}/s_t < 1$.

Part *v.* The effects on $s_{t,0}$ and $s_{t,0}/s_t$ from various parameters come from differentiation of (A.9) with $B^0_{t+1} = \beta$. Note that $s_{t,0}$ depends on τ_t only through the effect of taxes on s_t. Thus, it is straightforward to show that both $s_{t,0}$ and $s_{t,0}/s_t$ are increasing in τ_t.

The effects from τ_{t+1} and ϵ are given by:

$$\frac{\partial s_{t,0}}{\partial \tau_{t+1}} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\epsilon}{1+\gamma_0\epsilon}\left(1+\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}}\right) \leq 0,$$
$$\frac{\partial s_{t,0}}{\partial \epsilon} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\tau_{t+1}}{1+\gamma_0\epsilon}\left(\frac{1}{1+\gamma_0\epsilon}+\epsilon\frac{\partial \ln(s_t)}{\partial \epsilon}\right) \leq 0.$$

The first derivative can be signed since $\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} \geq -1$. It is straightforward to show that an increase in θ increases $s_{t,0}$.

To see the effect of τ_{t+1} on the ratio $s_{t,0}/s_t$ we have that:

$$\frac{\partial(s_{t,0}/s_t)}{\partial \tau_{t+1}} = -\frac{\beta}{1+\beta}\frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\frac{1}{\Theta_{s,t}}\frac{\partial \ln(\Theta_{s,t})}{\partial \tau_{t+1}} - \frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon},$$

where the derivative $\partial \ln(\Theta_{s,t})/\partial \tau_{t+1} = \partial \ln(s_t)/\partial \tau_{t+1}$ is defined above. We note that this may be negative or positive. For instance, if $\epsilon \approx 0$ an increase in future taxes increases this ratio. Conversely if $r_0 \approx 0$ then we would expect the opposite effect.

B Politico-economic equilibrium

B.1 Proof of proposition 2

By appendix A.2, all consumption functions are log-separable in the aggregate state s_{t-1}, and in all the Θ-coefficient functions the only *predetermined* variable that may enter the political problem as an endogenous state is the distribution of savings $s_{t-1,i}/s_{t-1}$. The current tax moves the aggregate state carried into the next period, s_t, but by the same

[39] The term whose sign determines this is slightly different

$$\frac{1}{1+\alpha\xi}\frac{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - 1.$$

log-separability its level drops out of the first order condition at $t+1$. The distribution chosen at t, $s_{t,i}/s_t$, depends on parameters and on the tax and the design anticipated for $t+1$ (part iii of proposition 1), but not on τ_t, so that the part of the law of motion (9) that carries the distribution does not respond to the current tax,

$$\frac{\partial\left(s_{t,i}/s_t\right)}{\partial\tau_t}=0, \qquad i>0.$$

The distribution is therefore not an *endogenous* state: changing τ_t has no strategic effect on future policy.

The tax at t maximizes (14) given the inherited distribution of savings and the continuation policy, and it satisfies the first order condition $\mathcal{F}_t(\tau_t;\boldsymbol{s}_{t-1})\leq 0$ where $\boldsymbol{s}_{t-1}$ is the vector of predetermined savings, $s_{t-1,i}/s_{t-1}$, and

$$\mathcal{F}_t=\omega\left[\gamma_0\mu_0\overbrace{\frac{\partial\ln\left(\Theta^0_{\tilde{c}_2,t}\right)}{\partial\tau_t}}^{=\mathcal{E}^0_{2,t}}+\sum_{i>0}\gamma_i\mu_i\overbrace{\frac{\partial\ln\left(\Theta^i_{c_2,t}\right)}{\partial\tau_t}}^{=\mathcal{E}^i_{2,t}}\right]$$
$$+\nu_t\left[\gamma_0\mu_0\underbrace{\left(\frac{\partial\ln\left(\Theta^0_{\tilde{c}_1,t}\right)}{\partial\tau_t}+\beta\frac{\partial\ln\left(\Theta^0_{\tilde{c}_2,t+1}\right)}{\partial\tau_t}\right)}_{=\mathcal{E}^0_{1,t}}+\sum_{i>0}\gamma_i\mu_i\underbrace{\left(\frac{\partial\ln\left(\Theta^i_{\tilde{c}_1,t}\right)}{\partial\tau_t}+\beta\frac{\partial\ln\left(\Theta^i_{c_2,t+1}\right)}{\partial\tau_t}\right)}_{=\mathcal{E}^i_{1,t}}\right],$$

where $\mathcal{E}^j_{x,t}$ denotes the marginal effect of the tax on the indirect utility of a household of type j in generation x and μ_j the propensity to vote of type j. This holds with equality for an interior solution. Using the equilibrium consumption functions of appendix A.1,

$$\mathcal{E}^0_{2,t}=\frac{(1-\alpha)\nu_t\epsilon}{1+\gamma_0\epsilon}\Theta_{h,t}^{1-\alpha}\left(1+(1-\alpha)\tau_t\frac{\partial\ln\left(\Theta_{h,t}\right)}{\partial\tau_t}\right)}{\frac{(\chi\eta_0)^{1+\xi}}{(1+\xi)X_0^\xi}\left(\frac{1-\alpha}{\Gamma_h^\alpha}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\Theta_{h,t}^{1-\alpha}} \tag{B.1}$$

$$\mathcal{E}^i_{2,t}=(1-\alpha)\frac{\partial\ln\left(\Theta_{h,t}\right)}{\partial\tau_t}+\frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s_{t-1,i}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)} \tag{B.2}$$

$$\mathcal{E}^0_{1,t}=-\frac{\beta}{1-\tau_t}\frac{\alpha(1+\xi)^2}{(1+\alpha\xi)^2} \tag{B.3}$$

$$\mathcal{E}^i_{1,t}=-\frac{1}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi}\left(1+\beta\frac{\alpha(1+\xi)}{1+\alpha\xi}\right). \tag{B.4}$$

Let $y_i\equiv\eta_i^{1+\xi}/(X_i^\xi\Gamma_h)=\eta_i h_{t-1,i}/h_{t-1}$ denote relative labor income and let ϕ_t denote the elasticity of revenue with respect to the tax rate by

$$\phi_t\equiv 1+(1-\alpha)\tau_t\frac{\partial\ln\left(\Theta_{h,t}\right)}{\partial\tau_t}=1-\frac{(1-\alpha)\xi}{1+\alpha\xi}\frac{\tau_t}{1-\tau_t},$$

so that $1-\phi_t$ is the elasticity of the tax base in absolute value. We can then rearrange (B.1) and (B.2)

$$\mathcal{E}^0_{2,t} = \frac{\phi_t \sigma^0_t}{\tau_t}, \qquad \mathcal{E}^i_{2,t} = \frac{\phi_t - 1 + \sigma^i_t}{\tau_t}, \tag{B.5}$$

where $\sigma^0_t \equiv b^0_t/\tilde{c}^0_{2,t}$ and $\sigma^i_t \equiv b^i_t/c^i_{2,t}$ are pension shares of consumption.[40] This states that a unit of tax raises every pension by ϕ_t/τ_t in proportion, and with log utility a retiree's support is that proportion times the pension's share of consumption, less, for formal retirees, the fall in the return on their savings.

Proposition 3. *Assume log-GHH preferences and hand-to-mouth informal households, with $\xi(1-2\alpha) \leq 1$ if $\gamma_0 > 0$. For every horizon T, the politico-economic equilibrium is unique.*

Proof of proposition 3. Let $G_t(\tau_t) \equiv \mathcal{F}_t(\tau_t; \boldsymbol{s}_{t-1}(\tau_t))$ denote the marginal effect of taxes on the political objective evaluated with past savings anticipating the candidate tax τ_t, i.e. with $s_{t-1,i}/s_{t-1}$ following (A.8). Along this path, the second term of (B.2) is positive, finite, and strictly decreasing in τ_t on $[0,1)$. A higher anticipated tax lowers the savings of a retiree below mean income, but its pension rises by more than its savings income falls, so the pension's share of its consumption rises less than in proportion to the tax. The first term of (B.2) and the young's terms (B.3) and (B.4) are non-positive multiples of $1/(1-\tau_t)$, and (B.4) is a negative multiple, strictly decreasing and unbounded below as $\tau_t \to 1$. The informal retirees' term (B.1) is the derivative of $\ln(e_t + b^0_t)$, where the endowment e_t does not depend on τ_t and, by (A.6), the pension b^0_t is proportional to $\tau_t(1-\tau_t)^m$ with $m \equiv (1-\alpha)\xi/(1+\alpha\xi)$. Its derivative in τ_t has the sign of $(e_t + b^0_t)\,\partial^2 b^0_t/\partial\tau_t^2 - (\partial b^0_t/\partial\tau_t)^2$, which is negative where b^0_t is concave in τ_t, on all of $[0,1)$ when $m \leq 1$, that is $\xi(1-2\alpha) \leq 1$. Every term enters G_t with a positive weight, so G_t is continuous and strictly decreasing on $[0,1)$, finite at zero and unbounded below as $\tau_t \to 1$.

At fixed shares, the second term of (B.2) is strictly decreasing in τ_t wherever the retiree's consumption is positive, so $\mathcal{F}_t(\cdot\,; \boldsymbol{s}_{t-1})$ is strictly decreasing and the political objective strictly concave in τ_t there. A tax τ^* therefore maximizes the objective at the shares $\boldsymbol{s}_{t-1}(\tau^*)$ if and only if $G_t(\tau^*) = 0$, or $\tau^* = 0$ and $G_t(0) \leq 0$. Exactly one tax does so, the zero of G_t if $G_t(0) > 0$ and $\tau_t = 0$ otherwise. Every other candidate of the test is a corner at which $G_t \neq 0$, so the objective at its own shares rises as the tax moves away from it, and it fails.

At T the young have no second period and their terms are (B.3)–(B.4) with $\beta = 0$ (appendix D), so G_T involves no state and neither does the tax at T. Given a continuation

[40] To see this, use that $b^i_t = R_t s_{t-1} \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0\epsilon} \tau_t(1 + \theta_t(y_i - 1))$.

that does not respond to the state, the young's terms at t take the form (B.3)–(B.4), and G_t again involves no state, so by induction this holds at every date. The continuation enters G_t only through the level of $\Theta_{h,t}$ in b_t^0, by (A.6). Backward induction from T therefore gives a unique equilibrium for every horizon, and without informal households G_t involves only τ_t, θ_t, ν_t and the parameters. □

It is straightforward to confirm that the argument carries over to the tax condition with the cost of section 7 at the inherited design (appendix C.3, where $\gamma_0 = 0$). The proposition does not extend to informal savers or to CRRA preferences. With either, the condition carries the response of the continuation policy to the state, which has no fixed sign, and under CRRA preferences every term carries a marginal-utility weight that moves with the tax. For these cases the solver records at every state and date how many candidates pass the test (appendix D).

Comparative statics. By the proof of proposition 3, G_t, and $\mathcal{F}_t$ at fixed shares, have strictly negative derivatives in τ_t, since that of (B.4) is strictly negative and those of the other terms are not positive. At an interior equilibrium a parameter therefore moves the tax in the direction in which it moves G_t, or $\mathcal{F}_t$ when the shares are held at their inherited values, by the implicit function theorem, and at the corner $\tau_t = 0$ a small change has no effect. The comparative statics are those of the tax at t given the continuation policy, which enters only through σ_t^0. Without a universal pension it does not enter, and they hold at every date of the equilibrium path.

ω and ν_t. The retirees' terms sum to a positive number at an interior equilibrium, since the young's are negative, so taxes rise with ω. The ratio ν_t weights the young's terms and also enters the universal pension. The pension b_t^0 is proportional to ν_t, more contributors per informal retiree, while the endowment is not, and both scale with the same power of s_{t-1}/ν_t (appendix A.1). Hence $\partial\sigma_t^0/\partial\nu_t = \sigma_t^0(1-\sigma_t^0)/\nu_t$ and $\partial\mathcal{E}_{2,t}^0/\partial\nu_t = \mathcal{E}_{2,t}^0(1-\sigma_t^0)/\nu_t$, while σ_t^i does not move, since formal pensions and capital income both scale with ν_t. Using $G_t = 0$ to substitute for the young's terms,

$$\begin{aligned}\frac{\partial G_t}{\partial \nu_t} &= \frac{\omega}{\nu_t}\gamma_0\mu_0\mathcal{E}_{2,t}^0\left(1-\sigma_t^0\right) + \gamma_0\mu_0\mathcal{E}_{1,t}^0 + \sum_{i>0}\gamma_i\mu_i\mathcal{E}_{1,t}^i \\ &= -\frac{\omega}{\nu_t}\left[\gamma_0\mu_0\mathcal{E}_{2,t}^0\sigma_t^0 + \sum_{i>0}\gamma_i\mu_i\mathcal{E}_{2,t}^i\right]. \qquad \text{(B.6)}\end{aligned}$$

Since $0 \le \sigma_t^0 \le 1$, the bracket is at least the smaller of the formal retirees' term $\sum_{i>0}\gamma_i\mu_i\mathcal{E}_{2,t}^i$ and the retirees' total, which is positive. Taxes therefore fall with ν_t whenever the aggregate marginal support of formal retirees is positive at the equilibrium,

which by (B.5) is

$$\sum_{i>0} \gamma_i \mu_i \mathcal{E}^i_{2,t} > 0 \quad \Longleftrightarrow \quad \frac{\sum_{i>0} \gamma_i \mu_i \sigma^i_t}{\sum_{i>0} \gamma_i \mu_i} > 1 - \phi_t : \tag{B.7}$$

the average share of the pension in the consumption of formal retirees, weighted by their propensities to vote, exceeds the elasticity of the tax base. Without a universal pension the formal retirees' term is the retirees' total and the condition always holds. With one it fails only if the tax is sustained by informal retirees against formal ones.

Income inequality. For any two types with $y_H > y_L$, raise η_H and lower η_L such that

$$\frac{\partial \eta_L}{\partial \eta_H} = -\frac{\gamma_H \left(\frac{\eta_H}{X_H}\right)^{\xi}}{\gamma_L \left(\frac{\eta_L}{X_L}\right)^{\xi}},$$

which keeps Γ_h, and so aggregate labor supply and savings, constant for given policies. Since $\partial y_m / \partial \eta_m = (1+\xi)(\eta_m / X_m)^{\xi} / \Gamma_h$, the change leaves every other y_j unchanged and satisfies $\gamma_H \, dy_H + \gamma_L \, dy_L = 0$: it transfers relative income from a poorer to a richer type at a constant mean. It reaches G_t only through the pension shares of formal retirees, since the young's terms and ϕ_t do not involve the y_i and σ^0_t involves them only through Γ_h. At consistent shares a formal retiree's pension share is a function of its relative income alone, and using $\gamma_L \, dy_L = -\gamma_H \, dy_H$, the transfer changes the retirees' terms by

$$\frac{\omega}{\tau_t} \gamma_H \, dy_H \left[\mu_H \frac{\partial \sigma^H_t}{\partial y_H} - \mu_L \frac{\partial \sigma^L_t}{\partial y_L} \right],$$

which is positive with equal propensities to vote, and in general whenever $\mu_H / \mu_L < (c^H_{2,t} / c^L_{2,t})^2$. Every spread of relative incomes built from such transfers therefore raises the tax.[41]

The design. The design θ_t pays current retirees, who anticipated it when they saved. Let $D_{t,i} \equiv \frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \tau_t \left(1 + \theta_t (y_i - 1)\right)$ such that we can write $c^i_{2,t} = R_t s_{t-1} D_{t,i}$. At consistent shares, (A.8) one period back, $\partial D_{t,i} / \partial \theta_t = \frac{\beta}{1+\beta} \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \tau_t (y_i - 1)$: consumption moves by the fraction $\beta/(1+\beta)$ of the change in the pension, savings having absorbed the rest. We then have

$$\begin{aligned} \frac{\partial \sigma^i_t}{\partial \theta_t} &= \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \tau_t \left(1 + \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \frac{\tau_t}{1+\beta}\right) \frac{y_i}{D^2_{t,i}} (y_i - 1), \\ \frac{\partial G_t}{\partial \theta_t} &= \omega \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \left(1 + \frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0 \epsilon} \frac{\tau_t}{1+\beta}\right) \sum_{i>0} \gamma_i \frac{\mu_i y_i}{D^2_{t,i}} (y_i - 1). \end{aligned} \tag{B.8}$$

[41] The result rests on the response of savings: with the shares held at their inherited values instead, the transfer would move σ^i_t only through the earnings-related part of the pension and would leave the tax unchanged at $\theta_t = 0$.

Since $\sum_{i>0}\gamma_i(y_i-1)=0$, the sum equals $\frac{1}{2}\sum_{i,j}\gamma_i\gamma_j\left(\frac{\mu_i y_i}{D_{t,i}^2}-\frac{\mu_j y_j}{D_{t,j}^2}\right)(y_i-y_j)$, and is negative when $\mu_i y_i/D_{t,i}^2$ falls with relative income and incomes are not all equal. Taxes therefore fall with θ_t whenever $\mu_i y_i/(c_{2,t}^i)^2$ is decreasing in relative income across formal types. With equal propensities, $D_{t,i}$ is affine in y_i at consistent shares, and $y_i/D_{t,i}^2$ falls with y_i wherever $D_{t,i}$ exceeds twice its value at $y_i=0$, which in levels is $\beta/(1+\beta)$ times the flat component of the pension, $(1-\theta_t)h_{t-1}\bar{b}_t$. The condition therefore holds across formal types if every formal retiree consumes more than $2\beta/(1+\beta)$ times this flat component. This is the case whenever no formal type borrows against its pension, since then $c_{2,t}^i\geq b_t^i\geq(1-\theta_t)h_{t-1}\bar{b}_t$, which exceeds the bound for $\beta<1$ and $\theta_t<1$, and the bound is zero at $\theta_t=1$. With the shares held at their inherited values instead, as for a design that retirees did not anticipate, the pension share moves only through the pension, and the weight on y_i-1 becomes $(s_{t-1,i}/s_{t-1})/(c_{2,t}^i)^2$. It need not fall with income, since by (A.8) the savings share rises with income faster than y_i when $\theta_t<1$; the sign is then open. A permanent change in the design also moves the continuation, which enters G_t only through σ_t^0 and so not at all without a universal pension.

The universal pension. The effect of ϵ works through retirees alone, since the young's terms do not depend on it. We sign it for a change that current retirees did not anticipate, with their savings shares at their inherited values, as at the reform date of section 5, and with $\Theta_{h,t}$, which moves with the universal pension the young anticipate for themselves, held fixed. Benefits are shares of revenue $\nu_t w_t h_t \tau_t$, $\epsilon/(1+\gamma_0\epsilon)$ per informal retiree and $(1+\theta_t(y_i-1))/(1+\gamma_0\epsilon)$ per formal retiree of type i, so a higher ϵ raises b_t^0 by $1/(\epsilon(1+\gamma_0\epsilon))$ in proportion and lowers every b_t^i by $\gamma_0/(1+\gamma_0\epsilon)$ in proportion, a pure transfer since there are γ_0 informal retirees per formal one. A pension share responds to a proportional change in the benefit by $\partial\sigma_t^j/\partial\ln b_t^j=\sigma_t^j(1-\sigma_t^j)$, since the rest of consumption does not move, and by (B.5) $\partial\mathcal{E}_{2,t}^0/\partial\sigma_t^0=\phi_t/\tau_t$ and $\partial\mathcal{E}_{2,t}^i/\partial\sigma_t^i=1/\tau_t$. Hence

$$\frac{\partial\mathcal{F}_t}{\partial\epsilon}=\frac{\omega\,\gamma_0}{\tau_t(1+\gamma_0\epsilon)}\left[\frac{\phi_t}{\epsilon}\,\mu_0\sigma_t^0\left(1-\sigma_t^0\right)-\sum_{i>0}\gamma_i\mu_i\sigma_t^i\left(1-\sigma_t^i\right)\right],$$

and taxes rise with ϵ if and only if the bracket is positive,

$$\frac{\phi_t}{\epsilon}\,\mu_0\sigma_t^0\left(1-\sigma_t^0\right)>\sum_{i>0}\gamma_i\mu_i\sigma_t^i\left(1-\sigma_t^i\right). \tag{B.9}$$

The share γ_0 multiplies the whole expression, because the loss of formal retirees is the gain of informal retirees times their number, and enters the bracket only through the consumption levels, so it does not determine the sign. The factor $\sigma(1-\sigma)$ is zero for a retiree without a pension and for one who lives on the pension alone, whose consumption and marginal benefit a higher ϵ raises in the same proportion, and is largest at one half. The factor $1/\epsilon$ is the asymmetry of the transfer: a unit of ϵ is a large proportional change

of a small universal pension and a small one of the formal pensions. With equal pension shares and equal propensities to vote the condition is $\epsilon < \phi_t$, and since informal retirees then consume ϵ times the average consumption of formal retirees, it requires them to be poorer by more than $1-\phi_t$, the elasticity of the tax base. The condition therefore holds when informal retirees are poorer than formal ones by more than this elasticity and the pension is a comparable share of their consumption, since a comparable share of a smaller consumption is a smaller pension, and fails when they live on the pension alone or are much richer than formal retirees. A permanent reform also lowers the hours of the young (part i of proposition 1), and with them the universal pension relative to the informal endowment, and a change that retirees anticipated would also move their savings shares; (B.9) leaves out both.

At the Argentine calibration of section 5, in 2010 before the reform, $\epsilon = 0.21$, $\phi_t = 0.976$, and the pension is 12.2% of the consumption of informal retirees and between 19.5% and 15.5% of that of formal retirees, falling with income. The left-hand side of (B.9) is then 0.498, three and a half times the right-hand side, 0.143. After the reform, at $\epsilon^U = 0.546$ and the post-reform tax of 13.8%, at which $\phi_t = 0.971$, the pension is 26.9% of the consumption of informal retirees and the two sides are 0.350 and 0.145. The calibration has informal savers, whose savings income falls with hours as that of formal retirees does, so their term carries a factor of one in place of ϕ_t, which raises the left-hand side before the reform to 0.511.

The representative-agent tax. With one type of formal household and no informal households, $J = 1$ and $\gamma_0 = 0$, relative income and the savings share are one, $y_1 = s_{t-1,1}/s_{t-1} = 1$, so the design drops out of (B.2): both components of the benefit pay the same household. The propensity to vote multiplies both generations' terms and cancels, and the first order condition is

$$\omega\left[\frac{1-\alpha}{\alpha+(1-\alpha)\tau_t} - \frac{(1-\alpha)\xi}{(1+\alpha\xi)(1-\tau_t)}\right] = \frac{\nu_t}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi}\left(1+\beta\frac{\alpha(1+\xi)}{1+\alpha\xi}\right).$$

With $\Psi \equiv (1+\xi)\left(1+\frac{\alpha\beta(1+\xi)}{1+\alpha\xi}\right)$, the condition is linear in τ_t once multiplied by $(1-\tau_t)(\alpha+(1-\alpha)\tau_t)$, and its unique solution is

$$\tau_t = \max\left\{0,\ \frac{\omega(1-\alpha)-\alpha\nu_t\Psi}{\omega(1-\alpha)(1+\xi)+(1-\alpha)\nu_t\Psi}\right\}. \tag{B.10}$$

Where it is positive, the tax rises with ω and falls with ν_t, α, β and ξ. With inelastic labor, $\xi = 0$, $\Psi = 1+\alpha\beta$ and (B.10) is the equilibrium tax of Gonzalez-Eiras and Niepelt (2008), their ratio of young to old households being ν_t and their per capita political weight of retirees ω. The Frisch elasticity lowers the tax because GHH preferences carry no income effect: the tax lowers hours, which erodes the base that pays retirees and

deepens the loss of the young, whereas under the separable logarithmic preferences of their section 3 a social security tax that is the only tax leaves hours unchanged. The same expression is the tax at the Bismarckian corner for any income distribution and any propensities to vote, as long as there are no informal households. At $\theta_t = 1$ benefits and, by part iv of proposition 1, savings are proportional to relative income, so every type has $\sigma_t^i = (1-\alpha)\tau_t/(\alpha+(1-\alpha)\tau_t)$ and the same marginal effects, and the propensities cancel. At the U.S. calibration of section 6 it gives a tax of 6.9% in 2020, against the calibrated 10.5%.

B.2 With log-GHH preferences and informal savings

The introduction of $s_{t-1,0}/s_{t-1}$ as an endogenous state variable complicates the model because this state variable not only depends on past choices (through the effect of τ_{t-1} on $\Theta_{s,t-1}$), but is also forward looking ($s_{t-1,0}/s_{t-1}$ depends explicitly on τ_t). To deal with this, let $dx_t/d\tau_t$ denote the effect in politico-economic equilibrium of changing the tax rate τ_t on the variable x_t accounting for the fact that a change in τ_t also changes the state $s_{t,0}/s_t$ which through the continuation policy affects the choice of τ_{t+1} as well. Specifically, for the state variable $s_{t,0}/s_t$ we have:

$$\frac{d(s_{t,0}/s_t)}{d\tau_t} = \frac{\partial(s_{t,0}/s_t)}{\partial\tau_t} + \frac{\partial(s_{t,0}/s_t)}{\partial\tau_{t+1}} \frac{d\tau_t^{+1}}{d(s_{t,0}/s_t)} \frac{d(s_{t,0}/s_t)}{d\tau_t},$$

where the partial derivatives of $s_{t,0}/s_t$ capture how the state reacts to policy in economic equilibrium and $\tau_t^{+1}(s_{t,0}/s_t)$ is the continuation policy that details how policy in $t+1$ adjusts when the endogenous state changes. When considering a marginal change in τ_t, the policymaker takes into account how this propagates via economic equilibrium and continuation policy functions into future periods in order to summarize the total impact on current relevant variables. The technical documentation derives a recursive structure for all the relevant derivatives required to evaluate the marginal effect of τ_t on the indirect utility of households (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

With the endogenous state $s_{t-1,0}/s_{t-1}$, the politico-economic equilibrium is now characterized by the first order condition:

$$\omega\left[\sum_{i\geq 0}\gamma_i\mu_i \overbrace{\frac{d\ln\left(c_{2,t}^i\right)}{d\tau_t}}^{=\mathcal{E}_{2,t}^i}\right] + \nu_t\left\{\sum_{i\geq 0}\gamma_i\mu_i \underbrace{\left(\frac{d\ln\left(\tilde{c}_{1,t}^i\right)}{d\tau_t} + \beta\frac{d\ln\left(c_{2,t+1}^i\right)}{d\tau_t}\right)}_{=\mathcal{E}_{1,t}^i}\right\} \leq 0,$$

Using equilibrium consumption functions from appendix A.1, we have the derivatives of

indirect utility for retired households:

$$\mathcal{E}^0_{2,t} = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}}{\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_t}{1+\gamma_0\epsilon}} \tag{B.11}$$

$$\mathcal{E}^i_{2,t} = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}. \tag{B.12}$$

We note that the marginal effect for formal retirees is the same as when continuation policies were independent of state variables. The only difference is that the change in the aggregate labor supply, $\frac{d\ln(h_t)}{d\tau_t}$, now includes strategic effects through future policies.

For the formal, working households, we can use the Euler equation to write the indirect utility as $\ln\left(\tilde{c}^i_{1,t}\right)(1+\beta) + \beta\ln(\beta R_{t+1})$. The marginal effect on indirect utility then consists of the two terms:

$$\begin{aligned}
\mathcal{E}^i_{1,t} =& (1+\beta)\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} + \beta\frac{d\ln(R_{t+1})}{d\tau_t} \\
\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} =& \frac{1+\xi}{\xi}\frac{d\ln(h_t)}{d\tau_t} + \frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta_{t+1}}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi} + \Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta_{t+1})}{1+\gamma_0\epsilon}} \\
\frac{d\ln(R_{t+1})}{d\tau_t} =& (1-\alpha)\left(\frac{d\ln(h_{t+1})}{d\ln(s_t)} - 1\right)\frac{d\ln(s_t)}{d\tau_t}.
\end{aligned}$$

where we use the shorthand $d\tau_{t+1}/d\tau_t = \frac{d\tau_t^{+1}}{d(s_{t,0}/s_t)}\frac{d(s_{t,0}/s_t)}{d\tau_t}$ to define the total effect of a change in the tax rate on future taxes. The technical documentation gives the computation of these derivatives in the numerical application (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

For the informal, working households, a similar Euler condition applies such that we can write indirect utility as $\ln\left(\tilde{c}^0_{1,t}\right)(1+\beta) + \beta\ln\left(\beta R^0_{t+1}\right)$. The marginal effect on indirect utility is then characterized by

$$\begin{aligned}
\mathcal{E}^0_{1,t} =& (1+\beta)\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t} + \beta\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t} \\
\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t} =& \frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\Theta_{s,t}\left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1}\frac{d\ln(s_t)}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon}\Theta_{s,t}} \\
\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t} =& (1-\alpha)\left(\frac{d\ln(h_{t+1})}{d\ln(s_t)} - 1\right)\frac{d\ln(s_t)}{d\tau_t}
\end{aligned}$$

B.3 With CRRA-GHH preferences and informal savings

With CRRA preferences it is no longer the case that future policy is independent of the capital stock. But conditional on this state variable, with GHH preferences it is still the case that the distribution of savings for formal workers, $\{s_{t-1,i}/s_{t-1}\}_{i>0}$, remains a function of s_{t-1} and τ_t and is thus not an independent state variable. Thus, the policy function can be written as a function of two endogenous states: s_{t-1} and $s_{t-1,0}/s_{t-1}$.

Compared to the log-GHH case, we adjust the definition of the politico-economic equilibrium derivative $dx_t/d\tau_t$ to include the effect through both states such that

$$\frac{dx_t}{d\tau_t} = \frac{\partial x_t}{\partial \tau_t} + \frac{\partial x_t}{\partial \tau_{t+1}} \left(\frac{\partial \tau_t^{+1}}{\partial s_t} \frac{ds_t}{d\tau_t} + \frac{\partial \tau_t^{+1}}{\partial (s_{t,0}/s_t)} \frac{d(s_{t,0}/s_t)}{d\tau_t} \right).$$

The first order condition identifying the politico-economic choice τ_t is now given by

$$\begin{aligned} &\omega \sum_{j\geq 0} \gamma_j \mu_j \underbrace{\left(c_{2,t}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(c_{2,t}^j\right)}{d\tau_t}}_{\equiv \mathcal{E}_{2,t}^j} \\ &+\nu_t \sum_{j\geq 0} \gamma_j \mu_j \underbrace{\left(\left(\tilde{c}_{1,t}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(\tilde{c}_{1,t}^j\right)}{d\tau_t} + \beta \left(c_{2,t+1}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(c_{2,t+1}^j\right)}{d\tau_t} \right)}_{\equiv \mathcal{E}_{1,t}^j} \leq 0, \end{aligned}$$

The log-derivatives $d\ln\left(c_{2,t}^j\right)/d\tau_t$ follow from the formulas for the log-GHH case and consumption levels $c_{2,t}^j$ depend on the state s_{t-1} and are given in appendix A.1.

For formal working households, we use the Euler equation to write indirect utility as

$$\begin{aligned} v_{1,t}^i =& (1+B_{t+1}) \frac{\left(\tilde{c}_{1,t}^i\right)^{1-1/\rho}}{1-1/\rho} = \frac{\left(\hat{c}_{1,t}^i\right)^{1-1/\rho}}{1-1/\rho} \\ \hat{c}_{1,t}^i =& \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} (1+B_{t+1})^{\frac{1}{\rho-1}} \mathcal{C}_{1,t}^i \\ \mathcal{C}_{1,t}^i =& \frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{1+\xi} + \Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta_{t+1})}{1+\gamma_0 \epsilon} \end{aligned}$$

The marginal effect of taxes can then be summed up by

$$\begin{aligned} \frac{dv_{1,t}^i}{d\tau_t} =& \left(\hat{c}_{1,t}^i\right)^{1-1/\rho} \frac{d\ln\left(\hat{c}_{1,t}^i\right)}{d\tau_t} \\ \frac{d\ln\left(\hat{c}_{1,t}^i\right)}{d\tau_t} =& \frac{1+\xi}{\xi} \frac{d\ln\left(h_t\right)}{d\tau_t} + \frac{B_{t+1}}{1+B_{t+1}} (1-\alpha) \left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)} - 1 \right) \frac{d\ln\left(s_t\right)}{d\tau_t} \end{aligned} \tag{B.13}$$

$$+\frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta_{t+1}}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi}+\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta_{t+1})}{1+\gamma_0\epsilon}} \tag{B.14}$$

For the young informal households, we can employ the Euler equation in a similar way and write:

$$\begin{aligned}
v_{1,t}^0 =& \frac{\left(\tilde{c}_{1,t}^0(1+B_{t+1}^0)^{\frac{\rho}{\rho-1}}\right)^{1-1/\rho}}{1-1/\rho} = \frac{\left(\hat{c}_{1,t}^0\right)^{1-1/\rho}}{1-1/\rho} \\
\hat{c}_{1,t}^0 =& \left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_s}(1+B_{t+1}^0)^{\frac{1}{\rho-1}}\mathcal{C}_{1,t}^0 \\
\mathcal{C}_{1,t}^0 =& \frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}
\end{aligned}$$

We then have that:

$$\frac{dv_{1,t}^0}{d\tau_t} = \left(\hat{c}_{1,t}^0\right)^{1-1/\rho}\frac{d\ln\left(\hat{c}_{1,t}^0\right)}{d\tau_t} \tag{B.15}$$

$$\begin{aligned}
\frac{d\ln\left(\hat{c}_{1,t}^0\right)}{d\tau_t} =& \frac{B_{t+1}^0}{1+B_{t+1}^0}(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t} \\
&+\frac{\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Theta_{s,t})}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}}
\end{aligned} \tag{B.16}$$

C Endogenous pension design

This appendix collects the derivations behind section 7: the corner that costless redistribution produces, the deadweight cost of redistribution, the equilibrium with the cost, and the state independence of the design choice under log preferences. Throughout there are no informal households, $\gamma_0 = 0$, so a retiree of type i receives the benefit $\nu_t w_t h_t \tau_t \left(1+\theta_t\left[y_i-1\right]\right)$, with y_i the relative labor income of appendix B.1 and $\sum_i \gamma_i y_i = 1$.

C.1 Costless redistribution

We consider two timings of the choice. If θ_t is chosen at t together with τ_t, it re-splits benefits that current retirees have already earned. By (11) the labor supply and savings of the young depend only on the design that applies to their own benefits, θ_{t+1}, so θ_t moves no decision and no state, and its marginal effect on the political objective (14) runs entirely through current retirees. With $c_{2,t}^i = R_t s_{t-1} D_{t,i}$ as in appendix B.1,

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega\,\frac{1-\alpha}{\alpha}\,\tau_t \sum_i \gamma_i \mu_i \frac{y_i-1}{D_{t,i}}, \qquad D_{t,i} = \frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\tau_t\left(1+\theta_t\left[y_i-1\right]\right). \tag{C.1}$$

The numerator is the change in the benefit of type i and averages to zero across types: a more Bismarckian split moves a given total from retirees below mean income to retirees above it. With equal propensities to vote the sum therefore weights the gains of the rich by a lower marginal utility than the losses of the poor.

Proposition 4. *Assume log-GHH preferences, no informal households and no deadweight cost of redistribution (section 7.1), and let the design θ_t that applies to the benefits paid at t be chosen at t together with τ_t. Let the propensities to vote be positive and not rise with relative income, $\mu_i \leq \mu_j$ whenever $y_i > y_j$, equal propensities included, and let relative incomes not all be equal. Then, for every $\omega > 0$ and $\tau_t > 0$, the derivative (C.1) is negative at every $\theta_t \in [0, 1]$ at which every retiree's consumption is positive, and the design chosen together with a positive tax is the Beveridgean corner, $\theta_t = 0$. This holds for all values of α, β, ξ, ν_t, γ_i, η_i and X_i, and whatever tax and design current retirees expected when they saved.*

Proof of proposition 4. Since $\sum_i \gamma_i(y_i - 1) = 0$, the sum in (C.1) equals $\frac{1}{2}\sum_{i,j} \gamma_i\gamma_j(y_i - y_j)\left(\mu_i/D_{t,i} - \mu_j/D_{t,j}\right)$. By (A.8) at $t-1$, the savings share of a current retiree rises with y_i with a slope of at least one, whatever tax and design generation $t-1$ anticipated for t, so $D_{t,i}$ rises with y_i. Where every retiree's consumption is positive, $\mu_i/D_{t,i}$ therefore falls with y_i, every term with $y_i \neq y_j$ is negative, and so is the derivative. Finally, $D_{t,i}$ falls with θ_t only for types below mean income, so the designs at which every retiree's consumption is positive form an interval that starts at $\theta_t = 0$, on which $\mathcal{W}_t$ is strictly decreasing. □

The proof uses the propensities to vote only through $\mu_i/D_{t,i}$, which orders the types as $\mu_i/c^i_{2,t}$ does. The derivative is therefore negative at every design at which propensities to vote rise with relative income, if at all, by proportionally less than consumption in retirement, $\mu_i/\mu_j < c^i_{2,t}/c^j_{2,t}$ whenever $y_i > y_j$. In the political objective μ_i is the weight of type i relative to its share of the population, of which turnout is the observable part. A responsiveness to policy that differed across types, the density of swing voters in probabilistic voting (Lindbeck and Weibull 1987), would multiply μ_i, so the threshold bounds the product of turnout and responsiveness, and the calibration measures only the first. Under CRRA-GHH preferences the design still enters only the consumption of current retirees, and relative savings keep the form of (A.8), so $\mu_i D_{t,i}^{-1/\rho}$ takes the place of $\mu_i/D_{t,i}$ and the threshold becomes $\mu_i/\mu_j < (c^i_{2,t}/c^j_{2,t})^{1/\rho}$.

Propensities to vote that rise with income can overturn the corner, and they do rise in the U.S. calibration of section 6, so there the sign is a quantitative matter. With the middle income group at the mean, the sign of (C.1) is set by the threshold for the high- and the low-income group, which binds at $\theta_t = 0$, where the ratio of their consumption in retirement is lowest, 3.50 in 2020. The corner therefore survives until the weight of the

high-income group is 3.5 times that of the low-income one at $\rho = 1$, twice the observed turnout ratio of 1.61. This critical ratio is 12.2 at $\rho = 0.5$ and 1.87 at $\rho = 2$. Along the projection a larger system compresses consumption in retirement, and at $\rho = 2$ the critical ratio falls to 1.80 from 2080 on, still above the observed ratio, so at every date and at each of the three elasticities the electorate that sets the design with the tax picks the Beveridgean corner.

If instead θ_{t+1} is chosen at t, the design no longer touches the benefits of current retirees but moves current labor supply and savings, directly and through the anticipated response of τ_{t+1}, and

$$\frac{\partial \mathcal{W}_t}{\partial \theta_{t+1}} = \sum_i \gamma_i \mu_i \left[\omega \left(1-\alpha\right) \frac{d \ln \Theta_{h,t}}{d\theta_{t+1}} + \nu_t \left((1+\beta) \frac{d \ln \tilde{c}^i_{1,t}}{d\theta_{t+1}} + \beta \frac{d \ln R_{t+1}}{d\theta_{t+1}} \right) \right], \tag{C.2}$$

where each total derivative includes the response of τ_{t+1}, and of θ_{t+2}, to the design chosen. Retirees are unanimous, since their term does not depend on i, and at a given anticipated tax they favor a more earnings-related system, which raises the return to an hour worked at t and with it the labor supply on which their benefits are levied. The young are divided between this stake and the redistribution of their own future benefits, the force of (C.1) one period ahead. At observed U.S. inequality and under log preferences the redistributive force dominates, and the choice is again the corner $\theta = 0$. Section 7 adopts this timing: it lets the design move labor supply and savings before it binds, without the commitment that a design chosen once and for all would require.

C.2 The deadweight cost of redistribution

This subsection derives the form of the cost in section 7.1, from the benefit formula to $f(\theta_t, \tau_t)$. The form follows from the tax component of the contribution and from Harberger's principle, applied on margins the model does not contain, rather than from a model of participation and formality, and its size is left to the calibration.

The implicit tax. A retiree of type i receives $\nu_t w_t h_t \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right)$ on past earnings $w_{t-1} y_i h_{t-1}$, that is $\Pi_t \tau_t \left[\theta_t + (1-\theta_t)/y_i\right]$ per unit of past earnings, where $\Pi_t \equiv \nu_t w_t h_t / (w_{t-1} h_{t-1})$ is the pay-as-you-go return factor. A purely earnings-related system that pays out the same revenue returns $\Pi_t \tau_t$ per unit of past earnings to every type. The shortfall of what type i receives from that benchmark is the implicit tax of (15), $t_i = \Pi_t \tau_t (1-\theta_t)(1 - 1/y_i)$ per unit of past earnings, which the flat component levies: a tax on types above mean income and a subsidy to types below it, proportional to the flat share $1-\theta_t$ and to the size of the system τ_t. Weighted by earnings, the implicit taxes are

a pure transfer,

$$\sum_i \gamma_i y_i t_i = \Pi_t \tau_t (1-\theta_t) \sum_i \gamma_i (y_i - 1) = 0, \tag{C.3}$$

since $\sum_i \gamma_i y_i = \sum_i \gamma_i = 1$: per unit of mean past earnings, what types above mean income pay, types below it receive. The implicit tax is the redistributive part of the tax component of the contribution in the sense of Summers (1989), the part of the payroll charge that buys the contributor no benefit.

Which margin. The implicit tax t_i is an average wedge: it compares what type i receives per unit of its earnings with the benchmark. The marginal wedge is a different object. The flat component depends on aggregate and not on own hours, so an extra unit of earnings raises a worker's own benefit by $\Pi_t \tau_t \theta_t$ whatever the type, and the marginal wedge of the flat component against the benchmark, $\Pi_t \tau_t (1-\theta_t)$, is common to all types. That wedge is already in the model. For the young at t, the marginal benefit of an hour, paid at $t+1$, enters the labor supply function (A.6) through the group $\frac{1-\alpha}{\alpha}\theta \tau_{t+1} \Gamma_{s,t}$, hours respond to it with the Frisch elasticity ξ, and GHH preferences carry no income effect through which the transfers between types could move hours. A cost built on the hours margin would count that response twice and could not depend on the spread of incomes. The cost therefore stands for margins the model does not contain and on which the average wedge is the relevant one, participation and formality: a worker who decides whether to bring earnings into the system, or into the formal sector, compares the earnings at stake with what they buy, so the wedge that matters is the average tax on those earnings (Saez 2002; Kleven and Kreiner 2006). Participation is the margin on which Disney (2004) finds the tax component to matter. The reporting of earnings, on which Kumler et al. (2020) find it, is a different margin: how much of a given wage to report responds to the marginal wedge $\Pi_t \tau_t (1-\theta_t)$, which does not vanish when incomes are equal, so a loss on that margin would not depend on the dispersion of relative incomes. The cost does not contain it, and we read the evidence on reporting as evidence that the tax component matters rather than as a margin the cost prices.

The Harberger loss. The implicit tax falls on the benefit paid at t per unit of earnings made at $t-1$, which a worker values at $1/R_t$ per unit when the earnings are made, as in the labor supply condition (6), so the wedge on a unit of earnings at $t-1$ is t_i/R_t. Per unit of mean past earnings, $w_{t-1}h_{t-1}$, the base of type i is its earnings, $e_i = y_i$. On each base the loss is a Harberger triangle (Harberger 1964). A base e that responds to its net-of-tax rate with elasticity ε shrinks under a small proportional wedge x by $\varepsilon x e$ to first order, and the units it loses were each worth between zero and x more than they cost to supply, $x/2$ on average, so the excess burden is the area of the triangle, $\frac{1}{2} x \cdot \varepsilon x e = \frac{1}{2} \varepsilon x^2 e$, to second order in x. Under a subsidy, $x < 0$, the base grows by units that cost between

zero and $-x$ more to supply than they are worth, and the same triangle measures the loss. Each base responds to its own wedge only, so the losses of the types add up, and with a common elasticity they sum, per unit of mean past earnings and valued at $t-1$, to

$$\begin{aligned}\frac{1}{2}\varepsilon\sum_i \gamma_i \left(\frac{t_i}{R_t}\right)^2 y_i &= \frac{1}{2}\varepsilon\left(\frac{\Pi_t}{R_t}\right)^2 \tau_t^2(1-\theta_t)^2 \sum_i \gamma_i y_i \left(1-\frac{1}{y_i}\right)^2 \\ &= \frac{1}{2}\varepsilon\left(\frac{\Pi_t}{R_t}\right)^2 \tau_t^2(1-\theta_t)^2\,\tilde{V}, \end{aligned} \tag{C.4}$$

since $y_i(1-1/y_i)^2 = (y_i-1)^2/y_i$, with $\tilde{V}$ the dispersion of relative incomes of (16). The loss is taken out of the benefits paid at t, where it is worth R_t times (C.4), and the revenue that pays the retirees of t is $\Pi_t\tau_t$ per unit of their mean past earnings. The share of revenue lost is therefore

$$\frac{R_t}{\Pi_t\tau_t}\,\frac{1}{2}\varepsilon\left(\frac{\Pi_t}{R_t}\right)^2 \tau_t^2(1-\theta_t)^2\,\tilde{V} = \frac{1}{2}\,\varepsilon\,\frac{\Pi_t}{R_t}\,\tau_t\,(1-\theta_t)^2\,\tilde{V}. \tag{C.5}$$

Each factor of the share comes from one of these steps. The rate t_i/R_t is linear in the flat share and in the size of the system, and the loss is quadratic in the rate, so its level (C.4) is quadratic in $1-\theta_t$ and in τ_t. Revenue is itself proportional to τ_t, so the share (C.5) is quadratic in $1-\theta_t$ and linear in τ_t, and each unit of redistribution costs more in a larger system. The dispersion $\tilde{V}$ comes from wedges that differ across types as $1-1/y_i$ on bases proportional to y_i, and it is zero when all incomes are equal, where flat and earnings-related benefits pay everyone the same. The factor Π_t/R_t multiplies every rate alike, so it enters the level of the loss squared and its share of revenue once.

Why the excess burden, and why against revenue. The cost includes no loss of revenue of first order in the wedges, and the transfer property (C.3) is the reason. To first order the wedge t_i/R_t changes the base of type i by $-\varepsilon y_i t_i/R_t$ per unit of mean past earnings, so the aggregate base on which contributions are levied changes by $-(\varepsilon/R_t)\sum_i \gamma_i y_i t_i = 0$. With a common elasticity, the base that the implicit tax drives out above mean income is the base that the subsidy draws in below it, and the first-order responses cancel in revenue. What remains is the excess burden (C.4), a welfare cost that arises on the bases of subsidized types as on those of taxed ones. Charging it against the revenue that pays retirees, as (16) does, is a choice of reduced form rather than a consequence of the argument. The choice keeps the structure of the model, since the equilibrium with the cost is the one without it with $\tau_t f$ in the place of τ_t in the benefits (appendix C.3), and it books the loss in the period of the benefits, where the implicit tax falls (section 7.1). That the subsidy to types below mean income costs as the tax on types above it does is a property of the reduced form rather than of the margins it stands

for: where participation is already taxed, lowering the participation tax of low-income workers can reduce the total loss (Saez 2002).

From the loss to f. The budget (16) takes the share (C.5) out of revenue in exponential form, $f(\theta_t, \tau_t) = \exp\!\left(-\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2\right)$, with the cost parameter λ in the place of $\varepsilon\Pi_t/R_t$. In the budget f multiplies revenue, so it scales both components of the benefit alike and $1-f$ is the share of revenue lost; at $\lambda = \varepsilon\Pi_t/R_t$ it equals (C.5) to first order. The exponential keeps f in $(0,1]$ for every $\lambda \geq 0$ and every size of the system, whereas the share (C.5) itself exceeds one in a large enough system, and at $\lambda = 0$ it gives back the model without the cost, $f = 1$. Along the U.S. path of section 7.2 at $\rho = 1$ the factor Π_t/R_t falls as the population ages, from 0.55 in 2020 to 0.47 in 2110, and the U.S. results there absorb it into a constant λ, which the online appendix also gives the UK and France. To carry the cost across countries, section 7.3 and appendix G keep the factor: country n faces $\lambda_{n,t} = \varepsilon\,\Pi_{n,t}/R_{n,t}$, with ε the elasticity of (C.4), the same in every country.

Properties of f. Since $\ln f = -\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2$,

$$\frac{\partial \ln f}{\partial \theta_t} = \lambda\,\tau_t\,\tilde{V}\,(1-\theta_t) \geq 0, \qquad \frac{\partial \ln f}{\partial \tau_t} = -\frac{1}{2}\,\lambda\,\tilde{V}\,(1-\theta_t)^2 \leq 0,$$

so a more earnings-related design loses a smaller share of revenue and a larger system a larger one. Both derivatives vanish at $\theta_t = 1$, where $f = 1$, and since $d\,[f(1+\ln f)] = f(2+\ln f)\,d\ln f$, so do those of the factor $f(1+\ln f)$ of the tax condition (C.6). The benefits and the tax, through which alone the design enters the equilibrium of appendix C.3, therefore have at the Bismarckian corner the values and slopes they have without the cost. Under log preferences the design chosen does not move the designs chosen later (proposition 5), so the slope of the political objective at the Bismarckian corner and the tax chosen there do not depend on λ, and the cost cannot by itself make the corner a local maximum of the political objective. Scaling both components alike is a choice. A cost on the flat component alone delivers the same endpoints and differs in the interior, but it loses the separate identification of θ and λ (appendix C.3). A cost on the design itself rather than on the transfer it performs, $f(\theta) = \phi + (1-\phi)\theta^p$, stays in place when a compressed income distribution removes the redistributive stakes; the online appendix reports it.

Magnitude. The shape of f follows from the tax component; its size does not. In the U.S. calibration for 2020 the return factor is $\Pi_t = 1.35$, close to ν_t (table 3), and the thirty-year return to which β is calibrated is $R_t = 2.44$ (appendix F), so that Π_t/R_t is about 0.55. Participation and taxable-income elasticities ε of 0.1 to 0.4 (Chetty et al. 2011; Saez et al. 2012) then correspond to $\lambda = \varepsilon\Pi_t/R_t$ between about 0.06 and 0.22. The calibrated values of table 4 are more than an order of magnitude above that range at

$\rho = 0.5$ and 1, 28 to 113 times as large at $\rho = 1$, and 6 to 25 times as large at $\rho = 2$, where λ is smallest, so λ also stands in for administrative, evasion and political costs of redistribution that the model does not contain.

C.3 The equilibrium with the cost

With the cost, a retiree of type i receives $f(\theta_t, \tau_t)\, \nu_t w_t h_t \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right)$, and the model without the cost is $f = 1$.

Economic equilibrium. In appendix A, τ_{t+1} enters only through the benefits it pays the current young, and τ_t through the benefits of current retirees and the net wage. The cost is therefore exactly the substitution of $\tau_{t+1} f(\theta_{t+1}, \tau_{t+1})$ for τ_{t+1} throughout appendix A, and of $\tau_t f(\theta_t, \tau_t)$ for τ_t in the benefits of current retirees, with the net wage $(1 - \tau_t) w_t$ unchanged. Here $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$, with $\mathcal{T}_{t+1}$ the equilibrium tax function below, so the young anticipate both the design and the size of the system that will pay them.

The tax condition. In the first order condition of appendix B.1, the substitution changes only the benefit term of $\mathcal{E}^i_{2,t}$, in which $\tau_t f$ replaces τ_t,

$$\mathcal{E}^i_{2,t} = (1 - \alpha) \frac{\partial \ln\left(\Theta_{h,t}\right)}{\partial \tau_t} + \frac{1 - \alpha}{\alpha} \left(1 + \theta_t \left[y_i - 1\right]\right) \frac{f\left(1 + \ln f\right)}{D_{t,i}}, \tag{C.6}$$

with $D_{t,i}$ evaluated at $\tau_t f$. The factor $f(1 + \ln f) = \partial(\tau_t f)/\partial \tau_t$, since $\ln f$ is linear in τ_t: a larger system is worth less to a retiree at the margin than the revenue it collects. The young's terms contain no policy but τ_t, so the condition involves θ_{t+1} nowhere, and the equilibrium tax at t is a function $\mathcal{T}_t(\theta_t)$ of the inherited design and of parameters and demographics alone, as in proposition 2.

Identification of θ. The benefit of type j per unit of mean past earnings, $\Pi_t \tau_t f \left(1 + \theta_t \left[y_j - 1\right]\right)$, is a line in relative earnings, and its earnings-linked share at mean earnings, the datum that identifies θ (appendix G.1), is θ_t, since f scales slope and intercept alike. The design is therefore identified separately from the cost, and $\theta^* = 0.544$ for the U.S. whatever λ. A cost on the flat component alone would give the share $\theta_t / \left(\theta_t + f(1 - \theta_t)\right)$, so that the same datum would imply a different θ for every value of the cost parameter.

The design condition. With the design chosen one period in advance, θ_{t+1} enters the objective through the split of the benefits of the young and through $\tau_{t+1} f(\theta_{t+1}, \tau_{t+1})$, with τ_{t+1} the anticipated equilibrium tax. Without the cost, a more Bismarckian design at a given anticipated tax shifts next period's benefits one for one from the flat to the earnings-related component, and the redistributive stake of the young dominates as in appendix C.1. With it, the design also raises the total the young will receive, since less

of the revenue is lost,

$$\frac{\partial f}{\partial \theta_{t+1}} = f\,\lambda\,\tau_{t+1}\,\tilde{V}\,(1-\theta_{t+1}) \geq 0, \tag{C.7}$$

which is what lets the design condition cross zero in the interior. The gain is zero at $\theta_{t+1} = 1$. It rises with the size of the system the young anticipate and with the dispersion of relative incomes if and only if $\ln f > -1$, and with the flat share $1-\theta_{t+1}$ if and only if $\ln f > -\frac{1}{2}$. The first condition holds wherever the anticipated tax is interior, since the benefit term of (C.6) carries the factor $1+\ln f$ and every other term of the tax condition is negative. At the 2020 tax rate the second holds at every design and every intertemporal elasticity: the loss is largest at a flat benefit, and there it is at most 26.6% of revenue, at $\rho = 0.5$ (table 4), less than the 39% at which $\ln f = -\frac{1}{2}$.

C.4 The design choice under log preferences

Proposition 5. *Assume log-GHH preferences and no informal households, and let the design be chosen one period in advance under the cost* (16)*. Then the political objective at t is additively separable,*

$$\mathcal{W}_t = \mathcal{A}_t\,(\tau_t, \theta_t) + \mathcal{B}_t\,(\theta_{t+1}) + \mathcal{C}_t\,(s_{t-1})\,, \tag{C.8}$$

where $\mathcal{C}_t$ depends on no policy. The tax chosen at t depends on the inherited design, θ_t, and on neither θ_{t+1} nor s_{t-1}; the design chosen at t maximizes $\mathcal{B}_t$ alone and depends on neither θ_t nor s_{t-1}.

Proof of proposition 5. By (A.6) with the substitution of appendix C.3, $\Theta_{h,t}$ is the product of a factor in τ_t and a factor in $(\tau_{t+1}, \theta_{t+1})$, and $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$, so $\ln h_t$ is the sum of a term in τ_t, a term in θ_{t+1} and a multiple of $\ln(s_{t-1}/\nu_t)$. By (A.5) and appendix A.1 the same holds for $\ln s_t$ and $\ln \tilde{c}^i_{1,t}$. The return $\ln R_{t+1}$ is affine in $\ln(s_t/\nu_{t+1})$ and $\ln \Theta_{h,t+1}$, which depends on τ_{t+1} and on the continuation $(\theta_{t+2}, \tau_{t+2})$. By backward induction from the terminal period, in which no design is chosen, the design chosen at $t+1$ depends on no state, so the continuation is a function of θ_{t+1} alone. Finally, $\ln c^i_{2,t}$ is $(1-\alpha)\ln h_t + \alpha \ln(s_{t-1}/\nu_t)$ plus $\ln D_{t,i}$, a function of (τ_t, θ_t) since by (A.8) at $t-1$ the savings shares of current retirees are. Every term of (14) is therefore a function of (τ_t, θ_t), of θ_{t+1} or of s_{t-1}, which is (C.8). The tax condition involves $\mathcal{A}_t$ alone, and the design chosen at t maximizes $\mathcal{B}_t$. □

The design chosen at t is thus the same whatever system the electorate inherits and whatever the capital stock, which is what section 4 uses: each date's choice is a search over a grid of designs with no state, the tax at each candidate design is found as in

proposition 2, and the condition $\theta_{2020} = \theta^*$ that calibrates λ in section 7.2 is a function of λ alone. The property is specific to log preferences. Under CRRA preferences the powers do not separate, both s_{t-1} and θ_t re-enter the design choice, and the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t) (Berg and Gonzalez-Eiras 2026).

D Terminal states and numerical checks

We consider two terminal states of the model: a finite-horizon version, in which the economy ends after $t = T$, and a steady-state version. The paper reports the finite-horizon solution, in periods of thirty years from 1960 to $T = 2200$ for the rich economies and from 1920 to $T = 2190$ for Argentina, with population growth held at its last projected rate after 2080 (2070 for Argentina). The appendix then reports three checks on the computed equilibria: how much they depend on the horizon and the terminal state, how many candidates and equilibria the solver finds, and how closely the policy functions solve their first order conditions between the nodes of their grids.

Finite horizon. In the finite horizon, consumption functions and indirect utility in economic equilibrium are those of the general case in every period $t < T$. For the young generations at time T, we simply say that

$$u_{T,j} = u_j\left(c^j_{1,T}, h_{T,j}\right).$$

In this case, the economic equilibrium at $t = T$ is the general formulation with $\beta = \tau_{t+1} = s_t = s_{t,j} = 0$. Specifically:

$$h_t = (\Gamma_h)^{\frac{1}{1+\alpha\xi}} \left((1-\alpha)(1-\tau_t)\right)^{\frac{\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}} \tag{D.1}$$

$$h_{t,i} = h_t \frac{(\eta_i/X_i)^\xi}{\Gamma_h} \tag{D.2}$$

$$\tilde{c}^i_{1,t} = \frac{1}{1+\xi} \frac{\eta_i^{1+\xi}}{X_i^\xi} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \tag{D.3}$$

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^\xi} \left(w^0_t\right)^{1+\xi}. \tag{D.4}$$

For the retired generation, the relevant consumption functions are unchanged.

For the politico-economic equilibrium, the marginal effect of τ_t on retirees is the same as for $t < T$. For working households, the effects are much simpler, namely:

$$\mathcal{E}^i_{1,t} = \left(\tilde{c}^i_{1,t}\right)^{1-1/\rho} \frac{1+\xi}{\xi} \frac{\partial \ln(h_t)}{\partial \tau_t} \tag{D.5}$$

$$\mathcal{E}^0_{1,t} = 0 \tag{D.6}$$

Steady state, infinite horizon. In the steady-state version, population growth stays at its last projected rate forever, and the equilibrium is the limit of finite-horizon equilibria as the horizon grows. We compute it as the finite horizon extended by 40 periods: the tax policy at the paper's T then differs from that of the period after by at most 1.1×10^{-6} over the states the path visits (table 6), so that T is the start of a steady state. In a steady state, where the tax and the design are constant, the equations for aggregate labor supply and savings combine into:

$$s^* = \Gamma_h^{1+\xi} \left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \right)^{1+\xi} \frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}} \right]^{\frac{1}{1-\alpha}} \tag{D.7}$$

$$\Gamma_s = \frac{1}{1+\xi} \frac{\Gamma_h B}{1 + B + \frac{1-\alpha}{\alpha} \frac{\tau}{1+\gamma_0\epsilon} (1 + \theta B)} \tag{D.8}$$

$$B = (\beta)^\rho (R^*)^{\rho-1}, \qquad R^* = \alpha \left(\frac{s^*}{\nu h^*} \right)^{\alpha-1}$$

$$\frac{s^*}{\nu h^*} = \frac{(s^*)^{\frac{1}{1+\xi}} \Gamma_s^{\frac{\xi}{1+\xi}}}{\nu \Gamma_h} = \left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \frac{\Gamma_s}{\nu} \right)^{\frac{1}{1-\alpha}} \tag{D.9}$$

We can use this to write the discount function as

$$B = (\beta)^\rho \left(\alpha \frac{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s}{(1-\alpha)(1-\tau)} \frac{\nu}{\Gamma_s} \right)^{\rho-1} \tag{D.10}$$

The horizon and the terminal state. Table 6 reports the paper's equilibria solved again with the backward induction started two periods earlier, three periods later and from the steady state: the U.S. with the design given and with the design chosen, and Argentina in its baseline and with the universal pension. Under log preferences the U.S. tax at a date before T cannot depend on the horizon, since without informal households the condition at each date does not involve the continuation (proposition 3). Under CRRA preferences and with informal savers the policy functions carry a state through which the horizon could reach the calibration year, but it matters little there. No tax rate in the calibration year changes by more than 0.03 p.p., no savings rate by more than 0.013 p.p. of GDP, and the design chosen in 2020 changes by less than 0.001. The largest change, 0.030 p.p., is in Argentina's response to the universal pension at $\rho = 0.5$ when the horizon is two periods shorter; in the U.S. it is 0.026 p.p., at $\rho = 0.5$ under the steady state. One period later the changes are of the same size, at most 0.035 p.p. in a tax rate.

Equilibrium counts. Proposition 3 gives one equilibrium under log-GHH preferences

Table 6: The horizon and the terminal condition

Outcome	ρ	**Paper's T**		**Shorter**		**Longer**		**Steady state**	
United States		2020	2050	2020	2050	2020	2050	2020	2050
Tax rate (%), design given	0.5	10.52	15.94	−0.010	−0.011	+0.010	+0.001	+0.026	+0.018
	1.0	10.52	15.69	0.000	0.000	0.000	0.000	0.000	0.000
	2.0	10.52	15.16	−0.005	0.000	0.000	−0.002	0.000	−0.002
Savings rate (% of GDP), design given	0.5	15.86	14.68	+0.005	−0.002	−0.003	−0.002	−0.012	−0.013
	1.0	15.33	14.19	0.000	0.000	0.000	0.000	0.000	0.000
	2.0	14.83	13.84	0.000	−0.001	0.000	0.000	0.000	0.000
Design θ, chosen	0.5	0.544	0.654	0.000	0.000	0.000	0.000	0.000	−0.001
	1.0	0.544	0.662	0.000	0.000	0.000	0.000	0.000	0.000
	2.0	0.544	0.663	0.000	0.000	0.000	0.000	0.000	0.000
Tax rate (%), design chosen	0.5	10.53	15.54	+0.011	+0.002	+0.002	−0.006	−0.006	−0.035
	1.0	10.52	15.26	0.000	0.000	0.000	0.000	0.000	0.000
	2.0	10.60	15.07	−0.001	0.000	0.000	0.000	0.000	0.000
Savings rate (% of GDP), design chosen	0.5	16.04	14.94	−0.004	−0.004	+0.001	+0.006	+0.009	+0.019
	1.0	15.42	14.34	0.000	0.000	0.000	0.000	0.000	0.000
	2.0	14.82	13.84	0.000	0.000	0.000	0.000	0.000	0.000
Argentina		2010	2040	2010	2040	2010	2040	2010	2040
Tax rate (%)	0.5	11.81	20.11	−0.005	+0.028	0.000	−0.001	−0.003	−0.005
	1.0	11.81	19.75	0.000	−0.003	0.000	0.000	0.000	0.000
	2.0	11.81	18.80	+0.001	−0.002	0.000	0.000	0.000	0.000
Savings rate (% of GDP)	0.5	13.83	12.33	−0.003	−0.011	0.000	0.000	+0.001	+0.001
	1.0	13.33	11.56	0.000	−0.001	0.000	0.000	0.000	0.000
	2.0	12.78	10.93	0.000	0.000	0.000	0.000	0.000	0.000
Universal pension: change in the tax rate (p.p.)	0.5	−0.42	+1.75	+0.030	−0.016	+0.001	−0.004	+0.004	−0.001
	1.0	+1.95	+3.47	−0.008	+0.021	0.000	0.000	0.000	0.000
	2.0	+2.03	+3.29	−0.006	−0.008	0.000	−0.001	0.000	−0.001

Note: The first two columns are levels at the paper's horizon, $T = 2200$ in the U.S. and 2190 in Argentina; the others are changes against them at the same date, with the backward induction started two periods earlier, three periods later, or from the steady state of appendix D. Tax and savings rates in percent, their changes in percentage points, the design in units of θ. The design is chosen one period ahead at the calibrated cost; in Argentina the universal pension is that of section 5, as a change against the baseline of the same horizon.

with no informal households, or with hand-to-mouth ones and $\xi(1 - 2\alpha) \leq 1$. With informal savers, as in every Argentine result, and under CRRA preferences we check uniqueness numerically. The candidates for the tax at a state are the two corners and every crossing of the first order condition. At every state and date the solver records their number and the number that pass the equilibrium test of section 4, and the exact recursion with the design chosen under CRRA preferences records the same for the design.

Every solve behind the paper's results records these counts, and in every one exactly one candidate passes at every state and date. Since the corners are always candidates, the count of candidates for the tax is at least two, and it never exceeds three, the corners and one crossing. The online appendix reports the counts by solve and file by file, for the results it adds as well.

Accuracy off the grid. The policy functions are computed at the nodes of a grid of states, and the equilibrium path enters each period between them. Table 7 compares the policy function at such states with the root of the period's first order condition there, located by the period's selection without the smoothing of the policy and against the solved continuation. On the path the two differ by about 0.001 p.p. of the tax rate at most in the U.S. and 0.01 p.p. in Argentina. Over the states the path visits, the difference stays at about 0.003 p.p. or less in the U.S. and below 0.02 p.p. in Argentina, except at $\rho = 0.5$ in the periods before 2010, where it reaches 0.22 p.p. There the policy, which the solver smooths across states, departs from the root at the nodes as much as between them. Larger differences arise only at states the path never visits. The condition carries derivatives in the tax that are computed numerically, and with derivatives converged on a finer tax grid the difference on the path is at most 0.003 p.p. in the U.S. and 0.014 p.p. in Argentina. Both solvers differentiate along $\ln(1-\tau)$, the U.S. one with a least-squares cubic spline whose knots are 0.04 apart in the tax and Argentina's with an interpolating spline, and the finer grid has 801 nodes in the U.S. and 401 in Argentina. Solved at every period with converged derivatives, the U.S. tax in 2020 is 10.52% at $\rho = 0.5$ and 2, as in the calibration, and the difference between $\theta = 0$ and $\theta = 1$ is 9.13 and 2.04 p.p., against 9.12 and 2.04 p.p. as solved. The U.S. results use a grid of 150 states; on one of 50 the difference on the path rises to 0.052 p.p. at $\rho = 0.5$.

Table 7: First order condition residuals off the grid

ρ	Grid	As solved				Converged derivatives			
		Path	Visited, max	Visited, 99%	All, max	Path	Visited, max	Visited, 99%	All, max
United States, CRRA preferences									
0.5	150	6.3×10^{-4}	0.0032	0.0027	0.48	0.0029	0.0051	0.0048	0.47
2.0	150	0.0010	0.0024	0.0022	0.41	0.0015	0.0016	0.0015	0.41
0.5	50	0.052	0.096	0.095	1.7	0.055	0.098	0.098	1.7
2.0	50	0.0059	0.017	0.016	1.8	0.0067	0.018	0.017	1.7
Argentina									
0.5	45×45	0.0064	0.22	0.14	0.35	0.0098	0.23	0.14	0.34
1.0	45	0.0012	0.0023	0.0021	0.0028	0.0024	0.0046	0.0039	0.0046
2.0	45×45	0.0082	0.014	0.011	0.014	0.014	0.029	0.025	0.029

Note: Each cell is the largest (or 99th percentile) absolute difference, in percentage points of the tax rate, between the root of the first order condition at a state and the policy function there, over the periods of the paper's horizon: at the state the equilibrium path enters each period with, at the centers of the state grid's cells within the range the path visits (padded by a fifth), and at the centers of all its cells. The grid column gives the nodes of the state grid. The first four columns evaluate the condition as the solver does, the last four with its derivatives in the tax converged (appendix D). Log preferences in the U.S. have no state.

E Calibration – Argentina

This appendix holds the data, the identifying formulas and the sources behind table 1, which section 5 summarizes.

E.1 The household survey

The Encuesta Permanente de Hogares (EPH) of the Instituto Nacional de Estadística y Censos (INDEC) is Argentina's main household survey of employment, hours and income, and the basis of its official labor market statistics. It is a continuous quarterly survey of the urban agglomerates, 31 metropolitan areas at present, in which households are drawn by a stratified multi-stage design. We use the last two quarters of 2006, a phase of sustained expansion after the recovery from the 2001–2002 crisis, at the start of the expansion of coverage under the amnesties (figure 5.1) and before the Great Recession. The survey asks respondents whether they contribute to social security, which separates formal from informal workers, and we keep those aged 25 to 60.

E.2 The pension system before and after the reform

Before the tax amnesties, retirees over the age of 70 who had made contributions for a minimum of 10 years were eligible for basic pension coverage. This coverage consisted of

70% of the minimum pension and a proportional amount from other contributions.[42] We calculate the pre-reform ϵ as

$$\epsilon = 0.7\left(1-\theta+\theta\frac{\eta_1 h_{t-1,1}}{h_{t-1}}\right)\frac{\beta^{5/30\,\frac{\text{LE}_{\text{men}}-5}{\text{LE}_{\text{men}}}}+\beta^{10/30\,\frac{\text{LE}_{\text{women}}-10}{\text{LE}_{\text{women}}}}}{2},$$

where $\text{LE}_{\text{men}} = 14.45$ and $\text{LE}_{\text{women}} = 22.55$ are the life expectancies of men and women, respectively, at their retirement ages (women retire at age 60, men at 65), such that the formula above corrects for the basic pension being received 5 or 10 years after the formal retirement age.[43] We set γ_0 so that informal households are 32% of all households, the share of the population over 65 without a pension before the reform, in 2004 (Rofman and Oliveri 2012); since the formal shares sum to one, $\gamma_0 = 0.32/0.68 = 0.47$. These households include those aged 65 to 69 with at least ten years of contributions, who would draw the basic pension at 70, the delay for which ϵ is discounted, and those with fewer than ten years, who would never draw it. Section 5 also reports the reform when every informal household is of the second kind, with $\epsilon = 0$ before the reform and the model recalibrated to the same targets. To identify the Bismarckian incentives, we use OECD data on the ratio of replacement rates, RR, at half mean and mean wages for males and females.[44] Specifically, we adjust θ to ensure that

$$RR \equiv \frac{\theta+(1-\theta)\frac{h_t}{\eta_3 h_{t,3}}}{\theta+(1-\theta)\frac{h_t}{\eta_1 h_{t,1}}} = \frac{0.678}{0.803},$$

where types 1 and 3, the first and the third quartiles, are those whose mean incomes are closest to half the formal average and to the average, at 0.46 and 0.97 times that average.

E.3 Technology, demographics, and the aggregate targets

The capital income share is that of the formal sector. In the model, informal households produce outside Y_t and K_t, and national accounts book the output of unincorporated household enterprises, where most informal work sits, as mixed income, which a share computed as one minus the wage bill over GDP counts entirely as capital (Gollin 2002).

[42] The basic pension was set by Law 24,347 of 1994 at 70% of the first pillar, denoted "prestación básica universal", plus the corresponding fully-funded or pay-as-you-go benefit. For simplicity we assume the latter on average was also 70% of the benefits received by workers that had at least 30 years of contributions. Thus, basic pensions were 70% of the benefits received by the least productive formal workers.

[43] To estimate life expectancy at retirement age for Argentina we conduct a regression analysis using data for men and women's life expectancies at birth and at age 65 for 40 OECD countries for the year 2000. We then use the fitted regression and data from the World Bank on male and female life expectancy at birth in Argentina in 2000.

[44] In a pure Bismarckian system the replacement rate would be the same at all wage levels, such that $RR = 1$, while in a pure Beveridgean it would double if we halve wages, implying $RR = 0.5$. Note that we combine men and women since we do not have data separately on the increase in coverage or government spending by gender.

The economy-wide share for Argentina is 0.43 (Frankema 2010). If a share y^0 of GDP is informal mixed income, the formal-sector share is $(0.43-y^0)/(1-y^0)$. Informal earnings of about 5% of GDP, as in the household survey, give 0.40, while shadow-economy estimates of 20–25% of GDP give 0.24 to 0.29. We take $\alpha = 0.35$, which also puts Argentina on the same footing as the Gollin-adjusted shares we use for the rich economies in appendix F. We set the Frisch elasticity of labor supply to $\xi = 0.30$. To calibrate β, we target the capital-output ratio. A model period is thirty years and capital depreciates fully between periods, so the capital stock per worker in place at t is the previous generation's savings, $k_t = s_{t-1}/\nu_t$, and with Cobb–Douglas production $K_t/Y_t = \alpha/R_t$ — a stock relative to one period, i.e. thirty years, of output. Thirty times that ratio is the capital-output ratio as the national accounts report it. Targeting it is therefore essentially equivalent to targeting the thirty-year interest rate, which for Argentina is not credibly measured. We target $K/Y = 3.23$, Argentina's ratio in 2010.[45] The model matches it with capital over formal output, K_t/Y_t, the ratio that fixes R_t; over GDP, formal plus informal output, its ratio is 2.99. We calibrate β and ϵ numerically, since β depends on the politico-economic equilibrium path for savings and labor supply and ϵ depends on β.

To construct the time series for ν_t we follow Gonzalez-Eiras and Niepelt (2008) and use thirty-year gross growth of the total population in the estimates and medium-variant projections of the United Nations' World Population Prospects 2024. Finally, we use social security spending before the 2001 crisis of 7.1% of GDP,[46] to calibrate ω. Since the payroll tax falls on formal labor income, which is a share $1-\alpha$ of formal output, and formal output is a share $1-y^0$ of GDP when informal output is a share y^0, spending of 7.1% of GDP corresponds to an equivalent social security tax rate of $\tau_{2010} = 0.071/[(1-\alpha)(1-y^0)]$. Informal output is the income of informal households, which the informal income condition in (E.1) fixes relative to formal income, so that $y^0 = 0.075$, between the survey's 5% and the shadow-economy estimates above, and $\tau_{2010} = 0.118$. This is the tax of the pre-reform economy in 2010, which we target at pre-crisis spending since contributive spending returned to that level after 2010. Savings and spending are reported over the same GDP.

[45] Capital stock and GDP at constant national prices (`rnna`, `rgdpna`) from the Penn World Table 11.0 (Feenstra et al. 2015), retrieved through FRED. We read it at the calibration year, where the tax rate, the replacement-rate ratio, the coverage share and the household survey are all measured, rather than averaging. Correcting for the shadow economy would raise the capital-output target, since the Penn World Table stock is economy-wide and the informal sector is labor-intensive. We keep the measured ratio, because no source measures an informal capital share and the stock also includes housing, which cuts the other way.

[46] Total public pension spending, national and provincial schemes, 7.0% and 7.2% of GDP in 2000 and 2001 (Cetrángolo and Grushka 2020, table 13). We use pre-crisis spending as it was relatively stable during that time, and after 2010 contributive pensions returned to their pre-crisis levels. Following the 2001 crisis, pension spending was reduced, to 5.2% of GDP in 2005, by only allowing minimum pensions to adjust for inflation.

E.4 Productivity and hours

We use η_j and X_j to match the income distribution across formal households and the income and working hours of informal households. Specifically, let z_j^η and z_j^x denote income and hours worked of group j relative to the population-weighted average over the formal households, so that $\sum_{i>0}\gamma_i z_i^\eta = \sum_{i>0}\gamma_i z_i^x = 1$; informal households are measured against the same formal average. With labor supply $h_{t,i}/h_t = (\eta_i/X_i)^\xi / \sum_{j>0}\gamma_j \eta_j^{1+\xi}/X_j^\xi$ for formal households (cf. proposition 1) and $h_{t,0} = (\eta_0 w_t^0/X_0)^\xi$ for informal households, where w_t^0 is the informal wage, we solve the system

$$
\begin{aligned}
z_i^\eta &= \frac{w_t(1-\tau_t)h_{t,i}\eta_i}{w_t(1-\tau_t)h_t} = \frac{\eta_i^{1+\xi}/X_i^\xi}{\sum_{j>0}\gamma_j\eta_j^{1+\xi}/X_j^\xi}, \qquad i>0, \\
z_i^x &= \frac{h_{t,i}}{\sum_{j>0}\gamma_j h_{t,j}} = \frac{(\eta_i/X_i)^\xi}{\sum_{j>0}\gamma_j(\eta_j/X_j)^\xi}, \qquad i>0, \\
z_0^\eta &= \frac{w_t^0 h_{t,0}\eta_0}{w_t(1-\tau_t)h_t}, \qquad z_0^x = \frac{h_{t,0}}{\sum_{j>0}\gamma_j h_{t,j}}.
\end{aligned}
\tag{E.1}
$$

Because only relative income and hours are observed, both the level of efficiency units and the unit in which hours are measured are free. We spend the first on the normalization $\sum_{i>0}\gamma_i\eta_i^{1+\xi}/X_i^\xi = 1$, and take the taste for leisure to be common across the formal groups, $X_i = X$ for $i>0$. The formal income conditions then invert in closed form, $\eta_i = (z_i^\eta)^{1/(1+\xi)} X^{\xi/(1+\xi)}$, and they satisfy the normalization at any X, so the level of X is what fixes the unit in which hours are counted. We pin it down by requiring average formal hours in the baseline year, $\sum_{i>0}\gamma_i h_{t,i}$, to equal the average workweek of formal workers in the survey, 42.5 hours expressed as a share of discretionary time. Relative hours across the formal quartiles are then a prediction rather than a target, rising with relative income at elasticity $\xi/(1+\xi)$, and the online appendix compares the prediction with the survey. The aggregate h_t weights hours by productivity and is not average formal hours; the two differ by the factor $\sum_{i>0}\gamma_i(\eta_i/X_i)^\xi$, which the informal hours condition in (E.1) carries. That condition and the informal income condition pin down η_0 and X_0 given the equilibrium h_t in the baseline year, and we solve for them jointly with β and ω at $X=1$. The level of X then follows in one step from the solved hours, and η_0 and X_0 are re-derived at it. That last step leaves the equilibrium untouched, since the informal household reaches every aggregate through $\eta_0^{1+\xi}/X_0^\xi$, which the unit of hours does not move.

E.5 The vector-X_i calibration

The alternative to a common taste for leisure treats the survey's relative hours as data at every point of the formal distribution: with both z_i^η and z_i^x imposed, (E.1) identifies one X_i per quartile, and the level of hours is then not identified. Since the productivity

distribution reaches every aggregate only through $\sum_{i>0} \gamma_i \eta_i^{1+\xi}/X_i^{\xi}$, which both calibrations normalize to one, β, ω and every equilibrium outcome of section 5, the reform at every ρ and its dependence on ϵ and θ included, are identical under the two; only the productivity and leisure parameters and the reading of relative hours differ. The online appendix reports that calibration against the one used here.

F Calibration – the U.S. and France

This appendix documents the data and the household heterogeneity behind the calibrations of the U.S. and France, whose parameters are in table 3. The U.S. is the calibrated economy of sections 6 and 7, and France's household data supply the counterfactuals of section 6. Both calibrations are also used in the online appendix, where France and the UK choose their designs under the U.S. cost of redistribution. The UK is calibrated in the same way, from its own household survey, in the online appendix. Each country has three income groups of men aged 25 to 60. The tables report, group by group, the population share γ_i, the taste for leisure X_i and the productivity η_i that the calibration assigns, and the propensity to vote μ_i taken from the data. Under the calibration of section 6 the taste for leisure is common across groups, so the X_i row repeats one number, fixed by the country's average workweek, and the income distribution is carried by η_i alone; the online appendix describes the alternative, with one X_i per group identified from relative hours.

The parameters in the lower panel of table 3 are common to the two countries. The discount factor β matches the U.S. thirty-year gross interest factor $R_{2020} = 2.443$, the gross interest rate net of average multifactor productivity growth estimated by Gonzalez-Eiras and Niepelt (2008), and is imposed on France. The Frisch elasticity is $\xi = 0.30$, as for Argentina, and the capital share $\alpha = 0.30$.[47] Pension spending is public cash spending on old-age and survivors pensions in 2020 from the OECD Social Expenditure Database, the measure of figure 1.1. The number of workers per retiree ν_t is thirty-year gross growth of the total population in the estimates and medium-variant projections of the United Nations' World Population Prospects 2024, corrected for France as described below. The online appendix reports the calibration at every intertemporal elasticity of the model's grid.

F.1 The U.S.

For the U.S. we use the March Supplement 2019 of the Current Population Survey (CPS). The low and medium income groups are cut so that their mean incomes are half the mean

[47] Averaging adjustments 2 and 3 of Gollin (2002) gives capital income shares close to 0.3 for France and the U.S.

and the mean, and the remaining households form the high income group. The cuts set the population shares and the productivities, not the design: θ is the earnings-linked share of the public pension under the pension rules of 2020 (OECD 2021), measured in the same way for every country but the UK (appendix G.1). The propensity to vote is from the Voting and Registration Supplement of the CPS of November 2020, which the U.S. Census Bureau tabulates in Voting and Registration in the Election of November 2020.

Table 8 shows the two features that set the U.S. apart. Productivity rises steeply across the groups, from 0.82 in the low group to 2.52 in the high one, and the propensity to vote rises with income, from 0.47 to 0.77, so the political objective weighs high-income households well above their population share. The low group holds 58% of households.

Table 8: Household heterogeneity – the U.S.

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	4.07	4.07	4.07	Average hours worked.
η_i	0.82	1.38	2.52	Income distribution.
μ_i	0.47	0.63	0.77	Voting propensity.

F.2 France

For France we use the 2018 wave of the Luxembourg Income Study. France's design, $\theta = 0.822$, does not depend on the income cuts either, so we cut the groups at the U.S. percentiles, which puts the two countries' household types in one-to-one correspondence and makes the counterfactuals of section 6 a type-by-type replacement. The propensity to vote is from the Comparative Study of Electoral Systems (CSES).[48] Since the normal retirement age of men retiring in 2020 was 64.5 years in France, against 66 in the U.S. (OECD 2021, figure 3.7), we take France's population growth in 2020 to be

$$\frac{\text{Population } 2020 \times (30 - 1.5)/30}{\text{Population } 1990 + \text{Population } 2020 \times 1.5/30},$$

and the other dates in the same way, with a gap of 1.5 years in 1990 and 2020 and of 1 year from 2050 on, the gap between the normal retirement ages of men entering the labor market in 2020, 66 against 67.

[48] https://cses.org/.

At the same percentiles France's productivity profile is flatter than the U.S. one (table 9): it runs from 1.09 to 2.49, a ratio of the high to the low group of 2.29 against 3.08 in the U.S. (table 3). The propensity to vote is nearly flat, between 0.81 and 0.86. The common taste for leisure is 6.48 against 4.07 in the U.S., which is how the calibration accounts for France's shorter workweek.

Table 9: Household heterogeneity – France

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	6.48	6.48	6.48	Average hours worked.
η_i	1.09	1.54	2.49	Income distribution.
μ_i	0.81	0.86	0.85	Voting propensity.

G The OECD cross-section

This appendix documents the cross-country tests of sections 6 and 7.3: the sample and the data, the U.S. at those data, where the cost parameter is calibrated, each country's political weight of the old, and the test under other population data and other measures of the design. Every economy is solved under log preferences, without the cost in section 6 and with it in section 7.3, where the cost parameter of country n is $\lambda_{n,t} = \varepsilon\,\Pi_{n,t}/R_{n,t}$ at each date. Exactly one candidate tax passes at every state and date of every solve (appendix D). The online appendix shows each variant in full, and the test without Luxembourg and with every country in the model at the U.S. political weight.

G.1 Sample and data

The sample is the 21 countries of figure 1.1: the OECD members before 2000 in which informal employment is below a quarter of employment and one public pay-as-you-go system pays the bulk of retirement benefits. The first condition leaves out Korea, Mexico and Turkey.[49] The second holds where funded tiers pay less than a quarter of mandatory

[49]Informal employment is the share of informal employment in total employment, indicator 8.3.1 of the Sustainable Development Goals, from ILOSTAT, in the latest year available, 2018 to 2020. It is 56% in Mexico, 31% in Turkey and 29% in Korea. The next are the UK and the Czech Republic, at 20% and 17%, where the measure mostly reflects self-employment that their pension systems cover. Canada, Japan, New Zealand and the U.S. have no such measure and stay in. Korea and Turkey also stand apart in their pension systems: Korea's public pension dates only from 1988, and Turkey's retirement age for men is 52.

benefits in 2020 and the public pension is not tested against them, and leaves out Australia, Denmark, Iceland, the Netherlands and Switzerland. Funded pensions are private saving in the model, so where they are large, or where the public pension falls as they rise, the public benefit formula alone does not describe how retirement income depends on earnings.

The design is read off the benefit formula, in the same way for every country except the UK. We fit a least-squares line $a + c\,z$ through the gross public pension of a man with a full career at earnings z of 0.5, 1 and 2 times average earnings, under the rules of 2020 (OECD 2021, table 4.2), and take the line's earnings-linked share at average earnings, $c/(a+c)$.[50] In the model this share is θ: a retiree's benefit is proportional to $\theta y_i + 1 - \theta$ (section 7.1), a line in relative earnings whose earnings-linked share at mean earnings is θ, with the cost of redistribution or without it (appendix C.3). A full-career worker at half of average earnings never draws the minimum top-ups and tested benefits that lie below his own pension, although many retirees do, and that spending is flat. We therefore multiply the share by $1 - s$, where s is the share of public old-age and survivors cash spending in 2020 that goes to the top-ups and tested benefits this worker does not reach.[51] The measure puts the U.S. at 0.544, France at 0.822 and the UK at 0.181. Ireland and New Zealand, whose public pensions are flat, are at about 0, and no country is at 1. The Bismarckian index of figure 1.1 is $I = 1/(2-\theta)$ of this measure.

The other inputs are measured in the same way for every country, the U.S. included. The number of workers per retiree ν_t is the thirty-year growth of the total population in the medium variant of the United Nations' World Population Prospects 2024, from 1990 to 2080, with 1960 at the value of 1990 and the periods after 2080 at the value of 2080, as in the calibrations of section 6, corrected for each country's gap to the U.S. in the normal retirement age of men as France's is in appendix F.[52] Pension spending is that of figure 1.1, public cash spending on old-age and survivors pensions in 2020, and the tax rate

[50] For New Zealand the lowest earnings level is 0.6 times average earnings, as in the source. The formula is that of a worker who enters the labor market in 2020, and it describes the pensions paid in 2020 unless a reform is phasing in. The exception is the UK, where most retirees of 2020 draw the basic State Pension and the earnings-related State Second Pension, which the flat new State Pension replaced from 2016. Its θ is the State Second Pension's share of spending on the State Pensions and Pension Credit in 2019–20.

[51] The spending is that booked inside public old-age and survivors cash in the OECD Social Expenditure Database, taken from national administrative data and from the programme lines of the database; the technical documentation gives the source of each country's figure (Berg and Gonzalez-Eiras 2026). Italy's and Portugal's figures include the top-ups of invalidity pensions, and Belgium's leave out its minimum-pension supplement. In Norway and Poland the worker at half of average earnings reaches the minimum pension, so the formula already contains it and s is zero; for Luxembourg no figure is available and s is set to zero. Across the 21 countries s is at most 0.082, in Portugal.

[52] The gap is that of men retiring in 2020 for 1990 and 2020, and that of men entering the labor market in 2020 from 2050 on (OECD 2021, figure 3.7). A country whose normal age is g years below the U.S. one has $\nu_t(30-g)/(30+\nu_t g)$, the rule of appendix F, and one whose age is g years above it has the mirror image, $(30\nu_t + g)/(30-g)$, which counts the years its older cohort works beyond the U.S. age. The U.S. ratio is unchanged, and France's is that of appendix F.

is spending over the labor share. The income distribution is that of the gross earnings of men in full-time employment in the OECD earnings database, in 2019 or the nearest year, through a lognormal fitted separately below and above the median to the ratios of the first and ninth deciles to the median. Households are grouped at the U.S. income percentiles of section 6, so the three groups hold 58%, 18% and 24% of households in every country. The propensities to vote are each country's own, from the Comparative Study of Electoral Systems (CSES): turnout in a national election by quintile of household income, averaged over the percentiles of each income group.[53] Luxembourg has no CSES survey; since it enforces compulsory voting, its three groups vote at the same rate. Everything else takes the U.S. values of section 6: $\alpha = \xi = 0.30$, a taste for leisure common to all groups, which under log preferences sets hours but no rate, and the discount factor of the calibration without the cost, $\beta = 0.65$, imposed as on France in appendix F. Table 10 lists the inputs that differ across countries, except the propensities to vote.

G.2 The U.S. at these data

At these inputs the U.S. keeps the design of section 7, $\theta = 0.544$, since the measure is read off the benefit formula, and differs from its calibration of section 6 in two respects. Its dispersion of relative incomes is 0.333, against 0.436 at the CPS groups: the OECD series covers men in full-time work only and is fitted to three points of the distribution, so it misses the part-time and part-year earners that the CPS records. And its propensities to vote are those of the CSES, which rise less steeply with income than those of the CPS (table 8). With β imposed, we calibrate ω to the U.S. tax rate and ε so that the electorate of 1990 chooses the observed design for 2020, as in section 7.2, which gives $\omega = 1.26$ and $\varepsilon = 13.09$. At these values the cost takes 2.6% of the revenue of 2020 at the observed design and would take 12.0% at a flat benefit, a little less than at $\rho = 1$ in table 4. In the model every country takes these values of β and ε and, in place of the U.S. political weight, $\omega = 1.20$, the mean of the weights that match each country's pension spending at its observed design (appendix G.3). Without the cost, the political weight that reproduces U.S. spending at these data is 1.19, against 1.17 in table 3, and the test of spending of section 6 gives every country the mean of the weights that reproduce each country's spending without the cost, 1.17 (figure 6.2).

G.3 Each country's political weight

The model with each country's own political weight, own ω for short, keeps the U.S. β and ε but calibrates each country's political weight of the old so that the country's tax rate of 2020 at its observed design matches its pension spending, as ω is for the U.S. in

[53] The election is the one nearest 2020, within 2015 to 2022, with at least 30 respondents in each quintile. For Spain and Portugal, where no election qualifies, the elections are those of 2008 and 2009.

section 6. Each electorate then chooses its design as in section 7.3, and spending is read on the path of that design, so it matches the data only where the predicted design is the observed one. Table 10 reports the political weights and the predicted designs. The weight of the old runs from 0.86 in Hungary to 1.46 in Austria, against 1.26 in the U.S.; its mean over the 21 countries, 1.20, is the weight every country takes in the model. The predicted designs run from 0 in Ireland, whose observed design is also at the Beveridgean corner, to 0.711 in Italy, a narrower range than the observed one, which reaches 0.997 in Hungary. For 17 of the 20 countries other than the U.S. the predicted design lies on the same side of the U.S. design as the observed one. The exceptions are Belgium, predicted above it at 0.605, and Hungary and Luxembourg, predicted below it at 0.522 and 0.473. The UK is predicted at 0.503 and France at 0.626. Calibrated in full from their own household data and given the U.S. cost parameter, the two choose designs close to these, 0.493 and 0.645 (online appendix). The predicted designs keep the relations of table 5 with population growth, spending and the Gini (the starred rows of panel A of table 11).

G.4 Other population data and other measures of the design

Table 11 sets the paper's test beside its variants, and figures G.1 and G.2 the predicted designs. Panel A changes the population data: workers per retiree are left uncorrected for the retirement age, or corrected for each country's gap to the U.S. in the effective age of labor-market exit in place of the normal age (OECD 2021, figure 6.14). With own ω the test hardly moves, since the political weights are calibrated to spending: no slope changes by more than 0.05, and the correlation between predicted and observed designs stays at 0.77. The model keeps the signs of the three relations that are significant in the data, and its slopes on the Gini stay insignificant, but its predicted designs move more. At the exit age the correlation is 0.68, against 0.72 at the normal age. Uncorrected, it falls to 0.36, because of Luxembourg. Its population grew to 1.65 times its 1990 size by 2020, which at the mean political weight supports a tax of only 5.3% at its observed design, about two fifths of the 12.9% its spending implies, and its electorate of 1990 chooses a design of 0.144 for 2020, against the observed 0.820. Luxembourg's normal retirement age is four years below the U.S. one, and the correction brings its workers per retiree in 2020 to 1.17 and its predicted design to 0.461. Without Luxembourg, on the 20 countries with a CSES survey, the correlation in the model is 0.74 with ν uncorrected, between the 0.69 of the exit age and the 0.76 of the normal age on the same countries (the last rows of panel A and the online appendix).

Panel B changes the measure of the design. Each measure is its own cross-section, the data and the model calibrated to it, at the paper's population data. The public tier on 23 countries extends the paper's measure, the public tier, to every country whose public pension is not tested against funded tiers, which adds the Netherlands and Switzerland.

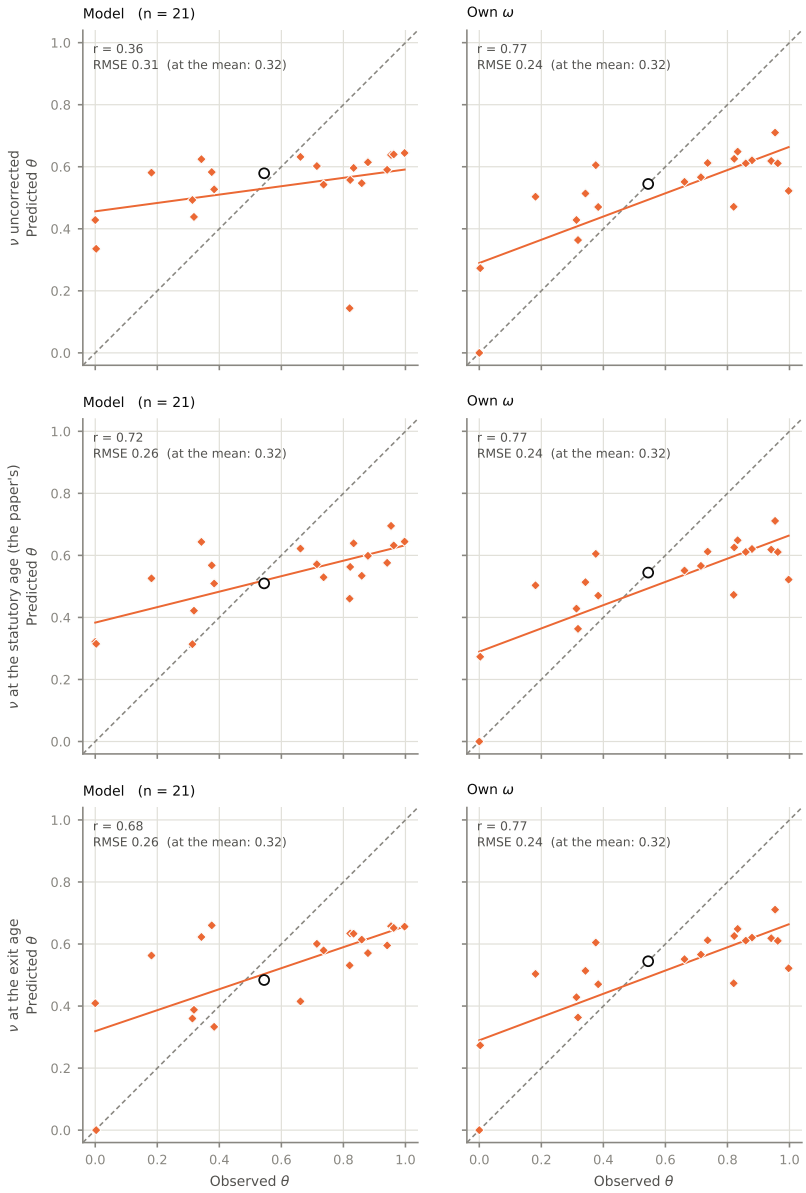
Table 10: The countries of the cross-country test, each calibrated to its pension spending

	Data					Model, own ω		
Country	ν_{2020}	$\tilde{V}$	spending	I	θ	ω	θ	I
Austria	1.08	0.161	14.3	0.791	0.736	1.456	0.612	0.720
Belgium	1.08	0.104	11.5	0.616	0.376	1.284	0.605	0.717
Canada	1.27	0.171	6.4	0.595	0.318	1.141	0.363	0.611
Czech Republic	0.87	0.170	8.6	0.603	0.342	0.896	0.514	0.673
Finland	1.03	0.094	12.6	0.944	0.941	1.308	0.619	0.724
France	1.04	0.150	14.4	0.849	0.822	1.416	0.626	0.728
Germany	1.03	0.148	11.2	0.778	0.714	1.190	0.566	0.697
Greece	0.79	0.225	17.9	0.857	0.833	1.266	0.649	0.740
Hungary	0.85	0.265	7.7	0.997	0.997	0.858	0.522	0.677
Ireland	1.42	0.240	3.3	0.500	0.000	1.026	0.000	0.500
Italy	0.80	0.134	17.4	0.956	0.954	1.275	0.711	0.776
Japan	0.96	0.124	9.6	0.747	0.662	1.032	0.551	0.690
Luxembourg	1.17	0.187	9.0	0.848	0.820	1.225	0.473	0.655
New Zealand	1.37	0.141	5.1	0.501	0.003	1.174	0.273	0.579
Norway	1.35	0.101	7.6	0.593	0.313	1.313	0.428	0.636
Poland	0.94	0.240	11.4	0.964	0.962	1.121	0.611	0.720
Portugal	0.99	0.249	13.6	0.892	0.879	1.296	0.621	0.725
Spain	1.13	0.183	12.8	0.877	0.859	1.430	0.611	0.720
Sweden	1.12	0.081	8.5	0.619	0.383	1.151	0.470	0.654
United Kingdom	1.17	0.185	7.8	0.550	0.181	1.167	0.503	0.668
United States*	1.34	0.333	7.4	0.687	0.544	1.257	0.544	0.687

Note: 21 countries. ν_{2020}, workers per retiree; $\tilde{V}$, the dispersion of relative incomes at the U.S. income percentiles; spending in % of GDP; θ and $I = 1/(2-\theta)$ as in appendix G.1. ω is calibrated to spending at the observed design (own ω, figure 7.3); the last two columns are the model's design for 2020, chosen in 1990, and its index. *The anchor, where ω and ε are calibrated.

All mandatory tiers takes the slope over all mandatory and quasi-mandatory tiers, public and funded (OECD 2021, table 4.1), with the same correction for top-ups, adds the spending of the funded tiers to the tax and covers all 26 countries in which informal employment is below a quarter of employment. The public tier on 23 countries leaves the test as it is, with correlations between predicted and observed designs of 0.69 in the model and 0.75 with own ω. All mandatory tiers keeps the signs of the significant slopes. Its data add both the funded tiers and the five countries where they are large or the public pension is tested against them, and in them the slope of the index on population growth is -0.38, against -0.44, while its slope on the Gini nearly vanishes, 0.26 against 0.88. The model predicts the design less well, with a correlation of 0.37, partly because of Iceland, which only this measure covers and which the model puts at 0.168 against an observed 0.813; without Iceland the correlation is 0.54. With own ω, where Iceland's political weight is calibrated to its spending, its design is 0.490 and the correlation is

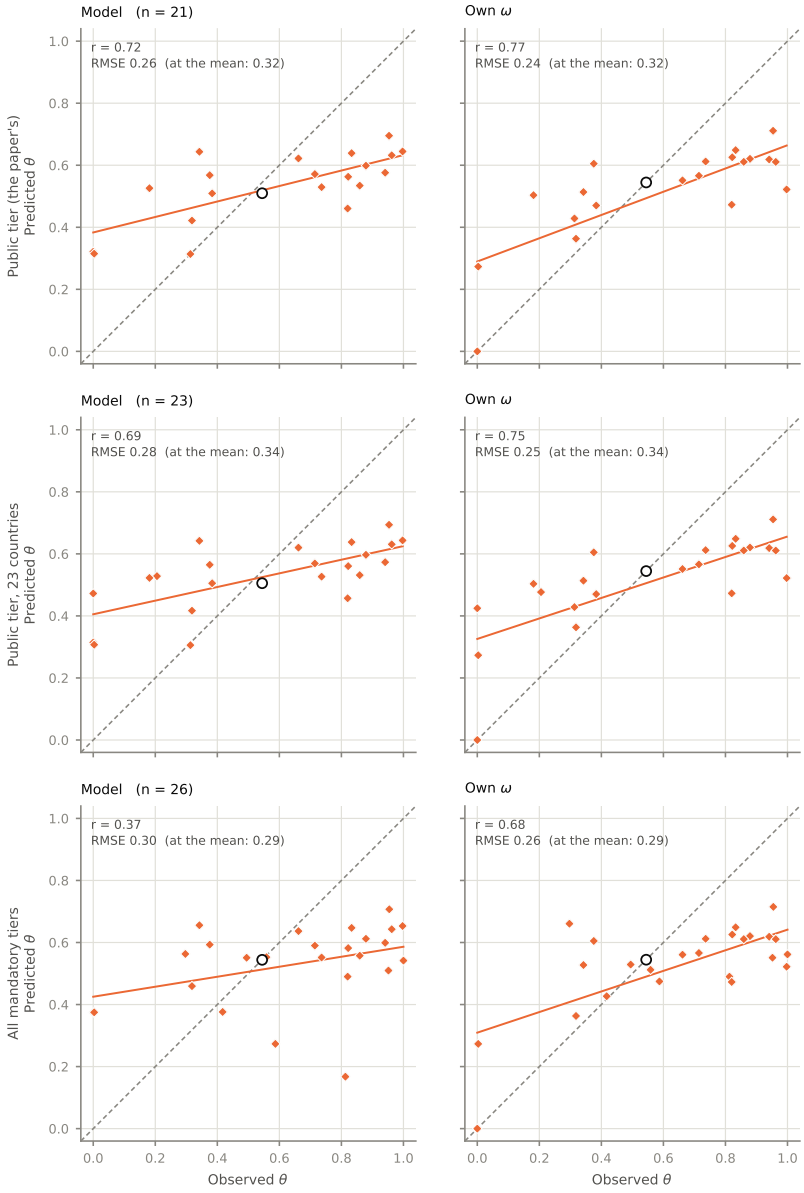
Figure G.1: The predicted and the observed design under three readings of the population data



Note: As figure 7.3, with ν corrected for each country's gap to the U.S. statutory retirement age (the paper's), uncorrected, or corrected for its gap in the effective age of labor-market exit. The U.S. is open; the solid lines are least-squares fits over the other countries.

0.68.

Figure G.2: The predicted and the observed design under three measures of the design



Note: As figure 7.3, under each measure of θ (appendix G.4), each its own cross-section with the U.S. anchor recalibrated under it, at the paper's ν. The U.S. is open; the solid lines are least-squares fits over the other countries.

Table 11: Figure 1.1's relations under other population data and other measures of the design

Variant		Spending on population growth	Spending on the Gini	Spending on I	I on population growth	I on the Gini
A. The population data, ν						
Data		−10.89	13.04	16.33	−0.44	0.88
		(4.11)	(25.40)	(3.91)	(0.18)	(1.06)
Uncorrected	Model	−15.36	−16.90	52.50	−0.26	−0.08
		(0.57)	(18.58)	(5.30)	(0.02)	(0.33)
	Own ω	−11.33	12.60	56.19	−0.22	0.31
		(4.08)	(25.55)	(6.32)	(0.06)	(0.40)
Statutory age*	Model	−14.70	−8.77	64.85	−0.22	0.08
		(2.71)	(22.90)	(5.86)	(0.04)	(0.33)
	Own ω	−11.34	12.59	56.20	−0.22	0.31
		(4.08)	(25.56)	(6.31)	(0.06)	(0.40)
Exit age	Model	−13.66	−20.02	62.34	−0.21	−0.10
		(4.21)	(27.54)	(4.72)	(0.06)	(0.42)
	Own ω	−11.34	12.58	56.24	−0.22	0.31
		(4.08)	(25.56)	(6.31)	(0.06)	(0.40)
Uncorrected, without Luxembourg	Data	−14.91	14.79	16.90	−0.77	0.80
		(4.92)	(26.28)	(3.94)	(0.18)	(1.10)
	Model	−16.34	−10.75	60.54	−0.23	0.08
		(0.59)	(16.91)	(6.24)	(0.02)	(0.26)
	Own ω	−15.82	14.10	56.33	−0.30	0.35
		(4.80)	(26.48)	(6.52)	(0.07)	(0.42)
B. The measure of the design, θ						
Public tier*	Data	−10.89	13.04	16.33	−0.44	0.88
		(4.11)	(25.40)	(3.91)	(0.18)	(1.06)
	Model	−14.70	−8.77	64.85	−0.22	0.08
		(2.71)	(22.90)	(5.86)	(0.04)	(0.33)
	Own ω	−11.34	12.59	56.20	−0.22	0.31
		(4.08)	(25.56)	(6.31)	(0.06)	(0.40)
Public tier, 23 countries	Data	−11.10	12.08	16.26	−0.45	0.87
		(4.05)	(24.53)	(3.44)	(0.18)	(1.07)
	Model	−14.72	−8.61	64.49	−0.22	0.10
		(2.67)	(21.75)	(5.84)	(0.04)	(0.31)
	Own ω	−11.55	11.45	57.27	−0.21	0.32
		(4.01)	(24.64)	(6.16)	(0.06)	(0.38)
All mandatory tiers	Data	−9.94	14.00	12.27	−0.38	0.26
		(3.36)	(20.88)	(3.84)	(0.16)	(0.94)
	Model	−16.15	−2.70	61.52	−0.24	0.21
		(2.45)	(22.03)	(5.45)	(0.04)	(0.33)
	Own ω	−10.35	13.46	55.26	−0.19	0.31
		(3.36)	(21.10)	(5.56)	(0.05)	(0.34)

Note: Least-squares slopes, standard errors in parentheses, the U.S. included. Panel A: ν corrected for each country's gap to the U.S. statutory retirement age, uncorrected, or corrected for its gap in the effective age of labor-market exit, on the 21 countries of the paper's measure; the data do not depend on ν. The last three rows leave out Luxembourg, which has no CSES survey, on 20 countries. Panel B: each measure of θ is its own cross-section, the data and the model calibrated to it, on its countries (public tier 21, public tier, 23 countries 23, all mandatory tiers 26), at the paper's ν. The model as in figure 7.2, own ω as in figure 7.3. *The paper's specification; its data and model rows are those of table 5.